

Beneficiary Yearbook

RESEARCH & ANALYSIS TRACK

A record of the individuals who trained, built, and interned across the Cohorts of 2024 and 2025. Projects centred on statistical analysis, modelling, and reporting.

105

BENEFICIARIES

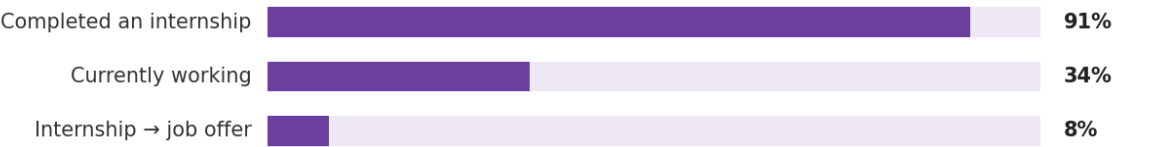
2

COHORTS

At a Glance



Program reach in three numbers



Internship figure is out of all 105 beneficiaries. “Currently working” is out of the 83 who reported a status, and “Internship → job offer” is out of the 38 who answered the follow-up form.

Who Trained Where

Beneficiaries by cohort year



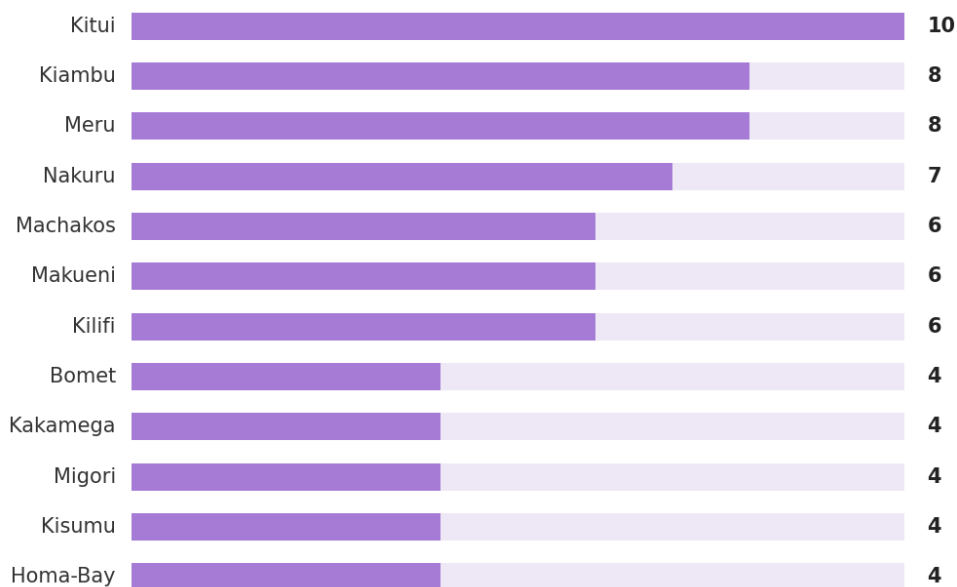
Regional training centre



Program tier



Home counties represented (top 12 of 34)



Where Skills Are Landing

28

Currently working or employed

19

In roles related to the ENGAGE sector

3

Internships converted to a job offer

Self-reported current employment status



Role relevance to the ENGAGE sector



Status is reported for 83 of 105 beneficiaries; role relevance and job-offer answers come from the 38 follow-up form responses received so far.

Research & Analysis

Projects centred on statistical analysis, modelling, and reporting.

57

beneficiaries

DeKUT (9) · SEKU (15) · JOOUST (9) · Kabarak (6) · MUST (15) · Pwani (3)



CAPSTONE Capstone Project

Diabetes Risk Prediction from Demographic and Lifestyle Factors

BACKGROUND & METHODOLOGY

This project builds and evaluates a logistic-regression model using BRFSS-style variables (age, BMI, general health, physical activity, blood pressure, cholesterol, etc.) to flag likely diabetes cases for earlier screening. Exploratory statistics and correlation analyses inform feature selection, and evaluation with accuracy/F1/ROC-AUC (up to 90.97 in reported tests) supports programmatic risk stratification in primary care.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model developed to predict diabetes risk based on demographic and symptom data achieved an overall accuracy of 78%, demonstrating good predictive performance for identifying individuals at risk of diabetes. The model showed strong discriminative ability with an ROC-AUC score of 0.9716, indicating excellent capability to distinguish between diabetic and non-diabetic cases. Class-specific metrics revealed balanced performance: for non-diabetic cases (Class 0), precision was 0.92 and recall was 0.87, while for diabetic cases (Class 1), precision was 0.89 and recall was 0.94, with F1-scores of 0.89 and 0.91 respectively. Key predictors identified through statistical testing and model coefficients included age, BMI, general health status, physical health, high blood pressure, and high cholesterol, all of which showed statistically significant associations with diabetes risk. The confusion matrix confirmed the model correctly classified 11,563 instances (6,316 true positives and 5,247 true negatives) out of 14,804 total test cases. These findings demonstrate the feasibility of using routinely collected demographic and clinical data to support early diabetes screening and risk stratification in primary healthcare settings, enabling timely interventions and personalized preventive care strategies.

INTERNSHIP Internship Experience

Nyeri County Referral Hospital — Cancer patient data analysis and clustering for public health interventions.

Dataset Used: Nyeri County Referral Hospital — Health Records, Cancer Registry. Included patient features such as age, sex, primary site, histology type, cancer stage, and patient status, serving as the foundation for risk assessment and cluster analysis.

METHODOLOGY

Cleaning and preprocessing cancer data, including handling missing values and encoding categorical features. Multiple clustering algorithms—K-means, DBSCAN, and Agglomerative Hierarchical Clustering—were applied to segment patients into meaningful groups. Statistical analysis and visualization techniques, using Python libraries like pandas, scikit-learn, matplotlib, and seaborn, supported the interpretation of cluster characteristics. Detailed reports and visualizations, such as dendrograms and cluster scatter plots, were created to communicate findings. The focus was on identifying high-risk profiles and common cancer trends to inform targeted healthcare interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis revealed distinct clusters highlighting high-risk patient profiles and demographics associated with advanced cancer stages. These insights enabled actionable recommendations for healthcare interventions, such as prioritizing resources for high-risk groups and tailoring preventive and treatment strategies. The project demonstrated the practical use of clustering in public health, providing evidence-based guidance for improving cancer care management and resource allocation.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Report On Clustering and Analysis of Cirrhosis Data for Enhanced Patient and Targeted Medical Interventions

BACKGROUND & METHODOLOGY

This report investigated a cirrhosis dataset to discover clinically meaningful patient subgroups that could inform targeted interventions. After data understanding and cleaning, numerical variables (e.g., age, bilirubin, cholesterol, albumin) were standardized and categorical fields prepared for analysis. Exploratory statistics and visualizations (histograms, box plots, and a correlation matrix) characterized distributions, outliers, and inter-variable relationships. Unsupervised learning with K-means identified three clusters; cluster centroids highlighted distinct risk profiles, including a high-risk group marked by elevated bilirubin and cholesterol. Principal Component Analysis (PCA) facilitated two-dimensional visualization of the clusters, and silhouette scoring assessed cluster separation. The findings provided an empirical basis to tailor medical resources and design interventions for specific patient profiles, supporting improved outcomes for individuals with or at risk of cirrhosis.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis successfully identified three distinct patient clusters using K-means clustering, with cluster centroids revealing defining characteristics such as elevated bilirubin and cholesterol levels in high-risk groups. Principal Component Analysis (PCA) was applied for dimensionality reduction, enabling two-dimensional visualization of cluster separation, while the Silhouette Score was calculated to assess clustering quality. Key findings highlighted high-risk patient subgroups characterized by significantly elevated levels of bilirubin, cholesterol, and other liver-related enzymes, providing actionable insights for healthcare providers to better allocate resources and design tailored interventions. The clustering results established a foundation for further research into underlying factors contributing to observed health profiles, ultimately supporting improved clinical decision-making and patient outcomes for individuals suffering from or at risk of cirrhosis.

INTERNSHIP Internship Experience

Nyeri Town Health Center — HIV testing, ART adherence, and patient care outcomes in maternity and CCC patients.

Dataset Used: Nyeri Town Health Centre, Maternity section and Comprehensive Care Centre (CCC). Two datasets were accessed: the –MAT TST– dataset with HIV testing and care records, and the CCC dataset containing patient ART adherence and viral load data, including demographics and treatment outcomes.

METHODOLOGY

The work involved preprocessing, cleaning, and encoding the datasets, followed by exploratory data analysis. Hierarchical clustering was applied to the –MAT TST– dataset to identify patterns in HIV testing and linkage to care, while K-Means clustering segmented patients in the CCC dataset based on ART adherence and viral load outcomes. Visualization techniques such as dendrograms, scatter plots, histograms, box plots, and PCA were used to interpret clusters. Statistical analyses, including correlation testing and hypothesis evaluation, supported the validity of the findings. Python libraries like Pandas, Matplotlib, Seaborn, and Scikit-learn were extensively employed to implement these processes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analyses revealed meaningful patient clusters with distinct health profiles. In the –MAT TST– dataset, groups were identified, such as younger clients who had not been tested and patients successfully linked to care, offering insights for targeted HIV testing interventions. The CCC dataset analysis highlighted three clusters of ART patients: those with good adherence and low viral loads, patients with missed appointments and high viral loads, and intermediate cases, guiding recommendations for personalized follow-up and care. These outcomes provided actionable insights for improving patient treatment adherence, viral load management, and public health strategies in HIV care.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Clustering health risk profiles in a population smoking dataset using K-Means

BACKGROUND & METHODOLOGY

This project analyzed a population health dataset (smoking.csv) to explore relationships among biometric, biochemical, dental, and behavioral factors and to segment individuals into risk profiles. Data were inspected for structure and missingness, categorical fields (gender, oral health, tartar) were label-encoded, and continuous variables were imputed with column means while categorical variables were imputed with modes. Exploratory analysis included histograms, boxplots with IQR-based outlier inspection, pairwise scatter plots for key cardiometabolic markers, and Pearson/Spearman correlation matrices (e.g., strong positive associations among triglycerides, LDL, and liver enzymes; HDL inversely related to LDL). Features were standardized, and K-Means clustering was applied; the elbow method indicated an optimal solution around $k = 3$, which was validated with a positive silhouette score and visualized using PCA. The resulting clusters were interpretable: (i) predominantly middle-aged females with lower waist and liver enzyme levels and low smoking prevalence; (ii) older males with elevated liver/kidney markers and moderate smoking prevalence; and (iii) younger males with higher anthropometrics and the highest smoking prevalence. These profiles suggested actionable public-health interventions, including smoking-cessation programs for high-prevalence clusters and targeted screening for groups exhibiting elevated biochemical risk

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully applied K-means clustering to segment a population health screening dataset containing cardiometabolic and behavioral risk indicators. Through comprehensive data cleaning including IQR outlier detection and mean/mode imputation, feature scaling, and validation using elbow and silhouette metrics, distinct patient risk profiles were identified and visualized via Principal Component Analysis (PCA). The analysis revealed that higher-risk clusters were characterized by elevated triglycerides, blood pressure, smoking status, and oral health indicators such as tartar and caries, while lower-risk profiles demonstrated more favorable health markers across laboratory and clinical variables including HDL/LDL cholesterol, liver enzymes, and anthropometric measures. These findings provide actionable insights for targeted tobacco-cessation outreach, prioritization of lifestyle counseling for high-risk groups, and more efficient allocation of preventive services in primary care programs. The project demonstrated the practical application of unsupervised machine learning in public health surveillance, establishing a foundation for evidence-based intervention planning and resource optimization in community health settings.

INTERNSHIP Internship Experience

Mt Kenya Sub-county hospital — Non-Communicable Diseases (NCDs) risk factors, patient clustering, and health outcome analysis.

Dataset Used: Mount Kenya Sub-County Hospital — patient records including age, blood pressure, chronic illness burden, and other demographic and clinical information. The datasets contained detailed clinical and demographic data, enabling the identification of patterns and risk factors among patients with NCDs. Information included age, blood pressure, comorbid conditions, and other relevant health metrics.

METHODOLOGY

Data cleaning and preprocessing were conducted to ensure accuracy and consistency. Exploratory data analysis (EDA) identified trends and patterns in patient health characteristics. Unsupervised clustering techniques, including K-means and hierarchical clustering, segmented patients into meaningful groups based on age, blood pressure, and chronic illness burden. Association rule mining was used to detect co-occurring conditions and relationships among risk factors. Visualizations and reports were produced to communicate findings effectively to healthcare professionals.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Three distinct patient clusters were identified, each with unique health profiles. Cluster 0 included younger patients with high blood pressure, Cluster 1 comprised patients with multiple chronic conditions requiring comprehensive care, and Cluster 2 consisted of older patients with well-controlled hypertension and stable health. These findings guided targeted interventions such as tailored health education, community-based support, and focused medical monitoring. Insights from the analysis informed strategies to improve NCD management and preventive care, highlighting the importance of data-driven decision-making in public health.

CURRENT STATUS

Not currently working

Sector relevance: Yes – Fully related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

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★ LED TO A JOB OFFER

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CURRENT STATUS

Data Engineer · Working / employed

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data visualization

Confidence applying ENGAGE skills: **5 / 5**



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Dataset Used: Nyeri County Referral Hospital — Health Records. patient information on age, sex, disease type (chronic and infectious), treatment outcomes, and discharge status, covering emergency, pediatric, and chronic disease management units.

METHODOLOGY

Cleaning and preprocessing patient data, handling missing values, encoding categorical features, and performing exploratory data analysis (EDA). Clustering techniques, particularly K-Means, were applied to segment patients based on risk factors, health outcomes, and age groups. Statistical analyses—including correlation, hypothesis testing, and descriptive statistics—were used to identify trends in mortality and morbidity. Visualizations were created using Python libraries (pandas, matplotlib, seaborn) to communicate patterns and findings effectively, supporting targeted public health interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis identified specific high-risk patient groups, enabling targeted public health interventions. Pediatric patients were found to be most affected by respiratory illnesses, while elderly patients exhibited multiple chronic conditions with higher mortality rates. These findings provided actionable insights for resource allocation, prioritization of care, and designing tailored strategies to reduce disease burden and improve health outcomes. Additionally, the work demonstrated the critical role of data-driven approaches in guiding hospital decision-making and optimizing healthcare delivery for diverse patient demographics.

CURRENT STATUS

Assistant Software Developer · Working / employed



CAPSTONE Capstone Project

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CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Diabetes Risk Prediction from Demographic and Lifestyle Factors

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model developed to predict diabetes risk based on demographic and symptom data achieved an overall accuracy of 78%, demonstrating good predictive performance for identifying individuals at risk of diabetes. The model showed strong discriminative ability with an ROC-AUC score of 0.9716, indicating excellent capability to distinguish between diabetic and non-diabetic cases. Class-specific metrics revealed balanced performance: for non-diabetic cases (Class 0), precision was 0.92 and recall was 0.87, while for diabetic cases (Class 1), precision was 0.89 and recall was 0.94, with F1-scores of 0.89 and 0.91 respectively. Key predictors identified through statistical testing and model coefficients included age, BMI, general health status, physical health, high blood pressure, and high cholesterol, all of which showed statistically significant associations with diabetes risk. The confusion matrix confirmed the model correctly classified 11,563 instances (6,316 true positives and 5,247 true negatives) out of 14,804 total test cases. These findings demonstrate the feasibility of using routinely collected demographic and clinical data to support early diabetes screening and risk stratification in primary healthcare settings, enabling timely interventions and personalized preventive care strategies.

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CURRENT STATUS

Not currently working



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INTERNSHIP Internship Experience

Nyeri County Government/Health Headquarters Department — Ambulance service patient transfers and emergency response patterns.

Dataset Used: Nyeri County Health Headquarters. Dataset contained records on patient transfers, including age, diagnosis, referring and referred facilities, allowing analysis of emergency and routine ambulance service trends.

METHODOLOGY

Collecting, cleaning, and preprocessing ambulance service data, handling missing values, and encoding categorical variables such as diagnosis and facility information. Data analysis included statistical testing (Chi-square, t-tests) and clustering using K-Means and Hierarchical methods to identify patterns in patient transfers and high-risk groups. Data visualization using histograms, heatmaps, dendrograms, and scatter plots communicated findings effectively. Predictive modeling with logistic regression was explored to forecast the likelihood of emergency transfers based on patient characteristics. Python, Excel, and Colab Notebooks were the main tools used throughout the analysis.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis identified three distinct patient clusters within the ambulance service data: younger patients primarily presenting with trauma cases, middle-aged patients with a mix of acute and chronic conditions, and elderly patients predominantly experiencing chronic illnesses such as hypertension and diabetes. The findings highlighted patterns in patient transfers, including which facilities most frequently referred or received patients, and which demographic groups were most at risk. Based on these insights, specific recommendations were made to optimize ambulance deployment, prioritize high-risk patient groups, and improve emergency response times. The results also informed strategic planning for resource allocation, ensuring that both trauma and chronic care needs were addressed efficiently across Nyeri County.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Diabetes Risk Prediction from Demographic and Lifestyle Factors

BACKGROUND & METHODOLOGY

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INTERNSHIP Internship Experience

Nyeri County Referral Hospital — Mortality and morbidity trends for patient demographics in a public health setting.

Dataset Used: Nyeri County Referral Hospital — Health Records. Patient information on age, sex, disease type (chronic and infectious), treatment outcomes, and discharge status, covering emergency, pediatric, and chronic disease management units.

METHODOLOGY

Cleaning and preprocessing patient data, handling missing values, encoding categorical features, and performing exploratory data analysis (EDA). Clustering techniques, particularly K-Means, were applied to segment patients based on risk factors, health outcomes, and age groups. Statistical analyses—including correlation, hypothesis testing, and descriptive statistics—were used to identify trends in mortality and morbidity. Visualizations were created using Python libraries (pandas, matplotlib, seaborn) to communicate patterns and findings effectively, supporting targeted public health interventions.

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The analysis identified specific high-risk patient groups, enabling targeted public health interventions. Pediatric patients were found to be most affected by respiratory illnesses, while elderly patients exhibited multiple chronic conditions with higher mortality rates. These findings provided actionable insights for resource allocation, prioritization of care, and designing tailored strategies to reduce disease burden and improve health outcomes. Additionally, the work demonstrated the critical role of data-driven approaches in guiding hospital decision-making and optimizing healthcare delivery for diverse patient demographics.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predicting Rhesus Risk of Incompatibility Using Machine Learning

BACKGROUND & METHODOLOGY

This capstone project utilizes logistic regression in developing a model for predicting chances of Rh incompatibility during pregnancies which poses severe threatening to both the mother and the baby. The issue appears when a Rh-negative mother is pregnant with a Rh-positive baby (which results in a condition known as hemolytic disease of the newborn, HDN). This project will use machine developed learning technologies for better identification, Coordinated individualized, and optimal handling of Rh incompatibility. The model coefficient can then predict the risk of Rh incompatibility with the help of the following comprehensive data: parents' Rh factor, antibody titer, pregnancy history revenge, sensitivity to prior pregnancies, demography and blood types. The data will be retrieved from clinical records and public health database; explore the data then the model will be developed using logistic regression model. Such steps include; feature selection, training the model, and using performance measures such as accuracy, precision, recall, and AUC-ROC. It is also an ethical study which incorporates such factors as data privacy and balance from bias. This study advances the field of maternal-fetal medicine and provides clear directions to prevent Rh disease and decrease threat to mothers and babies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the risk of Rh incompatibility during pregnancy using maternal and paternal Rh factors, antibody levels, previous pregnancy history, maternal age, ethnicity, socioeconomic status, and blood groups. The study used clinical records and public health data, applied data cleaning, preprocessing, feature selection, model training, and validation. The model achieved strong performance with an accuracy of 93%, precision of 96%, recall of 96%, F1-score of 96%, and AUC-ROC of 0.98, indicating high ability to identify pregnancies at risk of Rh incompatibility. The outcome shows that machine learning can support early detection, personalized maternal-fetal care, and timely interventions to reduce complications for mothers and babies.

INTERNSHIP Internship Experience

Snow Pharmacy-Kitui — Retail Sales Data Analysis and Inventory Management

Dataset Used: Snow Pharmacy sales and stock dataset, including monthly sales and medicine inventory. The dataset tracked monthly sales and inventory levels, highlighting discrepancies, duplicate entries, and expired stock. The aim was to analyze sales trends, improve stock management, and apply data science techniques such as logistic regression to predict high- and low-sales months.

METHODOLOGY

Data cleaning removed duplicates and corrected inconsistencies. Stock-taking ensured physical inventory matched system records. Data was digitized from printed reports and prepared for modeling in Excel and Jupyter Notebook. Logistic regression was applied to model sales trends, and teamwork supported efficient stock verification and data entry.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed inventory gaps and system inaccuracies. While the predictive model performed poorly, the internship provided practical experience in data cleaning, stock management, and applying Python for data analysis. Skills in technical data handling, teamwork, and problem-solving were enhanced, and operational challenges such as incomplete data and limited system access were identified.

CURRENT STATUS

Surveyor · Working / employed



CAPSTONE **Capstone Project**

Machine Learning Model for Predicting Teenage Pregnancy

BACKGROUND & METHODOLOGY

Teenage pregnancy has been a huge problem with building physical, mental and economic implications for teenage mother's future and that of their babies as well. The attributes collaborated for this study include age, residence, access to reproductive health care, education and teenage pregnancy. The findings for this study suggested that higher rates of pregnancy are related to rural area residence lower income level lesser education and issues related to the region. To solve this problem machine learning model i.e. Logistic regression algorithm was employed on the collected data. The performance of this model was calculated using three metrics which included accuracy, precision and F1-score values which indicated that this developed regression model is more accurate it holds the value of 63%.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict teenagers at risk of pregnancy using factors such as age, residence/region, poverty level, education level, and access to reproductive health care. The model was trained and evaluated using a dataset of 6,136 observations and achieved an accuracy of approximately 63%. The findings showed that region, education level, access to reproductive health care, and poverty level were key contributing factors, with the model intended to support early identification and guide interventions such as improved reproductive health access, sexual education, and targeted community support.

INTERNSHIP **Internship Experience**

Muthale Mission Hospital Kitui

CURRENT STATUS

Student · Not currently working



CAPSTONE Capstone Project

Predicting Fertility Rate Among Women Using Machine Learning

BACKGROUND & METHODOLOGY

This project explores the application of machine learning techniques to predict fertility rates among women, with a focus on identifying key demographic, medical, and lifestyle factors influencing reproductive outcomes. Using both primary and secondary data sources, variables such as age, body mass index (BMI), menstrual cycle regularity, history of miscarriages, previous pregnancies, chronic conditions, stress levels, and healthcare access were analyzed. A linear regression model was employed due to its simplicity, interpretability, and efficiency in handling multiple predictors. The data underwent extensive preprocessing, including handling missing values, normalization, and feature selection to ensure model reliability. Model performance was evaluated using metrics such as Adjusted R^2 and Mean Absolute Error (MAE). Results indicated that while the model captured some associations, the predictive power was limited, with an adjusted R^2 of 0.044 and an MAE of 2.88, suggesting that additional variables or advanced modeling techniques may be required to improve accuracy. Despite these limitations, the study provides valuable insights into factors affecting female fertility and demonstrates the potential of machine learning to support public health strategies, personalized medical interventions, and policy formulation in reproductive health.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a linear regression machine learning model to predict fertility rates among women using factors such as age, BMI, menstrual cycle irregularity, chronic condition, stress level, number of previous pregnancies, previous fertility treatments, exercise habits, and history of miscarriages. The study used secondary data, applied data cleaning and preprocessing steps such as handling missing values, checking duplicates, managing outliers, and preparing features for modelling. The model was evaluated using metrics such as Adjusted R-Squared and Mean Absolute Error, with results showing a low Adjusted R-Squared value of 0.044 and MAE of 2.88, indicating that the model had limited predictive strength but provided useful insight into how selected health and lifestyle factors may relate to fertility outcomes.

CURRENT STATUS

Not currently working



CAPSTONE **Capstone Project**

Machine Learning Model for Predicting Teenage Pregnancy

BACKGROUND & METHODOLOGY

Teenage pregnancy has been a huge problem with building physical, mental and economic implications for teenage mother's future and that of their babies as well. The attributes collaborated for this study include age, residence, access to reproductive health care, education and teenage pregnancy. The findings for this study suggested that higher rates of pregnancy are related to rural area residence lower income level lesser education and issues related to the region. To solve this problem machine learning model i.e. Logistic regression algorithm was employed on the collected data. The performance of this model was calculated using three metrics which included accuracy, precision and F1-score values which indicated that this developed regression model is more accurate it holds the value of 63%.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict teenagers at risk of pregnancy using factors such as age, residence/region, poverty level, education level, and access to reproductive health care. The model was trained and evaluated using a dataset of 6,136 observations and achieved an accuracy of approximately 63%. The findings showed that region, education level, access to reproductive health care, and poverty level were key contributing factors, with the model intended to support early identification and guide interventions such as improved reproductive health access, sexual education, and targeted community support.

INTERNSHIP **Internship Experience**

Makueni wote level 5 hospital — Tuberculosis Preventive Therapy (TPT) Implementation and Data Analysis

Dataset Used: Makueni County TPT program patient registers and health facility data (2021–2024). The dataset captured demographic, clinical, and treatment information for high-risk populations, including people living with HIV and close contacts of TB patients. The study assessed TPT uptake, adherence, and completion while identifying barriers to effective implementation, aiming to strengthen TB prevention strategies.

METHODOLOGY

The analysis involved systematic cleaning and preprocessing of patient register data to address missing entries and discrepancies. Descriptive and exploratory data analysis (EDA) techniques were applied to identify demographic trends, symptom profiles, and treatment patterns. Python software was employed to generate visualizations such as pie charts, bar graphs, and scatter plots for clear representation of data distributions, including gender, screening outcomes, TPT indications, regimen distribution, and months on treatment. The study also evaluated adherence, completion rates, and follow-up monitoring, while identifying gaps in data collection and program implementation. Insights were contextualized within national and WHO TPT guidelines to inform public health strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed that the majority of TPT participants were female, with a large proportion asymptomatic, indicating effective preventive screening measures. Most individuals received recommended regimens (3HP, 3RH, 6H), and treatment completion was achieved for a subset of participants, highlighting favorable adherence. Data gaps, such as missing follow-up records, underscored the need for improved monitoring systems. Key findings informed recommendations for targeted outreach, enhanced support for symptomatic individuals, regular follow-up at 6 and 12 months, and continuous training for healthcare workers. Overall, the study provided critical insights into TPT uptake, adherence, and outcomes, supporting evidence-based interventions to reduce TB incidence in high-risk populations in Makueni County.

CURRENT STATUS

Assistant Lab Engineer · Working / employed



CAPSTONE Capstone Project

Machine Learning to Predict Mental Health

BACKGROUND & METHODOLOGY

Mental health is one of the leading cause of suicide as the most of the victim know little about their mental health. This made us to come up with this project machine learning to predict mental health using logistic regression. The study was conducted under the guidance of the following objectives: Develop a model that effectively classifies individuals based on mental health indicators and associated features. The study involves p [reprocessing a data set comprising various mental health variables such as, demographic data, social media usage, physiological asses4ement, income, exercise frequency etc. The report details the data preparation steps such as handling missing values, checking for duplicates and implantation of logistic regression. We employed logistic regression due to its interpret-ability and efficiency in binary classification task (can be able to categorize data into one of two different classes or categories and it can be able to distinguish between the two categories based on the input variables). performance metrics including accuracy, precision and recall are evaluated to assess the model effectiveness. The project involved data collection in which we used secondary source of data. We used visualization such as scatter plot, bar chart, frequency distribution chart, pie chart and confusion metrics for our analysis. Logistic regression model demonstrated a moderate performance of 0.65. Findings demonstrate that logistic regression provides an approach for mental health predication indication is potential in supporting early intervention strategies. The report concludes with a recommendation of exploring feature engineering to improve the models performance and beater address the complexity of mental health.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict mental health status using variables such as age, gender, income, education level, employment status, social media usage, exercise frequency, sleep quality, marital status, housing situation, social support, and access to mental health resources. The model used secondary data, applied data cleaning, preprocessing, one-hot encoding, training/testing split, and evaluation using accuracy, precision, recall, and F1-score. The model achieved an accuracy of approximately 65%, showing moderate performance in classifying individuals with and without mental health issues. The outcome demonstrates that machine learning can support early identification of mental health risks and guide timely interventions, awareness creation, improved access to mental health services, and stronger support systems.

INTERNSHIP Internship Experience

Snow Pharmacy Limited- Kitui — Inventory and sales data management for operational efficiency

Dataset Used: Snow Pharmacy, store department sales records. The dataset captured monthly sales and stock information, revealing discrepancies such as missing items, duplicate entries, and incomplete records. These issues highlighted challenges in inventory tracking and data reliability within the pharmacy.

METHODOLOGY

Stocktaking to verify inventory, entering and cleaning data, removing duplicates, organizing medicines by expiry, and developing a logistic regression model to predict high and low sales months. Guidance from the host organization supported data management and visualization practices.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Enhanced inventory accuracy, improved organization of sales records, strengthened technical skills in Python and data cleaning, and developed teamwork abilities. Identified challenges like incomplete data, shared computer access, and difficulties in CSV transfer, with recommendations to improve data security and operational efficiency.

CURRENT STATUS

Production manager · Working / employed



CAPSTONE Capstone Project

Machine Learning to Predict Mental Health

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict mental health status using variables such as age, gender, income, education level, employment status, social media usage, exercise frequency, sleep quality, marital status, housing situation, social support, and access to mental health resources. The model used secondary data, applied data cleaning, preprocessing, one-hot encoding, training/testing split, and evaluation using accuracy, precision, recall, and F1-score. The model achieved an accuracy of approximately 65%, showing moderate performance in classifying individuals with and without mental health issues. The outcome demonstrates that machine learning can support early identification of mental health risks and guide timely interventions, awareness creation, improved access to mental health services, and stronger support systems.

INTERNSHIP Internship Experience

Muthale Mission Hospital — Public Health Data Management and Outpatient Disease Monitoring

Dataset Used: HIV, outpatient, and modality datasets from Maternal and Child Health (MCH), Comprehensive Care Clinic (CCC), reports office, and outpatient departments; Muthale Mission Hospital. The dataset comprised of patient records from HIV, outpatient, and modality services. It captured demographic information, clinical diagnoses, and follow-up data, enabling the categorization of diseases and analysis of trends by age to support healthcare planning and reporting.

METHODOLOGY

Tasks included entering patient data, filing and retrieving patient records, registering clients at the Comprehensive Care Clinic and outpatient departments, coding and indexing files, preparing patients for clinical procedures, and using software tools to organize and visualize data. Key systems used were Kenya Electronic Medical Records (KenyaEMR) for HIV clients, AIDS, Population and Health Integrated Assistance (APHCON) for patient data entry, Kenya Health Information System (KHIS) Tracker for monitoring morbidity and maternity patients, and KHIS Aggregate for generating facility reports. Guidance was provided by ENGAGE supervisors and the host organization to improve dataset structure and visualization.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully organized and analyzed, allowing clear categorization of outpatient diagnoses and visualization of disease patterns by age. This improved understanding of the health profile of patients, particularly for HIV and tuberculosis management, and supported accurate reporting and monitoring at the facility level. The process enhanced data quality, ensured proper patient record management, and provided actionable insights for healthcare planning and interventions.

CURRENT STATUS

Freelance · Not currently working

Sector relevance: Partially related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **3 / 5**



CAPSTONE Capstone Project

Machine Learning to Predict Mental Health

BACKGROUND & METHODOLOGY

Mental health is one of the leading cause of suicide as the most of the victim know little about their mental health. This made us to come up with this project machine learning to predict mental health using logistic regression. The study was conducted under the guidance of the following objectives: Develop a model that effectively classifies individuals based on mental health indicators and associated features. The study involves p [reprocessing a data set comprising various mental health variables such as, demographic data, social media usage, physiological asses4ement, income, exercise frequency etc. The report details the data preparation steps such as handling missing values, checking for duplicates and implantation of logistic regression. We employed logistic regression due to its interpret-ability and efficiency in binary classification task (can be able to categorize data into one of two different classes or categories and it can be able to distinguish between the two categories based on the input variables). performance metrics including accuracy, precision and recall are evaluated to assess the model effectiveness. The project involved data collection in which we used secondary source of data. We used visualization such as scatter plot, bar chart, frequency distribution chart, pie chart and confusion metrics for our analysis. Logistic regression model demonstrated a moderate performance of 0.65. Findings demonstrate that logistic regression provides an approach for mental health predication indication is potential in supporting early intervention strategies. The report concludes with a recommendation of exploring feature engineering to improve the models performance and beater address the complexity of mental health.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict mental health status using variables such as age, gender, income, education level, employment status, social media usage, exercise frequency, sleep quality, marital status, housing situation, social support, and access to mental health resources. The model used secondary data, applied data cleaning, preprocessing, one-hot encoding, training/testing split, and evaluation using accuracy, precision, recall, and F1-score. The model achieved an accuracy of approximately 65%, showing moderate performance in classifying individuals with and without mental health issues. The outcome demonstrates that machine learning can support early identification of mental health risks and guide timely interventions, awareness creation, improved access to mental health services, and stronger support systems.

INTERNSHIP Internship Experience

Makueni Subcounty Disease Surveillance(Makueni Referral Hospital) — Tuberculosis Preventive Therapy (TPT) Implementation and Data Analysis

Dataset Used: Makueni County TPT program patient registers and health facility data (2021–2024). The dataset captured demographic, clinical, and treatment information for high-risk populations, including people living with HIV and close contacts of TB patients. The study assessed TPT uptake, adherence, and completion while identifying barriers to effective implementation, aiming to strengthen TB prevention strategies.

METHODOLOGY

The analysis involved systematic cleaning and preprocessing of patient register data to address missing entries and discrepancies. Descriptive and exploratory data analysis (EDA) techniques were applied to identify demographic trends, symptom profiles, and treatment patterns. Python software was employed to generate visualizations such as pie charts, bar graphs, and scatter plots for clear representation of data distributions, including gender, screening outcomes, TPT indications, regimen distribution, and months on treatment. The study also evaluated adherence, completion rates, and follow-up monitoring, while identifying gaps in data collection and program implementation. Insights were contextualized within national and WHO TPT guidelines to inform public health strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed that the majority of TPT participants were female, with a large proportion asymptomatic, indicating effective preventive screening measures. Most individuals received recommended regimens (3HP, 3RH, 6H), and treatment completion was achieved for a subset of participants, highlighting favorable adherence. Data gaps, such as missing follow-up records, underscored the need for improved monitoring systems. Key findings informed recommendations for targeted outreach, enhanced support for symptomatic individuals, regular follow-up at 6 and 12 months, and continuous training for healthcare workers. Overall, the study provided critical insights into TPT uptake, adherence, and outcomes, supporting evidence-based interventions to reduce TB incidence in high-risk populations in Makueni County.

CURRENT STATUS

Relationship Officer · Working / employed



CAPSTONE Capstone Project

Machine Learning Model to Predict Potential Drug Users

BACKGROUND & METHODOLOGY

In the realm of substance abuse prevention, predictive models play a crucial role in identifying individuals at risk. This report investigates the efficacy of a logistic regression model designed to predict potential drug users based on various socio-demographic and behavioral features. The dataset utilized includes variables such as age, type of substance, education, and drug users of substance abuse. The logistic regression model was evaluated through standard performance metrics including accuracy, precision, recall, and the F1 score. The model achieved an accuracy of 0.60 and a precision of 0.60, indicating a moderate level of correct predictions among all classified instances. Notably, the model demonstrated a recall of 1.00, reflecting its perfect sensitivity in identifying actual drug users. The F1 score of 0.75 highlights a balanced performance between precision and recall. These results underscore the model's strong capability in detecting potential drug users, although the moderate accuracy and precision suggest that further refinement and feature enhancement are needed. This predictive model offers a foundation for developing targeted intervention strategies, with potential for improvement through advanced modeling techniques and additional data features.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict potential drug users using socio-demographic and behavioral variables such as age, gender, education level, socio-economic status, employment, income level, mental health status, self-esteem, type of substance, frequency of use, duration of use, and quantity of use. The study used secondary data, applied data cleaning, missing value handling, normalization, one-hot encoding, feature engineering, and an 80:20 training-testing split. The model achieved an accuracy of 0.60, precision of 0.60, recall of 1.00, and F1-score of 0.75, showing strong ability to identify actual drug users but with moderate precision due to false positives. The outcome suggests that machine learning can support early identification of at-risk individuals and guide targeted prevention and intervention strategies, while further improvements are needed through better features, threshold tuning, and alternative models.

INTERNSHIP Internship Experience

Mulutu Catholic Dispensary — Predictive Modeling for Childhood Vaccination Completion

Dataset Used: Mulutu Dispensary vaccination records, including demographic, family, and vaccination data (birth dose, 1st–3rd doses, MR1/MR2, vitamin A supplementation). The dataset comprised information on children's demographics, family status, residence, and vaccination history. The study aimed to assess vaccination completion rates, identify barriers to full immunization, and develop a machine learning model to predict which children are likely to complete their vaccination schedules. Insights were intended to support targeted interventions and public health strategies to improve coverage.

METHODOLOGY

Data cleaning and preprocessing addressed missing values, duplicates, and inconsistencies. Exploratory data analysis was performed to examine vaccination completion trends across gender, family status, and residence. Logistic regression was used to develop a predictive model for vaccination completion, with the dataset split into training and testing sets and evaluated using cross-validation. Key features were analyzed for importance, and model performance metrics—accuracy, precision, recall, and F1-score—were calculated. Data visualization was performed using Matplotlib and Seaborn to illustrate vaccination patterns and identify gaps in coverage.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The model achieved high recall (1.00) and an overall accuracy of 85%, indicating strong predictive ability for children who completed vaccinations. Feature analysis highlighted family status, residence, and previous vaccination status as the most influential predictors. About 60% of children had completed their vaccination schedules, leaving 40% requiring targeted interventions. The findings informed recommendations for addressing class imbalance, improving outreach, optimizing model performance, enhancing feature selection, engaging communities, and collaborating with public health officials to increase vaccination coverage among children at Mulutu Dispensary.

CURRENT STATUS

Survey Officer (Intern) · Working / employed

Sector relevance: Not related

SKILLS IN USE

Data collection

Confidence applying ENGAGE skills: 3 / 5



CAPSTONE Capstone Project

Predicting Fertility Rate Among Women Using Machine Learning

BACKGROUND & METHODOLOGY

This project explores the application of machine learning techniques to predict fertility rates among women, with a focus on identifying key demographic, medical, and lifestyle factors influencing reproductive outcomes. Using both primary and secondary data sources, variables such as age, body mass index (BMI), menstrual cycle regularity, history of miscarriages, previous pregnancies, chronic conditions, stress levels, and healthcare access were analyzed. A linear regression model was employed due to its simplicity, interpretability, and efficiency in handling multiple predictors. The data underwent extensive preprocessing, including handling missing values, normalization, and feature selection to ensure model reliability. Model performance was evaluated using metrics such as Adjusted R^2 and Mean Absolute Error (MAE). Results indicated that while the model captured some associations, the predictive power was limited, with an adjusted R^2 of 0.044 and an MAE of 2.88, suggesting that additional variables or advanced modeling techniques may be required to improve accuracy. Despite these limitations, the study provides valuable insights into factors affecting female fertility and demonstrates the potential of machine learning to support public health strategies, personalized medical interventions, and policy formulation in reproductive health.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a linear regression machine learning model to predict fertility rates among women using factors such as age, BMI, menstrual cycle irregularity, chronic condition, stress level, number of previous pregnancies, previous fertility treatments, exercise habits, and history of miscarriages. The study used secondary data, applied data cleaning and preprocessing steps such as handling missing values, checking duplicates, managing outliers, and preparing features for modelling. The model was evaluated using metrics such as Adjusted R-Squared and Mean Absolute Error, with results showing a low Adjusted R-Squared value of 0.044 and MAE of 2.88, indicating that the model had limited predictive strength but provided useful insight into how selected health and lifestyle factors may relate to fertility outcomes.

INTERNSHIP Internship Experience

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METHODOLOGY

Data cleaning and preprocessing addressed missing values, duplicates, and inconsistencies. Exploratory data analysis was performed to examine vaccination completion trends across gender, family status, and residence. Logistic regression was used to develop a predictive model for vaccination completion, with the dataset split into training and testing sets and evaluated using cross-validation. Key features were analyzed for importance, and model performance metrics—accuracy, precision, recall, and F1-score—were calculated. Data visualization was performed using Matplotlib and Seaborn to illustrate vaccination patterns and identify gaps in coverage.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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CURRENT STATUS

Business person · Working / employed



CAPSTONE Capstone Project

Predicting Fertility Rate Among Women Using Machine Learning

BACKGROUND & METHODOLOGY

This project explores the application of machine learning techniques to predict fertility rates among women, with a focus on identifying key demographic, medical, and lifestyle factors influencing reproductive outcomes. Using both primary and secondary data sources, variables such as age, body mass index (BMI), menstrual cycle regularity, history of miscarriages, previous pregnancies, chronic conditions, stress levels, and healthcare access were analyzed. A linear regression model was employed due to its simplicity, interpretability, and efficiency in handling multiple predictors. The data underwent extensive preprocessing, including handling missing values, normalization, and feature selection to ensure model reliability. Model performance was evaluated using metrics such as Adjusted R^2 and Mean Absolute Error (MAE). Results indicated that while the model captured some associations, the predictive power was limited, with an adjusted R^2 of 0.044 and an MAE of 2.88, suggesting that additional variables or advanced modeling techniques may be required to improve accuracy. Despite these limitations, the study provides valuable insights into factors affecting female fertility and demonstrates the potential of machine learning to support public health strategies, personalized medical interventions, and policy formulation in reproductive health.

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INTERNSHIP Internship Experience

Makueni Hospital — Tuberculosis Preventive Therapy (TPT) Implementation and Data Analysis

Dataset Used: Makueni County TPT program patient registers and health facility data (2021–2024). The dataset captured demographic, clinical, and treatment information for high-risk populations, including people living with HIV and close contacts of TB patients. The study assessed TPT uptake, adherence, and completion while identifying barriers to effective implementation, aiming to strengthen TB prevention strategies.

METHODOLOGY

The analysis involved systematic cleaning and preprocessing of patient register data to address missing entries and discrepancies. Descriptive and exploratory data analysis (EDA) techniques were applied to identify demographic trends, symptom profiles, and treatment patterns. Python software was employed to generate visualizations such as pie charts, bar graphs, and scatter plots for clear representation of data distributions, including gender, screening outcomes, TPT indications, regimen distribution, and months on treatment. The study also evaluated adherence, completion rates, and follow-up monitoring, while identifying gaps in data collection and program implementation. Insights were contextualized within national and WHO TPT guidelines to inform public health strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed that the majority of TPT participants were female, with a large proportion asymptomatic, indicating effective preventive screening measures. Most individuals received recommended regimens (3HP, 3RH, 6H), and treatment completion was achieved for a subset of participants, highlighting favorable adherence. Data gaps, such as missing follow-up records, underscored the need for improved monitoring systems. Key findings informed recommendations for targeted outreach, enhanced support for symptomatic individuals, regular follow-up at 6 and 12 months, and continuous training for healthcare workers. Overall, the study provided critical insights into TPT uptake, adherence, and outcomes, supporting evidence-based interventions to reduce TB incidence in high-risk populations in Makueni County.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Machine Learning Model to Predict Potential Drug Users

BACKGROUND & METHODOLOGY

In the realm of substance abuse prevention, predictive models play a crucial role in identifying individuals at risk. This report investigates the efficacy of a logistic regression model designed to predict potential drug users based on various socio-demographic and behavioral features. The dataset utilized includes variables such as age, type of substance, education, and drug users of substance abuse. The logistic regression model was evaluated through standard performance metrics including accuracy, precision, recall, and the F1 score. The model achieved an accuracy of 0.60 and a precision of 0.60, indicating a moderate level of correct predictions among all classified instances. Notably, the model demonstrated a recall of 1.00, reflecting its perfect sensitivity in identifying actual drug users. The F1 score of 0.75 highlights a balanced performance between precision and recall. These results underscore the model's strong capability in detecting potential drug users, although the moderate accuracy and precision suggest that further refinement and feature enhancement are needed. This predictive model offers a foundation for developing targeted intervention strategies, with potential for improvement through advanced modeling techniques and additional data features.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict potential drug users using socio-demographic and behavioral variables such as age, gender, education level, socio-economic status, employment, income level, mental health status, self-esteem, type of substance, frequency of use, duration of use, and quantity of use. The study used secondary data, applied data cleaning, missing value handling, normalization, one-hot encoding, feature engineering, and an 80:20 training-testing split. The model achieved an accuracy of 0.60, precision of 0.60, recall of 1.00, and F1-score of 0.75, showing strong ability to identify actual drug users but with moderate precision due to false positives. The outcome suggests that machine learning can support early identification of at-risk individuals and guide targeted prevention and intervention strategies, while further improvements are needed through better features, threshold tuning, and alternative models.

INTERNSHIP Internship Experience

World Vision Kenya-Mutomo — Child Nutrition and Malnutrition Intervention

Dataset Used: World Vision program data and community health surveys in Mutomo, Kitui County, Kenya, including Mid-Upper Arm Circumference (MUAC), immunization status, breastfeeding, deworming, age, and gender. The dataset captured health and nutritional information for children under five in Mutomo, a semi-arid region with high malnutrition rates. MUAC (Mid-Upper Arm Circumference) was used to assess acute malnutrition. The study evaluated malnutrition prevalence, determinants, and the impact of World Vision interventions such as food distribution, healthcare services, and educational campaigns. Logistic regression was applied to identify key predictors of malnutrition.

METHODOLOGY

Data cleaning addressed missing or inaccurate entries, duplicates, and inconsistencies. Exploratory data analysis and descriptive statistics were performed to assess malnutrition prevalence and identify correlations with immunization, breastfeeding, and access to clean water. Logistic regression modeling determined predictors of malnutrition. Visualization tools were used to communicate findings, while program activities—food distribution, healthcare support, and caregiver education—were analyzed to evaluate their effectiveness in reducing malnutrition rates.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis showed that children under five were the most affected by malnutrition, with immunization, breastfeeding practices, and access to healthcare and clean water significantly influencing nutritional outcomes. World Vision's interventions led to measurable improvements in child health, reducing severe malnutrition and stunting. The findings informed recommendations for expanding healthcare access, enhancing educational campaigns, and developing sustainable food programs to sustain gains and further reduce malnutrition in the region.

CURRENT STATUS

Agricultural Technician · Working / employed



CAPSTONE Capstone Project

Machine Learning Model to Predict Potential Drug Users

BACKGROUND & METHODOLOGY

In the realm of substance abuse prevention, predictive models play a crucial role in identifying individuals at risk. This report investigates the efficacy of a logistic regression model designed to predict potential drug users based on various socio-demographic and behavioral features. The dataset utilized includes variables such as age, type of substance, education, and drug users of substance abuse. The logistic regression model was evaluated through standard performance metrics including accuracy, precision, recall, and the F1 score. The model achieved an accuracy of 0.60 and a precision of 0.60, indicating a moderate level of correct predictions among all classified instances. Notably, the model demonstrated a recall of 1.00, reflecting its perfect sensitivity in identifying actual drug users. The F1 score of 0.75 highlights a balanced performance between precision and recall. These results underscore the model's strong capability in detecting potential drug users, although the moderate accuracy and precision suggest that further refinement and feature enhancement are needed. This predictive model offers a foundation for developing targeted intervention strategies, with potential for improvement through advanced modeling techniques and additional data features.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict potential drug users using socio-demographic and behavioral variables such as age, gender, education level, socio-economic status, employment, income level, mental health status, self-esteem, type of substance, frequency of use, duration of use, and quantity of use. The study used secondary data, applied data cleaning, missing value handling, normalization, one-hot encoding, feature engineering, and an 80:20 training-testing split. The model achieved an accuracy of 0.60, precision of 0.60, recall of 1.00, and F1-score of 0.75, showing strong ability to identify actual drug users but with moderate precision due to false positives. The outcome suggests that machine learning can support early identification of at-risk individuals and guide targeted prevention and intervention strategies, while further improvements are needed through better features, threshold tuning, and alternative models.

INTERNSHIP Internship Experience

Makueni Hospital — Tuberculosis Preventive Therapy (TPT) Implementation and Data Analysis

Dataset Used: Makueni County TPT program patient registers and health facility data (2021–2024). The dataset captured demographic, clinical, and treatment information for high-risk populations, including people living with HIV and close contacts of TB patients. The study assessed TPT uptake, adherence, and completion while identifying barriers to effective implementation, aiming to strengthen TB prevention strategies.

METHODOLOGY

The analysis involved systematic cleaning and preprocessing of patient register data to address missing entries and discrepancies. Descriptive and exploratory data analysis (EDA) techniques were applied to identify demographic trends, symptom profiles, and treatment patterns. Python software was employed to generate visualizations such as pie charts, bar graphs, and scatter plots for clear representation of data distributions, including gender, screening outcomes, TPT indications, regimen distribution, and months on treatment. The study also evaluated adherence, completion rates, and follow-up monitoring, while identifying gaps in data collection and program implementation. Insights were contextualized within national and WHO TPT guidelines to inform public health strategies.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Machine Learning Model for Predicting Teenage Pregnancy

BACKGROUND & METHODOLOGY

Teenage pregnancy has been a huge problem with building physical, mental and economic implications for teenage mother's future and that of their babies as well. The attributes collaborated for this study include age, residence, access to reproductive health care, education and teenage pregnancy. The findings for this study suggested that higher rates of pregnancy are related to rural area residence lower income level lesser education and issues related to the region. To solve this problem machine learning model i.e. Logistic regression algorithm was employed on the collected data. The performance of this model was calculated using three metrics which included accuracy, precision and F1-score values which indicated that this developed regression model is more accurate it holds the value of 63%.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict teenagers at risk of pregnancy using factors such as age, residence/region, poverty level, education level, and access to reproductive health care. The model was trained and evaluated using a dataset of 6,136 observations and achieved an accuracy of approximately 63%. The findings showed that region, education level, access to reproductive health care, and poverty level were key contributing factors, with the model intended to support early identification and guide interventions such as improved reproductive health access, sexual education, and targeted community support.

INTERNSHIP Internship Experience

Snow Pharmacy Limited- Kitui — Inventory and sales data management for operational efficiency

Dataset Used: Snow Pharmacy, store department sales records. The dataset captured monthly sales and stock information, revealing discrepancies such as missing items, duplicate entries, and incomplete records. These issues highlighted challenges in inventory tracking and data reliability within the pharmacy.

METHODOLOGY

Played a crucial role in supporting the company with stock taking by counting all the available medicine in the store and entering it to the system. I also helped in data cleaning and removing duplicates and arranging the medicine in terms of expiry so that the short term should be Infront to be sold faster.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Enhanced inventory accuracy, improved organization of sales records, strengthened technical skills in Python and data cleaning, and developed teamwork abilities. Identified challenges like incomplete data, shared computer access, and difficulties in CSV transfer, with recommendations to improve data security and operational efficiency.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predicting Rhesus Risk of Incompatibility Using Machine Learning

BACKGROUND & METHODOLOGY

This capstone project utilizes logistic regression in developing a model for predicting chances of Rh incompatibility during pregnancies which poses severe threatening to both the mother and the baby. The issue appears when a Rh-negative mother is pregnant with a Rh-positive baby (which results in a condition known as hemolytic disease of the newborn, HDN). This project will use machine developed learning technologies for better identification, Coordinated individualized, and optimal handling of Rh incompatibility. The model coefficient can then predict the risk of Rh incompatibility with the help of the following comprehensive data: parents' Rh factor, antibody titer, pregnancy history revenge, sensitivity to prior pregnancies, demography and blood types. The data will be retrieved from clinical records and public health database; explore the data then the model will be developed using logistic regression model. Such steps include; feature selection, training the model, and using performance measures such as accuracy, precision, recall, and AUC-ROC. It is also an ethical study which incorporates such factors as data privacy and balance from bias. This study advances the field of maternal-fetal medicine and provides clear directions to prevent Rh disease and decrease threat to mothers and babies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the risk of Rh incompatibility during pregnancy using maternal and paternal Rh factors, antibody levels, previous pregnancy history, maternal age, ethnicity, socioeconomic status, and blood groups. The study used clinical records and public health data, applied data cleaning, preprocessing, feature selection, model training, and validation. The model achieved strong performance with an accuracy of 93%, precision of 96%, recall of 96%, F1-score of 96%, and AUC-ROC of 0.98, indicating high ability to identify pregnancies at risk of Rh incompatibility. The outcome shows that machine learning can support early detection, personalized maternal-fetal care, and timely interventions to reduce complications for mothers and babies.

CURRENT STATUS

Customer Care · Working / employed



CAPSTONE Capstone Project

Predicting Rhesus Risk of Incompatibility Using Machine Learning

BACKGROUND & METHODOLOGY

This capstone project utilizes logistic regression in developing a model for predicting chances of Rh incompatibility during pregnancies which poses severe threatening to both the mother and the baby. The issue appears when a Rh-negative mother is pregnant with a Rh-positive baby (which results in a condition known as hemolytic disease of the newborn, HDN). This project will use machine developed learning technologies for better identification, Coordinated individualized, and optimal handling of Rh incompatibility. The model coefficient can then predict the risk of Rh incompatibility with the help of the following comprehensive data: parents' Rh factor, antibody titer, pregnancy history revenge, sensitivity to prior pregnancies, demography and blood types. The data will be retrieved from clinical records and public health database; explore the data then the model will be developed using logistic regression model. Such steps include; feature selection, training the model, and using performance measures such as accuracy, precision, recall, and AUC-ROC. It is also an ethical study which incorporates such factors as data privacy and balance from bias. This study advances the field of maternal-fetal medicine and provides clear directions to prevent Rh disease and decrease threat to mothers and babies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the risk of Rh incompatibility during pregnancy using maternal and paternal Rh factors, antibody levels, previous pregnancy history, maternal age, ethnicity, socioeconomic status, and blood groups. The study used clinical records and public health data, applied data cleaning, preprocessing, feature selection, model training, and validation. The model achieved strong performance with an accuracy of 93%, precision of 96%, recall of 96%, F1-score of 96%, and AUC-ROC of 0.98, indicating high ability to identify pregnancies at risk of Rh incompatibility. The outcome shows that machine learning can support early detection, personalized maternal-fetal care, and timely interventions to reduce complications for mothers and babies.

INTERNSHIP Internship Experience

World Vision Kenya - Mutomo — Child Nutrition and Malnutrition Intervention

Dataset Used: World Vision program data and community health surveys in Mutomo, Kitui County, Kenya, including Mid-Upper Arm Circumference (MUAC), immunization status, breastfeeding, deworming, age, and gender. The dataset captured health and nutritional information for children under five in Mutomo, a semi-arid region with high malnutrition rates. MUAC (Mid-Upper Arm Circumference) was used to assess acute malnutrition. The study evaluated malnutrition prevalence, determinants, and the impact of World Vision interventions such as food distribution, healthcare services, and educational campaigns. Logistic regression was applied to identify key predictors of malnutrition.

METHODOLOGY

Data cleaning addressed missing or inaccurate entries, duplicates, and inconsistencies. Exploratory data analysis and descriptive statistics were performed to assess malnutrition prevalence and identify correlations with immunization, breastfeeding, and access to clean water. Logistic regression modeling determined predictors of malnutrition. Visualization tools were used to communicate findings, while program activities—food distribution, healthcare support, and caregiver education—were analyzed to evaluate their effectiveness in reducing malnutrition rates.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis showed that children under five were the most affected by malnutrition, with immunization, breastfeeding practices, and access to healthcare and clean water significantly influencing nutritional outcomes. World Vision's interventions led to measurable improvements in child health, reducing severe malnutrition and stunting. The findings informed recommendations for expanding healthcare access, enhancing educational campaigns, and developing sustainable food programs to sustain gains and further reduce malnutrition in the region.



CAPSTONE Capstone Project

Factors Influencing Child Participation in the Polio Campaign of August 2013

BACKGROUND & METHODOLOGY

Polio, a highly contagious viral disease, continues to threaten global health, particularly in regions with low vaccination coverage. Despite significant progress towards eradication, understanding the factors influencing vaccine uptake is crucial for the success of vaccination campaigns. This study focuses on the August 2013 polio vaccination campaign in Bungoma County, Kenya, aiming to identify key predictors of child participation in the campaign. Using a dataset comprising 768 observations, various demographic, health, and socioeconomic variables were analysed, including diarrhoea status, height, residence, sex, age, education level, and wealth index.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict child participation in the August 2013 polio vaccination campaign in Bungoma County, Kenya. The model achieved an overall accuracy of 86% on the test set, with weighted average precision of 0.85, recall of 0.86, and F1-score of 0.84, demonstrating strong predictive performance for binary classification. Key predictors identified included wealth index (children from wealthier households had higher participation odds), residence (urban children showed higher participation than rural counterparts), age (younger children were more likely to be vaccinated), and parental education level (higher education correlated with increased participation). The confusion matrix $\begin{bmatrix} 121 & 4 \\ 17 & 10 \end{bmatrix}$ revealed the model correctly classified 121 true positives and 10 true negatives, with only 4 false positives and 17 false negatives. These findings provide actionable insights for public health planning, highlighting the need for targeted interventions to address socioeconomic and geographic disparities in vaccination access. The project demonstrated the practical utility of logistic regression in public health research and established a framework for using predictive analytics to optimize vaccination campaign strategies and improve coverage in underserved communities.

INTERNSHIP Internship Experience

Center for Health Solutions(CHS) — Health data management, analysis, and predictive modeling for HIV/AIDS, ART, and DREAMS interventions.

Dataset Used: Datasets from CHS including Active on ART Patient Linelist (Aug 2024), DREAMS enrollment and intervention data, HTS prediction report, and Kenya EMR system extracts. The datasets included patient-level and program-level information on ART adherence, viral load suppression, DREAMS vulnerability status, and HIV testing services. These were used to build predictive models, support reporting to KHIS, and strengthen decision-making for HIV prevention and treatment programs.

METHODOLOGY

Engaged in both manual and digital data collection through hospital registers and Kenya EMR; performed data cleaning in Excel and Python, including deduplication and imputation; accessed and extracted data from the EMR, warehouse, and KHIS; created pivot tables, visualizations, and dashboards; applied logistic regression and random forest to build predictive models for ART viral suppression, DREAMS vulnerability status, and HIV risk; deployed models on Streamlit for real-time analysis; learned database management with MySQL; participated in site visits, PEPFAR/CDC meetings, and community safe-space reviews; and collaborated with multidisciplinary teams on quality assurance and reporting.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Delivered predictive models highlighting viral load suppression gaps among OVCs and key populations; developed a DREAMS model identifying high-vulnerability adolescent girls based on socioeconomic and health factors; created an HIV prediction tool linking demographic and risk features to HIV status; produced dashboards and visualizations supporting program monitoring; improved data quality for CHS reporting to KHIS and donor partners; and strengthened CHS capacity to use predictive analytics for targeted interventions in ART adherence, DREAMS support, and HIV prevention strategies.

CURRENT STATUS

Programs Manager · Working / employed



CAPSTONE Capstone Project

Risk Factors of Occurrence in Cholera Outbreak

BACKGROUND & METHODOLOGY

This project investigates the predictors of cholera occurrence/morbidity in a recent outbreak using a comprehensive dataset of cholera outbreak dataset encompassing various risk factors. The primary objective is to identify significant predictors contributing to the occurrence of cholera, thereby informing targeted public health interventions. The dataset includes demographic information such as age, sex, and income, as well as details on water and sanitation facilities, sub-county locations, and the presence of informal settlements. The analysis is divided into two main components. First, descriptive statistics are employed to summarize the demographic characteristics of the affected population. Table 1 presents measures of central tendency for variables such as age, sex, and income, providing a foundational understanding of the study population. Additionally, there are pie charts and histograms used to visualize the distribution of the demographic characteristics. Following this, a bivariate analysis in Table 2 examines the relationship between each predictor and cholera morbidity through 2x2 contingency tables, allowing for preliminary identification of associations. For the core analysis, logistic regression is utilized to model the likelihood of cholera incidence based on multiple predictors, including sex, age, sub-county, water and sanitation conditions, income, and the presence of informal settlements. This approach enables the assessment of the impact of each factor while controlling for others, revealing the relative significance of each predictor in influencing cholera outcomes. The methodology used to conduct the project include data preprocessing, machine learning models like logistic regression, model training and evaluation metrics. Key findings indicate that factors such as inadequate water and sanitation facilities, as well as living in informal settlements, are strongly associated with higher morbidity rates. The logistic regression model further refines these associations, highlighting the critical role of environmental and socio-economic factors in cholera outbreaks. These insights are crucial for developing targeted interventions to mitigate future outbreaks and improve public health strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model developed to predict cholera occurrence achieved an overall accuracy of 68.75%, demonstrating reasonable predictive performance for binary classification in public health settings. The confusion matrix revealed 37 true positives, 29 true negatives, 18 false positives, and 12 false negatives, with precision values of 71% for negative predictions and 67% for positive predictions, and recall values of 62% and 76% respectively. Chi-square hypothesis testing identified strong statistically significant associations between cholera occurrence and key predictors including sub-county location ($p=4.44e-10$), water source access ($p=2.91e-12$), sanitation conditions ($p=6.24e-25$), informal settlement residence ($p=2.92e-07$), and age group ($p=6.846e-16$), while income showed a weaker but significant association ($p=0.0396$) and sex showed no significant association ($p=0.263$). The model demonstrated slightly better performance in detecting positive cholera cases, which is clinically valuable for early case identification. Key findings emphasized that inadequate water and sanitation infrastructure, living in informal settlements, and specific geographic locations are critical determinants of cholera risk, providing actionable evidence for targeted public health interventions. The project successfully demonstrated the practical application of association rule mining, statistical testing, and predictive modeling in resource-limited healthcare settings, establishing a foundation for data-driven cholera prevention strategies, improved resource allocation, and strengthened emergency response planning in endemic regions.

INTERNSHIP Internship Experience

Center for Health Solutions(CHS) — HIV treatment outcomes, HIV prevention for vulnerable populations, and HIV testing prediction.

Dataset Used: Comprehensive Care Centre (ART program), DREAMS Safe Spaces (Karapul and Wagai), and CHS Data Management Unit.

Datasets capture a mix of clinical, demographic, behavioral, and intervention-related data. They support predictive modeling for ART adherence and viral suppression, vulnerability assessment among AGYW, and HIV testing risk stratification.

METHODOLOGY

Conducted data collection using EMR systems, applied data quality assurance, and analyzed HIV/TB program data with Python and Excel. Designed dashboards in Power BI to track key health indicators. Collaborated with multidisciplinary teams on project planning, reporting (via CRIS), and community outreach. Addressed data gaps and EMR technical issues through validation, training, and backup systems.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Improved data accuracy, enhanced reporting reliability, strengthened diagnostic and monitoring processes, and gained technical, soft, and public health skills applicable to real-world interventions.

CURRENT STATUS

Data and Web Development Intern · Working / employed



CAPSTONE Capstone Project

Factors Influencing Child Participation in the Polio Campaign of August 2013

BACKGROUND & METHODOLOGY

Polio, a highly contagious viral disease, continues to threaten global health, particularly in regions with low vaccination coverage. Despite significant progress towards eradication, understanding the factors influencing vaccine uptake is crucial for the success of vaccination campaigns. This study focuses on the August 2013 polio vaccination campaign in Bungoma County, Kenya, aiming to identify key predictors of child participation in the campaign. Using a dataset comprising 768 observations, various demographic, health, and socioeconomic variables were analysed, including diarrhoea status, height, residence, sex, age, education level, and wealth index.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict child participation in the August 2013 polio vaccination campaign in Bungoma County, Kenya. The model achieved an overall accuracy of 86% on the test set, with weighted average precision of 0.85, recall of 0.86, and F1-score of 0.84, demonstrating strong predictive performance for binary classification. Key predictors identified included wealth index (children from wealthier households had higher participation odds), residence (urban children showed higher participation than rural counterparts), age (younger children were more likely to be vaccinated), and parental education level (higher education correlated with increased participation). The confusion matrix $\begin{bmatrix} 121 & 4 \\ 17 & 10 \end{bmatrix}$ revealed the model correctly classified 121 true positives and 10 true negatives, with only 4 false positives and 17 false negatives. These findings provide actionable insights for public health planning, highlighting the need for targeted interventions to address socioeconomic and geographic disparities in vaccination access. The project demonstrated the practical utility of logistic regression in public health research and established a framework for using predictive analytics to optimize vaccination campaign strategies and improve coverage in underserved communities.

INTERNSHIP Internship Experience

Kakamega County General Teaching and Referral Hospital — Cancer epidemiology focusing on prevalence, survival, and treatment outcomes.

Dataset Used: Cancer Center via ENGAGE project in collaboration with Kakamega General Hospital. Patient-level records with demographics, cancer stage, treatment history, and survival outcomes

METHODOLOGY

Analyzed cancer dataset to study incidence, survival rates, and treatment outcomes; performed data cleaning, EDA, predictive modeling (logistic regression), and visualizations using Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn) to uncover patterns and support cancer care decision-making.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified high-risk demographic groups and gaps in cancer screening; built models to predict cancer stages; improved data quality and reporting; enhanced technical (data analysis, visualization, predictive modeling) and soft skills (communication, teamwork, time management); contributed insights for targeted cancer prevention and treatment strategies.

CURRENT STATUS

Data collection · Not currently working

Sector relevance: Not related

SKILLS IN USE

Data collection

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Risk Factors of Occurrence in Cholera Outbreak

BACKGROUND & METHODOLOGY

This project investigates the predictors of cholera occurrence/morbidity in a recent outbreak using a comprehensive dataset of cholera outbreak dataset encompassing various risk factors. The primary objective is to identify significant predictors contributing to the occurrence of cholera, thereby informing targeted public health interventions. The dataset includes demographic information such as age, sex, and income, as well as details on water and sanitation facilities, sub-county locations, and the presence of informal settlements. The analysis is divided into two main components. First, descriptive statistics are employed to summarize the demographic characteristics of the affected population. Table 1 presents measures of central tendency for variables such as age, sex, and income, providing a foundational understanding of the study population. Additionally, there are pie charts and histograms used to visualize the distribution of the demographic characteristics. Following this, a bivariate analysis in Table 2 examines the relationship between each predictor and cholera morbidity through 2x2 contingency tables, allowing for preliminary identification of associations. For the core analysis, logistic regression is utilized to model the likelihood of cholera incidence based on multiple predictors, including sex, age, sub-county, water and sanitation conditions, income, and the presence of informal settlements. This approach enables the assessment of the impact of each factor while controlling for others, revealing the relative significance of each predictor in influencing cholera outcomes. The methodology used to conduct the project include data preprocessing, machine learning models like logistic regression, model training and evaluation metrics. Key findings indicate that factors such as inadequate water and sanitation facilities, as well as living in informal settlements, are strongly associated with higher morbidity rates. The logistic regression model further refines these associations, highlighting the critical role of environmental and socio-economic factors in cholera outbreaks. These insights are crucial for developing targeted interventions to mitigate future outbreaks and improve public health strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model developed to predict cholera occurrence achieved an overall accuracy of 68.75%, demonstrating reasonable predictive performance for binary classification in public health settings. The confusion matrix revealed 37 true positives, 29 true negatives, 18 false positives, and 12 false negatives, with precision values of 71% for negative predictions and 67% for positive predictions, and recall values of 62% and 76% respectively. Chi-square hypothesis testing identified strong statistically significant associations between cholera occurrence and key predictors including sub-county location ($p=4.44e-10$), water source access ($p=2.91e-12$), sanitation conditions ($p=6.24e-25$), informal settlement residence ($p=2.92e-07$), and age group ($p=6.846e-16$), while income showed a weaker but significant association ($p=0.0396$) and sex showed no significant association ($p=0.263$). The model demonstrated slightly better performance in detecting positive cholera cases, which is clinically valuable for early case identification. Key findings emphasized that inadequate water and sanitation infrastructure, living in informal settlements, and specific geographic locations are critical determinants of cholera risk, providing actionable evidence for targeted public health interventions. The project successfully demonstrated the practical application of association rule mining, statistical testing, and predictive modeling in resource-limited healthcare settings, establishing a foundation for data-driven cholera prevention strategies, improved resource allocation, and strengthened emergency response planning in endemic regions.

INTERNSHIP Internship Experience

Sirisia Sub-County Hospital — Hospital-based health data focusing on child, men's, and maternal health outcomes.

Dataset Used: Central Health Records Department, Sirisia Sub-County Hospital. The datasets consist of hospital-generated routine health records covering pediatric, male, and maternal populations. They provide demographic, diagnostic, and treatment-related data for use in predictive health analytics and service delivery improvement

METHODOLOGY

Analyzed pediatric, male ward, and antenatal datasets at Sirisia Hospital; applied Python/Excel for data cleaning, EDA, predictive modeling, and visualization to predict child diseases, NCDs, and ANC malaria prevention. At Chulaimbo Hospital, used Python (Pandas, NumPy, Scikit-learn), Streamlit, and GitHub to build and deploy predictive health models with interactive web apps.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified seasonal child disease patterns, high NCD prevalence in men, and links between ANC visits and malaria prevention; improved hospital planning for resources and outreach; enhanced technical skills (Python, ML, Streamlit, GitHub, visualization) and soft skills (communication, problem-solving, independence); contributed actionable insights for community health programs.

CURRENT STATUS

Data Analysis Trainer · Working / employed



CAPSTONE Capstone Project

Risk Factors of Occurrence in Cholera Outbreak

BACKGROUND & METHODOLOGY

This project investigates the predictors of cholera occurrence/morbidity in a recent outbreak using a comprehensive dataset of cholera outbreak dataset encompassing various risk factors. The primary objective is to identify significant predictors contributing to the occurrence of cholera, thereby informing targeted public health interventions. The dataset includes demographic information such as age, sex, and income, as well as details on water and sanitation facilities, sub-county locations, and the presence of informal settlements. The analysis is divided into two main components. First, descriptive statistics are employed to summarize the demographic characteristics of the affected population. Table 1 presents measures of central tendency for variables such as age, sex, and income, providing a foundational understanding of the study population. Additionally, there are pie charts and histograms used to visualize the distribution of the demographic characteristics. Following this, a bivariate analysis in Table 2 examines the relationship between each predictor and cholera morbidity through 2x2 contingency tables, allowing for preliminary identification of associations. For the core analysis, logistic regression is utilized to model the likelihood of cholera incidence based on multiple predictors, including sex, age, sub-county, water and sanitation conditions, income, and the presence of informal settlements. This approach enables the assessment of the impact of each factor while controlling for others, revealing the relative significance of each predictor in influencing cholera outcomes. The methodology used to conduct the project include data preprocessing, machine learning models like logistic regression, model training and evaluation metrics. Key findings indicate that factors such as inadequate water and sanitation facilities, as well as living in informal settlements, are strongly associated with higher morbidity rates. The logistic regression model further refines these associations, highlighting the critical role of environmental and socio-economic factors in cholera outbreaks. These insights are crucial for developing targeted interventions to mitigate future outbreaks and improve public health strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model developed to predict cholera occurrence achieved an overall accuracy of 68.75%, demonstrating reasonable predictive performance for binary classification in public health settings. The confusion matrix revealed 37 true positives, 29 true negatives, 18 false positives, and 12 false negatives, with precision values of 71% for negative predictions and 67% for positive predictions, and recall values of 62% and 76% respectively. Chi-square hypothesis testing identified strong statistically significant associations between cholera occurrence and key predictors including sub-county location ($p=4.44e-10$), water source access ($p=2.91e-12$), sanitation conditions ($p=6.24e-25$), informal settlement residence ($p=2.92e-07$), and age group ($p=6.846e-16$), while income showed a weaker but significant association ($p=0.0396$) and sex showed no significant association ($p=0.263$). The model demonstrated slightly better performance in detecting positive cholera cases, which is clinically valuable for early case identification. Key findings emphasized that inadequate water and sanitation infrastructure, living in informal settlements, and specific geographic locations are critical determinants of cholera risk, providing actionable evidence for targeted public health interventions. The project successfully demonstrated the practical application of association rule mining, statistical testing, and predictive modeling in resource-limited healthcare settings, establishing a foundation for data-driven cholera prevention strategies, improved resource allocation, and strengthened emergency response planning in endemic regions.

INTERNSHIP Internship Experience

Homabay County Teaching and Referral Hospital — Communicable & non-communicable disease management (HIV, TB, malaria, diabetes, cancer).

Dataset Used: Hospital departments (CCC, Chest, Pediatrics/KEMRI, Special Clinic, Oncology). Rich patient-level datasets with demographic, diagnostic, treatment, and outcome variables aimed at predictive modeling, clinical decision support, and public health improvements.

METHODOLOGY

Analyzed HIV, chest, malaria, diabetes, and oncology datasets; cleaned and digitized data; conducted EDA, visualizations, and predictive modeling (logistic regression, random forest, linear regression); deployed user-friendly apps for real-time predictions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified key disease predictors and mortality risks; improved diagnostic accuracy and resource planning; developed deployable predictive tools; strengthened technical (Python, ML, visualization, app deployment) and soft skills (teamwork, communication, time management).

CURRENT STATUS

Network Engineer · Working / employed



CAPSTONE Capstone Project

Factors Influencing Child Participation in the Polio Campaign of August 2013

BACKGROUND & METHODOLOGY

Polio, a highly contagious viral disease, continues to threaten global health, particularly in regions with low vaccination coverage. Despite significant progress towards eradication, understanding the factors influencing vaccine uptake is crucial for the success of vaccination campaigns. This study focuses on the August 2013 polio vaccination campaign in Bungoma County, Kenya, aiming to identify key predictors of child participation in the campaign. Using a dataset comprising 768 observations, various demographic, health, and socioeconomic variables were analysed, including diarrhoea status, height, residence, sex, age, education level, and wealth index.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict child participation in the August 2013 polio vaccination campaign in Bungoma County, Kenya. The model achieved an overall accuracy of 86% on the test set, with weighted average precision of 0.85, recall of 0.86, and F1-score of 0.84, demonstrating strong predictive performance for binary classification. Key predictors identified included wealth index (children from wealthier households had higher participation odds), residence (urban children showed higher participation than rural counterparts), age (younger children were more likely to be vaccinated), and parental education level (higher education correlated with increased participation). The confusion matrix $\begin{bmatrix} 121 & 4 \\ 17 & 10 \end{bmatrix}$ revealed the model correctly classified 121 true positives and 10 true negatives, with only 4 false positives and 17 false negatives. These findings provide actionable insights for public health planning, highlighting the need for targeted interventions to address socioeconomic and geographic disparities in vaccination access. The project demonstrated the practical utility of logistic regression in public health research and established a framework for using predictive analytics to optimize vaccination campaign strategies and improve coverage in underserved communities.

INTERNSHIP Internship Experience

Chulaimbo County Hospital — Public health data analytics and predictive modeling in HIV, TB, and outpatient care.

Dataset Used: Hospital datasets from Chulaimbo County Hospital (CCC, TB clinic, outpatient register, and lab). Analysis, modeling, and deployment of healthcare data to support clinical decision-making and improve patient outcomes.

METHODOLOGY

Activities included transcribing and collecting data from physical registers, cleaning and preprocessing datasets, and engineering features for analysis. Machine learning models were selected and trained using Python (Scikit-learn), with Random Forest applied to HIV progression prediction and Logistic Regression used for TB outcomes. Data visualization was carried out through bar graphs, line graphs, feature importance charts, and confusion matrices to communicate results clearly. Predictive models were deployed using Streamlit, allowing healthcare workers to input real-time patient data and receive immediate outcome predictions. Version control was maintained through GitHub to enable documentation and future refinements.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The activities resulted in functional predictive models that supported healthcare teams in identifying patient risks and monitoring treatment adherence. Models provided insights into disease prevalence, treatment outcomes, and patient risk factors, while visualizations enhanced understanding of health trends. Deployment into user-friendly web applications increased accessibility, enabling clinicians to make timely, data-driven decisions without technical expertise. The outcomes also included strengthened data quality, better clinical decision support, and enhanced public health insight into HIV and TB management. Additionally, the internship fostered acquisition of technical, analytical, and soft skills, bridging the gap between data science and real-world healthcare practice.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predictors of Mortality During Cholera Outbreak

BACKGROUND & METHODOLOGY

Cholera continues to be a major public health concern, particularly in areas where water quality, sanitation, and health care infrastructures are poor. This study seeks to determine the predictors of cholera mortality during an outbreak based on demographic, environmental, and geographic factors. We therefore apply the logistic regression, Random Forest, and K-Nearest Neighbors to a real dataset from the recent cholera outbreak. This latter one aims to evaluate the influence that several factors have, such as age, sex, income, and water quality, on the availability of sanitation facilities and whether one lives in an informal settlement with respect to cholera death determination. Our analysis starts with a close look at the Exploratory Data Analysis, followed by bivariate analysis of different individual predictors of mortality. Then the logistic regression model will be applied for identification of significant predictors, controlling for other variables. Finally, models like Random Forest and KNN will be applied for better predictability and the choice of more impactful factors causing mortality. The results confirm that the environmental factors, such as water quality and sanitation, and demographic variates, such as age and income, are critical to influence the risk of mortality. The models also illustrate the importance of the geographic location of living in informal settlements on the likelihood of death during the cholera outbreak. The work advances our understanding of the key predictors of cholera mortality, and thus, it will be helpful in allowing development of targeted public health interventions that can minimize the outcome of future cholera outbreaks. Identification of highly vulnerable populations and factors significantly increasing the risk of mortality will feed into improved cholera management and prevention strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully developed and evaluated a logistic regression model to predict mortality during a cholera outbreak using demographic, environmental, and geographic factors. The model achieved an accuracy of 83%, with an F1-score of 0.90 and recall of 0.85, demonstrating strong predictive performance for binary classification of survival outcomes. Key predictors identified included age, sanitation quality, income level, and informal settlement status, with older individuals and those living in areas with poor sanitation experiencing significantly higher mortality rates. The analysis confirmed that inadequate sanitation facilities and lower socio-economic status were strongly associated with increased odds of mortality, highlighting the critical role of environmental and social determinants in cholera outcomes. The model incorporated class weighting (class_weight='balanced') to address dataset imbalance, ensuring reliable predictions for the minority mortality class. Feature selection using Recursive Feature Elimination (RFE) helped identify the top 10 most influential predictors, improving model interpretability and performance. These findings provide actionable evidence for public health officials to prioritize high-risk populations, target sanitation improvements in vulnerable communities, and allocate resources more effectively during cholera outbreaks. The project demonstrated the practical application of supervised machine learning techniques in epidemiological surveillance and outbreak response planning, establishing a foundation for data-driven decision-making in resource-limited public health settings.

INTERNSHIP Internship Experience

Center for Health Solutions(CHS) — Empowering adolescent girls and young women to achieve an AIDS-free future.

Dataset Used: The Determined, Resilient, Empowered, AIDS-free, Mentored, and Safe (DREAMS) partnership, initiated by PEPFAR and global partners. DREAMS is a multisectoral program designed to reduce HIV infections among adolescent girls and young women in sub-Saharan Africa by integrating biomedical, behavioral, and structural interventions that address social, economic, and health-related vulnerabilities.

METHODOLOGY

Implemented HIV prevention initiatives such as HIV testing, PrEP provision, and linkage to care; supported educational retention through school fees and mentorship; delivered economic empowerment programs including vocational training and financial literacy; strengthened families through parenting programs; addressed gender-based violence; and engaged communities to reduce stigma while promoting positive male involvement.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Achieved significant reductions in new HIV infections among adolescent girls and young women, improved retention in school, increased access to health and prevention services, enhanced resilience through economic opportunities, and fostered safer environments that support long-term empowerment and health equity.

CURRENT STATUS

Administrator · Working / employed

Sector relevance: Not related

SKILLS IN USE

Data collection · Data cleaning & wrangling · Communication & teamwork

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

Predictors of Mortality During Cholera Outbreak

BACKGROUND & METHODOLOGY

Cholera continues to be a major public health concern, particularly in areas where water quality, sanitation, and health care infrastructures are poor. This study seeks to determine the predictors of cholera mortality during an outbreak based on demographic, environmental, and geographic factors. We therefore apply the logistic regression, Random Forest, and K-Nearest Neighbors to a real dataset from the recent cholera outbreak. This latter one aims to evaluate the influence that several factors have, such as age, sex, income, and water quality, on the availability of sanitation facilities and whether one lives in an informal settlement with respect to cholera death determination. Our analysis starts with a close look at the Exploratory Data Analysis, followed by bivariate analysis of different individual predictors of mortality. Then the logistic regression model will be applied for identification of significant predictors, controlling for other variables. Finally, models like Random Forest and KNN will be applied for better predictability and the choice of more impactful factors causing mortality. The results confirm that the environmental factors, such as water quality and sanitation, and demographic variates, such as age and income, are critical to influence the risk of mortality. The models also illustrate the importance of the geographic location of living in informal settlements on the likelihood of death during the cholera outbreak. The work advances our understanding of the key predictors of cholera mortality, and thus, it will be helpful in allowing development of targeted public health interventions that can minimize the outcome of future cholera outbreaks. Identification of highly vulnerable populations and factors significantly increasing the risk of mortality will feed into improved cholera management and prevention strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully developed and evaluated a logistic regression model to predict mortality during a cholera outbreak using demographic, environmental, and geographic factors. The model achieved an accuracy of 83%, with an F1-score of 0.90 and recall of 0.85, demonstrating strong predictive performance for binary classification of survival outcomes. Key predictors identified included age, sanitation quality, income level, and informal settlement status, with older individuals and those living in areas with poor sanitation experiencing significantly higher mortality rates. The analysis confirmed that inadequate sanitation facilities and lower socio-economic status were strongly associated with increased odds of mortality, highlighting the critical role of environmental and social determinants in cholera outcomes. The model incorporated class weighting (class_weight='balanced') to address dataset imbalance, ensuring reliable predictions for the minority mortality class. Feature selection using Recursive Feature Elimination (RFE) helped identify the top 10 most influential predictors, improving model interpretability and performance. These findings provide actionable evidence for public health officials to prioritize high-risk populations, target sanitation improvements in vulnerable communities, and allocate resources more effectively during cholera outbreaks. The project demonstrated the practical application of supervised machine learning techniques in epidemiological surveillance and outbreak response planning, establishing a foundation for data-driven decision-making in resource-limited public health settings.

INTERNSHIP Internship Experience

Homabay Teaching and Referral Hospital — Application of predictive analytics and machine learning for early disease detection, treatment monitoring, and outcome prediction across major public health concerns including HIV, tuberculosis, malaria, chronic diseases, and cancer.

Dataset Used: Health facility datasets from Comprehensive Care Clinic (CCC), Chest Clinic, Pediatric Ward (MVIP), Special Clinic, and Oncology Department, comprising demographic, clinical, laboratory, and treatment history data. The datasets were applied to train and validate machine learning models for risk classification, survival prediction, and treatment outcome monitoring. Predictive tools were designed to support clinical decision-making, optimize resource allocation, and improve patient survival in resource-constrained healthcare settings.

METHODOLOGY

Activities involved collecting, cleaning, and harmonizing datasets from multiple hospital departments, followed by feature engineering and selection to identify key predictors of disease outcomes. Machine learning models including Logistic Regression, Random Forest, Linear Regression, and Survival Analysis were applied for risk classification, treatment monitoring, and survival prediction. Model performance was evaluated using accuracy, precision, recall, F1-score, and AUC to ensure reliability. Real-time predictive tools were deployed using Streamlit, accompanied by interactive dashboards and visualizations to support healthcare workers in clinical decision-making.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Outcomes achieved included early detection of high-risk patients, improved triaging, and timely interventions across HIV, TB, malaria, diabetes, hypertension, and cancer. The models highlighted adherence, co-infections, and comorbidities as critical predictors, enabling more targeted and efficient interventions. These insights contributed to reduced mortality, improved patient outcomes, optimized resource allocation, and strengthened institutional capacity through the integration of predictive analytics into routine healthcare practice.



CAPSTONE Capstone Project

Predictors of Mortality During Cholera Outbreak

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully developed and evaluated a logistic regression model to predict mortality during a cholera outbreak using demographic, environmental, and geographic factors. The model achieved an accuracy of 83%, with an F1-score of 0.90 and recall of 0.85, demonstrating strong predictive performance for binary classification of survival outcomes. Key predictors identified included age, sanitation quality, income level, and informal settlement status, with older individuals and those living in areas with poor sanitation experiencing significantly higher mortality rates. The analysis confirmed that inadequate sanitation facilities and lower socio-economic status were strongly associated with increased odds of mortality, highlighting the critical role of environmental and social determinants in cholera outcomes. The model incorporated class weighting (class_weight='balanced') to address dataset imbalance, ensuring reliable predictions for the minority mortality class. Feature selection using Recursive Feature Elimination (RFE) helped identify the top 10 most influential predictors, improving model interpretability and performance. These findings provide actionable evidence for public health officials to prioritize high-risk populations, target sanitation improvements in vulnerable communities, and allocate resources more effectively during cholera outbreaks. The project demonstrated the practical application of supervised machine learning techniques in epidemiological surveillance and outbreak response planning, establishing a foundation for data-driven decision-making in resource-limited public health settings.

INTERNSHIP Internship Experience

Kakamega County General Teaching and Referral Hospital — Predictive healthcare analytics for cancer diagnosis and patient management

Dataset Used: Kakamega General Hospital Cancer Register dataset. The dataset included patient demographic and clinical information, used to analyze patterns, identify correlations, and develop predictive models for cancer staging. The goal was to improve early detection, personalized treatment, and public health outcomes.

METHODOLOGY

Tasks included cleaning and preprocessing the dataset, handling missing values and inconsistencies, conducting exploratory data analysis to identify trends and correlations, creating new features to improve model performance, and developing predictive models using machine learning algorithms such as logistic regression, decision trees, and random forests. Models were evaluated using metrics like accuracy, precision, recall, and F1-score. Data visualizations such as charts, histograms, and scatter plots were created to communicate findings, and tools including Python, R, and Excel were employed. The predictive insights were designed to support cancer diagnosis and treatment planning.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The internship resulted in a predictive model capable of accurately determining cancer stage from the register data, supporting early detection and personalized treatment. The findings provided insights into cancer risk factors and trends, contributing to public health interventions in Kakamega. The work strengthened technical, analytical, and soft skills while demonstrating the impact of data-driven decision-making in improving patient outcomes and clinical workflows.

CURRENT STATUS

Not currently working



CAPSTONE **Capstone Project**

Predicting Weight Category of Children Using Machine Learning

BACKGROUND & METHODOLOGY

This project developed a machine learning model to predict the weight category of children aged 0–5 years using routine child health data from Nakuru County Referral and Teaching Hospital. The dataset was obtained from Maternal Child Health Growth and Monitoring records and contained 1,113 child records. The model used features such as age, sex, weight, height, exclusive breastfeeding status, and Vitamin A supplementation to classify children into weight categories such as underweight, normal weight, overweight, obese, and severe underweight. The project addressed the need for early identification of malnutrition and obesity risks, since traditional health assessments may miss early warning signs. Data preprocessing included removing records with missing values and encoding categorical variables before training a supervised machine learning model.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully produced a Random Forest model that predicts children's weight categories with an accuracy of 79%. The outcome shows that machine learning can support early detection of child growth and nutrition issues using data already collected in health clinics. Key predictors such as height and breastfeeding status were found to be important in determining a child's weight category. The model can help healthcare workers improve routine child health assessments, identify children at risk of undernutrition or overnutrition earlier, and guide timely interventions. However, the project noted limitations such as possible class imbalance and the absence of genetic, nutrition, and socio-economic factors, which could be added in future work to improve accuracy and generalizability.

INTERNSHIP **Internship Experience**

Nakuru County Referral and Teaching Hospital — Mother and Child Health; Child Growth and Monitoring

Dataset Used: Hospital registers (physical records). Included prenatal care, vaccination schedules, maternal health practices, and child growth indicators like height, weight, and nutrition status

METHODOLOGY

The team accessed and reviewed physical health records, analyzed maternal and child health indicators, and examined growth trends to identify stunting and malnutrition patterns. They explored vaccination coverage and effectiveness and documented patterns for future recommendations.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Key insights revealed gaps in maternal care and child nutrition. Stunted growth and malnutrition were highlighted, along with trends in immunization uptake. Recommendations were proposed for digitization of records, improved data accessibility, and enhanced monitoring of maternal and child health interventions.

CURRENT STATUS

Student · Working / employed



CAPSTONE **Capstone Project**

Data-Driven Public Health: Analyzing MOH Datasets Using Machine Learning Approaches

BACKGROUND & METHODOLOGY

The project explores the application of data science and machine learning techniques in analyzing public health data sourced from Kenya's Ministry of Health (MOH) datasets, including MOH 705A & B (Outpatient), MOH 717 (Service Workload), and MOH 731 series (HIV Testing, PMTCT, and Post-Exposure Prophylaxis). The team focused particularly on respiratory disease data due to its high prevalence in the study area. The project involved data collection, cleaning, visualization, and exploratory data analysis to generate meaningful insights for health decision-making. The findings underscore the importance of proper data management in informing health policies and improving service delivery outcomes.

INTERNSHIP **Internship Experience**

Bondeni Sub-county Hospital — Maternal, Child, and Outpatient Health Data

Dataset Used: Kenya EMR System and MOH registers (MOH 204A, 204B, 405, 510, 705A/B, 711, PNC). Patient visits, antenatal and postnatal care, maternal health services, birth outcomes, and pediatric care

METHODOLOGY

The team collected, cleaned, and analyzed EMR datasets, performed descriptive statistics, correlation analyses, and K-Means clustering to identify high-risk maternal and child health groups. They focused on identifying gaps in hypertensive disorders monitoring and evaluating antenatal care service uptake.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed positive antenatal care uptake but gaps in risk assessment for hypertensive disorders in pregnant women. Insights emphasized the need for AI-driven risk assessment models, improved data integration, and enhanced maternal health interventions.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predicting Weight Category of Children Using Machine Learning

BACKGROUND & METHODOLOGY

This project developed a machine learning model to predict the weight category of children aged 0–5 years using routine child health data from Nakuru County Referral and Teaching Hospital. The dataset was obtained from Maternal Child Health Growth and Monitoring records and contained 1,113 child records. The model used features such as age, sex, weight, height, exclusive breastfeeding status, and Vitamin A supplementation to classify children into weight categories such as underweight, normal weight, overweight, obese, and severe underweight. The project addressed the need for early identification of malnutrition and obesity risks, since traditional health assessments may miss early warning signs. Data preprocessing included removing records with missing values and encoding categorical variables before training a supervised machine learning model.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully produced a Random Forest model that predicts children's weight categories with an accuracy of 79%. The outcome shows that machine learning can support early detection of child growth and nutrition issues using data already collected in health clinics. Key predictors such as height and breastfeeding status were found to be important in determining a child's weight category. The model can help healthcare workers improve routine child health assessments, identify children at risk of undernutrition or overnutrition earlier, and guide timely interventions. However, the project noted limitations such as possible class imbalance and the absence of genetic, nutrition, and socio-economic factors, which could be added in future work to improve accuracy and generalizability.

INTERNSHIP Internship Experience

Nakuru County Referral and Teaching Hospital — Mother and Child Health; Child Growth and Monitoring

Dataset Used: Hospital registers (physical records). Included prenatal care, vaccination schedules, maternal health practices, and child growth indicators like height, weight, and nutrition status

METHODOLOGY

The team accessed and reviewed physical health records, analyzed maternal and child health indicators, and examined growth trends to identify stunting and malnutrition patterns. They explored vaccination coverage and effectiveness and documented patterns for future recommendations.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Key insights revealed gaps in maternal care and child nutrition. Stunted growth and malnutrition were highlighted, along with trends in immunization uptake. Recommendations were proposed for digitization of records, improved data accessibility, and enhanced monitoring of maternal and child health interventions.

CURRENT STATUS

Developer advocate · Working / employed



CAPSTONE Capstone Project

Data-Driven Public Health: Analyzing MOH Datasets Using Machine Learning Approaches

BACKGROUND & METHODOLOGY

The project explores the application of data science and machine learning techniques in analyzing public health data sourced from Kenya's Ministry of Health (MOH) datasets, including MOH 705A & B (Outpatient), MOH 717 (Service Workload), and MOH 731 series (HIV Testing, PMTCT, and Post-Exposure Prophylaxis). The team focused particularly on respiratory disease data due to its high prevalence in the study area. The project involved data collection, cleaning, visualization, and exploratory data analysis to generate meaningful insights for health decision-making. The findings underscore the importance of proper data management in informing health policies and improving service delivery outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict child participation in the August 2013 polio vaccination campaign in Bungoma County, Kenya. The model achieved an overall accuracy of 86% on the test set, with weighted average precision of 0.85, recall of 0.86, and F1-score of 0.84, demonstrating strong predictive performance for binary classification. Key predictors identified included wealth index (children from wealthier households had higher participation odds), residence (urban children showed higher participation than rural counterparts), age (younger children were more likely to be vaccinated), and parental education level (higher education correlated with increased participation). The confusion matrix $\begin{bmatrix} 121 & 4 \\ 17 & 10 \end{bmatrix}$ revealed the model correctly classified 121 true positives and 10 true negatives, with only 4 false positives and 17 false negatives. These findings provide actionable evidence for public health planning, highlighting the need for targeted interventions to address socioeconomic and geographic disparities in vaccination access. The project demonstrated the practical utility of logistic regression in public health research and established a framework for using predictive analytics to optimize vaccination campaign strategies and improve coverage in underserved communities.

INTERNSHIP Internship Experience

Bondeni Sub-county Level 4 Hospital — Maternal, Child, and Outpatient Health Data

Dataset Used: Hospital Records including KeMR System, ANC, CCC, Maternity, and Bin Card Inventories. Patient visits, antenatal and postnatal care, maternal health services, birth outcomes, and pediatric care

METHODOLOGY

The team collected, cleaned, and analyzed EMR datasets, performed descriptive statistics, correlation analyses, and K-Means clustering to identify high-risk maternal and child health groups. They focused on identifying gaps in hypertensive disorders monitoring and evaluating antenatal care service uptake.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed positive antenatal care uptake but gaps in risk assessment for hypertensive disorders in pregnant women. Insights emphasized the need for AI-driven risk assessment models, improved data integration, and enhanced maternal health interventions.

CURRENT STATUS

Not currently working

Sector relevance: Partially related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · Natural Language Processing (NLP) · Generative AI / LLMs · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predicting Weight Category of Children Using Machine Learning

BACKGROUND & METHODOLOGY

This project developed a machine learning model to predict the weight category of children aged 0–5 years using routine child health data from Nakuru County Referral and Teaching Hospital. The dataset was obtained from Maternal Child Health Growth and Monitoring records and contained 1,113 child records. The model used features such as age, sex, weight, height, exclusive breastfeeding status, and Vitamin A supplementation to classify children into weight categories such as underweight, normal weight, overweight, obese, and severe underweight. The project addressed the need for early identification of malnutrition and obesity risks, since traditional health assessments may miss early warning signs. Data preprocessing included removing records with missing values and encoding categorical variables before training a supervised machine learning model.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully produced a Random Forest model that predicts children's weight categories with an accuracy of 79%. The outcome shows that machine learning can support early detection of child growth and nutrition issues using data already collected in health clinics. Key predictors such as height and breastfeeding status were found to be important in determining a child's weight category. The model can help healthcare workers improve routine child health assessments, identify children at risk of undernutrition or overnutrition earlier, and guide timely interventions. However, the project noted limitations such as possible class imbalance and the absence of genetic, nutrition, and socio-economic factors, which could be added in future work to improve accuracy and generalizability.

INTERNSHIP Internship Experience

Nakuru County Teaching & Referral Hospital — Mother and Child Health; Child Growth and Monitoring

★ LED TO A JOB OFFER

Dataset Used: Hospital registers (physical records). Included prenatal care, vaccination schedules, maternal health practices, and child growth indicators like height, weight, and nutrition status

METHODOLOGY

The team accessed and reviewed physical health records, analyzed maternal and child health indicators, and examined growth trends to identify stunting and malnutrition patterns. They explored vaccination coverage and effectiveness and documented patterns for future recommendations.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Key insights revealed gaps in maternal care and child nutrition. Stunted growth and malnutrition were highlighted, along with trends in immunization uptake. Recommendations were proposed for digitization of records, improved data accessibility, and enhanced monitoring of maternal and child health interventions.

CURRENT STATUS

Intern at ENGAGE · Working / employed

Sector relevance: Partially related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: 4 / 5



CAPSTONE Capstone Project

Data-Driven Public Health: Analyzing MOH Datasets Using Machine Learning Approaches

BACKGROUND & METHODOLOGY

The project explores the application of data science and machine learning techniques in analyzing public health data sourced from Kenya's Ministry of Health (MOH) datasets, including MOH 705A & B (Outpatient), MOH 717 (Service Workload), and MOH 731 series (HIV Testing, PMTCT, and Post-Exposure Prophylaxis). The team focused particularly on respiratory disease data due to its high prevalence in the study area. The project involved data collection, cleaning, visualization, and exploratory data analysis to generate meaningful insights for health decision-making. The findings underscore the importance of proper data management in informing health policies and improving service delivery outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict child participation in the August 2013 polio vaccination campaign in Bungoma County, Kenya. The model achieved an overall accuracy of 86% on the test set, with weighted average precision of 0.85, recall of 0.86, and F1-score of 0.84, demonstrating strong predictive performance for binary classification. Key predictors identified included wealth index (children from wealthier households had higher participation odds), residence (urban children showed higher participation than rural counterparts), age (younger children were more likely to be vaccinated), and parental education level (higher education correlated with increased participation). The confusion matrix $\begin{bmatrix} 121 & 4 \\ 17 & 10 \end{bmatrix}$ revealed the model correctly classified 121 true positives and 10 true negatives, with only 4 false positives and 17 false negatives. These findings provide actionable evidence for public health planning, highlighting the need for targeted interventions to address socioeconomic and geographic disparities in vaccination access. The project demonstrated the practical utility of logistic regression in public health research and established a framework for using predictive analytics to optimize vaccination campaign strategies and improve coverage in underserved communities.

INTERNSHIP Internship Experience

Nakuru County Referral and Teaching Hospital — Maternal, Child, and Outpatient Health Data

Dataset Used: Kenya EMR System and MOH registers (MOH 204A, 204B, 405, 510, 705A/B, 711, PNC). Patient visits, antenatal and postnatal care, maternal health services, birth outcomes, and pediatric care

METHODOLOGY

The team collected, cleaned, and analyzed EMR datasets, performed descriptive statistics, correlation analyses, and K-Means clustering to identify high-risk maternal and child health groups. They focused on identifying gaps in hypertensive disorders monitoring and evaluating antenatal care service uptake.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed positive antenatal care uptake but gaps in risk assessment for hypertensive disorders in pregnant women. Insights emphasized the need for AI-driven risk assessment models, improved data integration, and enhanced maternal health interventions.

CURRENT STATUS

Not currently working

**CAPSTONE** **Capstone Project***Predictive Modeling of Heart Disease in Kenya Using Machine Learning***BACKGROUND & METHODOLOGY**

Heart disease is a major public health concern in Kenya, accounting for approximately 10–15% of all deaths and ranking among the leading causes of mortality after infectious diseases. Cardiovascular disease (CVD) encompasses a range of conditions affecting the heart and blood vessels, often linked to risk factors such as high blood pressure, high cholesterol, diabetes, smoking, and physical inactivity. This study aims to identify the key risk factors of heart disease in Kenya and develop a predictive machine learning model to assist in early detection and prevention. Using secondary data obtained from Kaggle, logistic regression was employed for classification due to its effectiveness in analyzing relationships between medical factors and disease presence. Model evaluation was performed using confusion matrix, precision, recall, F1-score, and accuracy metrics. The model achieved an accuracy of 83%, demonstrating strong reliability in predicting heart disease outcomes. This research underscores the potential of machine learning in supporting healthcare professionals through early risk identification and improved allocation of limited health resources in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the presence of heart disease using medical and lifestyle-related risk factors. The study used secondary data from Kaggle and focused on cardiovascular disease indicators such as age, chest pain, blood pressure, maximum heart rate, abnormal heartbeat, exercise-induced angina, cholesterol-related factors, diabetes, smoking, physical inactivity, and stress. The model was evaluated using a confusion matrix, classification report, F1-score, and accuracy score, achieving an overall accuracy of 83%, with precision of 0.90, recall of 0.76, and F1-score of 0.82 for class 0. The outcome shows that machine learning can support early identification of individuals at risk of heart disease, improve clinical decision-making, and guide prevention and management strategies, while further validation and improvement are recommended to increase recall and capture more true positive cases.

INTERNSHIP **Internship Experience***Lifecare Hospital — Patient health records and insurance claim data analysis.*

Dataset Used: LifeCare Hospital, Medical Records Department and Front Office. Deceased patient files and inpatient/outpatient medical records used for data entry, analysis, and predictive modeling.

METHODOLOGY

Worked with both inpatient and outpatient files, focusing on data entry, analysis, and prediction. Handled patient data related to the duration of stay and common diagnoses such as hypertension and diabetes, aiming to provide insights into hospital operations and patient outcomes. Participated in analyzing claim rejection patterns for chronic diseases treatment and recommending ways to reduce insurance claim rejections, specifically focusing on repeated treatments within a month.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The intern contributed to predicting hospital stay durations, identifying diagnoses associated with prolonged stays, and developing a method to flag multiple treatments to reduce chronic disease claim rejections. Technical skills gained included data cleaning, visualization, predictive modeling, and understanding insurance claim processes. Soft skills acquired encompassed time management, teamwork, communication, and professional handling of confidential patient data. The internship also provided insights into public health challenges and proposed improvements for patient workflow efficiency.

CURRENT STATUS**Consultant · Working / employed**



CAPSTONE Capstone Project

Predicting Hair Loss Using Machine Learning

BACKGROUND & METHODOLOGY

Alopecia (hair loss) is a prevalent dermatological condition with rising incidence in Kenya, especially among women aged 15–49, and substantial psychosocial and quality-of-life impacts. This study develops a machine-learning approach to predict hair-loss occurrence and explore contributing factors spanning hormonal changes, autoimmune and chronic conditions, scalp infections, hereditary patterns, medications, and lifestyle practices. The workflow comprised data loading and exploration, preprocessing, feature selection, train—test splitting, model fitting, and evaluation. The final classifier achieved 97% overall accuracy, with particularly strong performance for two major classes (class 0: precision 97%, recall 100%; class 1: precision 100%, recall 82%), indicating reliable discrimination for the most represented categories while highlighting room to improve recall for underrepresented classes (e.g., class 3). Findings suggest the model can support early risk flagging and targeted counseling, while future work should expand and rebalance datasets, incorporate richer clinical and behavioral variables, and assess generalizability across populations. The study underscores the potential of predictive analytics to inform screening and intervention strategies for women in the Kenyan context.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a machine learning classification model to predict the occurrence of hair loss among females aged 15–49 years by analyzing factors associated with alopecia, including hormonal changes, medical conditions, chronic illnesses, scalp infections, hereditary factors, lifestyle practices, stress levels, environmental factors, and medication-related causes. The study involved loading and exploring the dataset, preprocessing the data, selecting features, splitting the data, fitting and testing the model, and evaluating model performance using accuracy and classification metrics. The model achieved a reported accuracy of 97%, with strong precision and recall for the main classes, showing that machine learning can help identify individuals at risk of hair loss and support early awareness, prevention, and further investigation of related health or lifestyle factors.

INTERNSHIP Internship Experience

Meru University Medical and Research Center — Student health records analysis and predictive modeling of health conditions.

Dataset Used: Meru University Medical and Research Center, Health Record Office. Health data of first- to fourth-year students, including diagnoses, vital signs, and reproductive health information, used for data cleaning, visualization, and predictive modeling.

METHODOLOGY

Applying artificial intelligence and machine learning techniques to collect, clean, analyze, and visualize student health data. Data was extracted from hospital health record books and entered into Excel. Analysis and modeling were conducted using Google Colab with Python libraries (Pandas, NumPy, Matplotlib, Scikit-learn). Logistic regression models were applied to classify health outcomes such as high blood pressure. Projects involved analyzing triage records and exploring semester-wise student diagnoses to identify trends and correlations between gender and prevalent conditions. Heatmaps and visualizations were created to highlight patterns across academic years.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Produced predictive models for student health conditions and identified leading diagnoses such as Vaginal Candidiasis, pregnancy, and UTI among female students, and urethritis, wounds, and dental issues among male students. Technical skills developed included data cleaning, visualization, model selection, and deployment of an AI health application. Additional competencies included adaptation to the work environment, communication, discipline, and time management. Public health insights highlighted the prevalence of reproductive system conditions among students, supporting the recommendation for awareness campaigns and targeted interventions.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

A Machine Learning Approach to Predicting Fetal Health Outcomes

BACKGROUND & METHODOLOGY

Fetal health refers to the well-being and proper development of a baby during pregnancy. In Kenya, numerous challenges such as preterm birth, birth defects, fetal anemia, fetal distress, macrosomia, and umbilical cord complications threaten fetal health. Traditionally, obstetricians rely on cardiotocography (CTG) to monitor fetal heart rate and uterine contractions, but this manual interpretation is often time-consuming and unreliable. This project proposes a computer-aided system utilizing a supervised machine learning approach—specifically, the Logistic Regression model—to classify fetal health data and predict the fetus's condition accurately. The study involved data importation, exploration, visualization, feature selection using correlation heat maps, model training, and evaluation. The model achieved a 100% performance rate, demonstrating its potential to improve early detection of fetal health issues and assist healthcare professionals in making faster, data-driven decisions to ensure better prenatal outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a supervised machine learning model using logistic regression to classify fetal health and support early detection of fetal well-being concerns. The study focused on fetal health problems such as preterm birth, fetal distress, fetal anemia, birth defects, macrosomia, and umbilical cord complications, with emphasis on improving traditional cardiotocography-based assessment, which can be time-consuming and unreliable. The team loaded and explored the dataset, visualized patterns, selected relevant features using correlation analysis and a heatmap, split the data for training and testing, fitted the logistic regression model, and evaluated its performance. The model reportedly achieved 100% performance, showing that machine learning can support faster and more reliable classification of fetal health status and assist healthcare providers in identifying at-risk pregnancies for timely intervention.

INTERNSHIP Internship Experience

Meru University Medical and Research Center — Student health records analysis and predictive modeling of health conditions.

Dataset Used: Meru University Medical and Research Center, Records Office. Health data of first-year to fourth-year students, including triage records and diagnoses grouped into infectious, respiratory, reproductive/urinary, skin/soft tissue, neurological, oral/dental, eye/ENT, and dermatological conditions, used for data cleaning, visualization, and predictive modeling

METHODOLOGY

Collecting, cleaning, analyzing, and visualizing student health data, as well as deploying predictive models using artificial intelligence and machine learning techniques. Data from first-year triage records and semester-wise student diagnosis records were processed in Excel, analyzed and modeled in Google Colab, and deployed on Hugging Face. Projects included building a predictive model for high blood pressure using pulse rate, temperature, BMI, and gender, as well as exploring semester-wise health trends to identify the most prevalent conditions among male and female students. Heatmaps and correlation analyses were conducted to illustrate patterns across academic years.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Produced predictive models for high blood pressure and identified leading health conditions, including Vaginal Candidiasis, pregnancy, and UTI among female students, and urethritis, wounds, and dental issues among male students. Technical skills developed included data collection, cleaning, visualization, modeling, and deployment of an AI-based health application. Soft skills acquired encompassed teamwork, communication, time management, and professional etiquette. Public health insights highlighted the prevalence of reproductive system conditions, emphasizing the need for awareness campaigns and interventions to curb the spread of infections.

CURRENT STATUS

Ethical Hacker · Not currently working

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP) · Prompt engineering · Deep Learning · Generative AI / LLMs · Ethics & bias in AI · Freelancing in data science / building a portfolio · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

A Machine Learning Approach to Predicting Fetal Health Outcomes

BACKGROUND & METHODOLOGY

Fetal health refers to the well-being and proper development of a baby during pregnancy. In Kenya, numerous challenges such as preterm birth, birth defects, fetal anemia, fetal distress, macrosomia, and umbilical cord complications threaten fetal health. Traditionally, obstetricians rely on cardiotocography (CTG) to monitor fetal heart rate and uterine contractions, but this manual interpretation is often time-consuming and unreliable. This project proposes a computer-aided system utilizing a supervised machine learning approach—specifically, the Logistic Regression model—to classify fetal health data and predict the fetus's condition accurately. The study involved data importation, exploration, visualization, feature selection using correlation heat maps, model training, and evaluation. The model achieved a 100% performance rate, demonstrating its potential to improve early detection of fetal health issues and assist healthcare professionals in making faster, data-driven decisions to ensure better prenatal outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a supervised machine learning model using logistic regression to classify fetal health and support early detection of fetal well-being concerns. The study focused on fetal health problems such as preterm birth, fetal distress, fetal anemia, birth defects, macrosomia, and umbilical cord complications, with emphasis on improving traditional cardiotocography-based assessment, which can be time-consuming and unreliable. The team loaded and explored the dataset, visualized patterns, selected relevant features using correlation analysis and a heatmap, split the data for training and testing, fitted the logistic regression model, and evaluated its performance. The model reportedly achieved 100% performance, showing that machine learning can support faster and more reliable classification of fetal health status and assist healthcare providers in identifying at-risk pregnancies for timely intervention.

INTERNSHIP Internship Experience

Mikinduri District Hospital — Maternal health data analysis and predictive modeling of pregnancy-related outcomes.

Dataset Used: Mikinduri Sub-county hospital, maternal health department. Maternal health data including parity, weight, height, gravidae, and blood pressure, used for data cleaning, visualization, and predictive modeling to assess risks such as overweight and gestational diabetes during pregnancy.

METHODOLOGY

Collecting, cleaning, analyzing, and visualizing maternal health data, as well as developing predictive models to forecast health outcomes. Data was gathered from hospital records, entered into software tools such as Excel and Python, and analyzed using statistical methods. Visualizations such as charts and graphs were created to effectively communicate insights. Projects focused on identifying trends related to overweight, gestational diabetes, and other pregnancy-related risks, and on recommending preventive measures through data-driven analysis.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Produced insights into maternal health risks, including the impact of overweight on premature birth and gestational diabetes. Technical skills developed included data collection, analysis, visualization, and predictive modeling using Python and Excel. Soft skills acquired encompassed teamwork, communication, time management, and professional etiquette. Public health insights emphasized the importance of data-driven interventions to track trends, identify at-risk populations, and guide preventive health measures, supporting improved maternal health outcomes in the community.

CURRENT STATUS

Systems Engineer · Working / employed



CAPSTONE Capstone Project

Predicting Hair Loss Using Machine Learning

BACKGROUND & METHODOLOGY

Alopecia (hair loss) is a prevalent dermatological condition with rising incidence in Kenya, especially among women aged 15–49, and substantial psychosocial and quality-of-life impacts. This study develops a machine-learning approach to predict hair-loss occurrence and explore contributing factors spanning hormonal changes, autoimmune and chronic conditions, scalp infections, hereditary patterns, medications, and lifestyle practices. The workflow comprised data loading and exploration, preprocessing, feature selection, train—test splitting, model fitting, and evaluation. The final classifier achieved 97% overall accuracy, with particularly strong performance for two major classes (class 0: precision 97%, recall 100%; class 1: precision 100%, recall 82%), indicating reliable discrimination for the most represented categories while highlighting room to improve recall for underrepresented classes (e.g., class 3). Findings suggest the model can support early risk flagging and targeted counseling, while future work should expand and rebalance datasets, incorporate richer clinical and behavioral variables, and assess generalizability across populations. The study underscores the potential of predictive analytics to inform screening and intervention strategies for women in the Kenyan context.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a machine learning classification model to predict the occurrence of hair loss among females aged 15–49 years by analyzing factors associated with alopecia, including hormonal changes, medical conditions, chronic illnesses, scalp infections, hereditary factors, lifestyle practices, stress levels, environmental factors, and medication-related causes. The study involved loading and exploring the dataset, preprocessing the data, selecting features, splitting the data, fitting and testing the model, and evaluating model performance using accuracy and classification metrics. The model achieved a reported accuracy of 97%, with strong precision and recall for the main classes, showing that machine learning can help identify individuals at risk of hair loss and support early awareness, prevention, and further investigation of related health or lifestyle factors.

INTERNSHIP Internship Experience

Meru Hospice — Palliative care and cancer patient outcomes.

Dataset Used: Cancer patient records at Meru Hospice. Dataset containing patient demographics, cancer types, treatment details, mortality, and lifestyle factors such as alcohol and substance use.

METHODOLOGY

Collecting and organizing patient records into Excel, cleaning data by addressing missing or inconsistent information, and preparing it for analysis. Performing exploratory data analysis using Python in Google Colab to examine trends in demographics, cancer types, treatment efficacy, and mortality. Creating visualizations, such as bar charts and line graphs, and developing predictive models to forecast patient outcomes based on historical data.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Improved dataset accuracy and reliability for future research. Identified effective palliative care interventions, highlighting the impact of pain management, psychological support, and home visits on patient quality of life. Produced visual and predictive insights that informed hospice decision-making, prioritized high-risk patients, and revealed prevalent cancers in the region, contributing to evidence-based strategies for enhancing public health and patient care.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predicting Hair Loss Using Machine Learning

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a machine learning classification model to predict the occurrence of hair loss among females aged 15–49 years by analyzing factors associated with alopecia, including hormonal changes, medical conditions, chronic illnesses, scalp infections, hereditary factors, lifestyle practices, stress levels, environmental factors, and medication-related causes. The study involved loading and exploring the dataset, preprocessing the data, selecting features, splitting the data, fitting and testing the model, and evaluating model performance using accuracy and classification metrics. The model achieved a reported accuracy of 97%, with strong precision and recall for the main classes, showing that machine learning can help identify individuals at risk of hair loss and support early awareness, prevention, and further investigation of related health or lifestyle factors.

INTERNSHIP Internship Experience

CIFORD KENYA — Water access, sanitation, and community financial practices analysis.

Dataset Used: CIFORD Kenya; datasets included Meru Baseline Survey, Water Tanks Project data, and Tharaka Administrators Survey.

Community-level data on water access, sanitation practices, health outcomes, and financial record-keeping, used for cleaning, analysis, visualization, and descriptive modeling.

METHODOLOGY

Cleaning, analyzing, and visualizing datasets related to water access, sanitation, and financial practices. Data from the Meru Baseline Survey, Water Tanks Project, and Tharaka Administrators Survey were processed using Excel and Google Colab. Analytical tasks involved descriptive statistics and variable extraction, while visualizations were created using Python libraries such as Pandas, Matplotlib, NumPy, and Seaborn. Projects focused on assessing water tank effectiveness, identifying challenges in water access and sanitation, and evaluating community financial record-keeping practices. Results were shared with team members to inform decision-making and public health interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified critical issues including limited water access, inadequate sanitation, and gaps in financial management and record-keeping within communities. Technical skills developed included data cleaning, descriptive analysis, and visualization using Python and Excel. Soft skills acquired encompassed teamwork, communication, time management, and professional etiquette. Public health insights highlighted the connection between water and sanitation access and community health outcomes, emphasizing the need for improved water management and hygiene practices.

CURRENT STATUS

General worker · Not currently working

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Predictive Modeling for Gestational Diabetes Using Machine Learning

BACKGROUND & METHODOLOGY

Gestational diabetes is a public health concern that occurs during pregnancy when the body cannot produce sufficient insulin to meet increased demands, leading to high blood sugar levels. Although it typically resolves after childbirth, it significantly increases the risk of developing Type 2 diabetes later in life. The study aims to identify risk factors for gestational diabetes among pregnant women and develop a predictive machine learning model for early detection. Using secondary data from Kaggle, logistic regression was employed due to its effectiveness in handling binary outcomes. Key variables included pregnancies, glucose levels, blood pressure, insulin, BMI, and age. The developed model achieved 77.9% accuracy, with precision of 0.81, recall of 0.86, and an F1-score of 0.83. This predictive model facilitates early identification of at-risk women, enabling timely monitoring and preventive interventions to improve maternal and fetal health outcomes and reduce healthcare costs associated with gestational diabetes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict gestational diabetes among pregnant women using variables such as pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and outcome. The study used secondary data from Kaggle, applied model training and evaluation, and assessed performance using precision, recall, F1-score, and confusion matrix. The model achieved an accuracy of 77.9%, with precision of 0.81, recall of 0.86, and F1-score of 0.83 for identifying non-diabetic cases, showing good predictive performance. The outcome demonstrates that machine learning can support early detection of gestational diabetes, helping healthcare providers identify at-risk women, improve monitoring, guide preventive interventions, and promote better maternal and fetal health outcomes.

INTERNSHIP Internship Experience

Lifecare Hospital — Hospital patient records analysis and predictive modeling of inpatient outcomes and insurance claim patterns.

Dataset Used: LifeCare Hospital, Medical Records Department (MRD) and Front Office. Deceased patient files (RIP files) and inpatient/outpatient medical records used for data entry, analysis, visualization, and predictive modeling to support hospital operations and reduce insurance claim rejections.

METHODOLOGY

Collected, cleaned, analyzed, and visualized inpatient and outpatient records to predict hospital stay durations and assess insurance claim rejections for chronic diseases. Data was processed using Excel and Google Colab, with statistical analysis and predictive modeling performed in Python (Pandas, Matplotlib). Visualizations and trend analyses were created using Streamlit. Projects involved developing predictive models for length of hospital stay and automating the identification of multiple treatments within a month to improve understanding of claim rejection patterns.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Insights were generated on common diagnoses associated with prolonged hospital stays, such as hypertension and diabetes, and methods to flag repeated treatments contributing to claim rejections were identified. Technical skills gained included data cleaning, analysis, visualization, predictive modeling, and understanding insurance claim processes. Soft skills developed included time management, teamwork, communication, and professional handling of confidential patient data. Public health insights highlighted the importance of data analysis in improving hospital resource management and addressing chronic disease care challenges.

CURRENT STATUS

Not currently working



CAPSTONE **Capstone Project**

Predictive Modeling for Gestational Diabetes Using Machine Learning

BACKGROUND & METHODOLOGY

Gestational diabetes is a public health concern that occurs during pregnancy when the body cannot produce sufficient insulin to meet increased demands, leading to high blood sugar levels. Although it typically resolves after childbirth, it significantly increases the risk of developing Type 2 diabetes later in life. The study aims to identify risk factors for gestational diabetes among pregnant women and develop a predictive machine learning model for early detection. Using secondary data from Kaggle, logistic regression was employed due to its effectiveness in handling binary outcomes. Key variables included pregnancies, glucose levels, blood pressure, insulin, BMI, and age. The developed model achieved 77.9% accuracy, with precision of 0.81, recall of 0.86, and an F1-score of 0.83. This predictive model facilitates early identification of at-risk women, enabling timely monitoring and preventive interventions to improve maternal and fetal health outcomes and reduce healthcare costs associated with gestational diabetes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict gestational diabetes among pregnant women using variables such as pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and outcome. The study used secondary data from Kaggle, applied model training and evaluation, and assessed performance using precision, recall, F1-score, and confusion matrix. The model achieved an accuracy of 77.9%, with precision of 0.81, recall of 0.86, and F1-score of 0.83 for identifying non-diabetic cases, showing good predictive performance. The outcome demonstrates that machine learning can support early detection of gestational diabetes, helping healthcare providers identify at-risk women, improve monitoring, guide preventive interventions, and promote better maternal and fetal health outcomes.

INTERNSHIP **Internship Experience**

Meru Hospice — Palliative care and cancer patient data analysis.

Dataset Used: Meru Hospice, patient records. Cancer patient data including demographic information, medical history, symptoms, treatments, and outcomes, used for data cleaning, analysis, visualization, and predictive modeling to support palliative care services.

METHODOLOGY

Collected, cleaned, and prepared cancer patient records for analysis. Performed statistical analysis and created visualizations to identify trends and patterns. Applied machine learning algorithms for predictive modeling of patient outcomes. Used Python (Pandas) and R for data processing, analysis, and visualization, focusing on improving palliative care delivery.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified trends in patient demographics, treatments, and symptom management affecting palliative care. Developed skills in data cleaning, analysis, visualization, statistical methods, and predictive modeling. Gained soft skills in teamwork, communication, time management, and professional conduct. Generated public health insights on palliative care challenges, health disparities, quality of care, ethical considerations, and the impact of interventions.

CURRENT STATUS

Data Analyst · Working / employed



CAPSTONE Capstone Project

Predictive Modeling of Heart Disease in Kenya Using Machine Learning

BACKGROUND & METHODOLOGY

Heart disease is a major public health concern in Kenya, accounting for approximately 10–15% of all deaths and ranking among the leading causes of mortality after infectious diseases. Cardiovascular disease (CVD) encompasses a range of conditions affecting the heart and blood vessels, often linked to risk factors such as high blood pressure, high cholesterol, diabetes, smoking, and physical inactivity. This study aims to identify the key risk factors of heart disease in Kenya and develop a predictive machine learning model to assist in early detection and prevention. Using secondary data obtained from Kaggle, logistic regression was employed for classification due to its effectiveness in analyzing relationships between medical factors and disease presence. Model evaluation was performed using confusion matrix, precision, recall, F1-score, and accuracy metrics. The model achieved an accuracy of 83%, demonstrating strong reliability in predicting heart disease outcomes. This research underscores the potential of machine learning in supporting healthcare professionals through early risk identification and improved allocation of limited health resources in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the presence of heart disease using medical and lifestyle-related risk factors. The study used secondary data from Kaggle and focused on cardiovascular disease indicators such as age, chest pain, blood pressure, maximum heart rate, abnormal heartbeat, exercise-induced angina, cholesterol-related factors, diabetes, smoking, physical inactivity, and stress. The model was evaluated using a confusion matrix, classification report, F1-score, and accuracy score, achieving an overall accuracy of 83%, with precision of 0.90, recall of 0.76, and F1-score of 0.82 for class 0. The outcome shows that machine learning can support early identification of individuals at risk of heart disease, improve clinical decision-making, and guide prevention and management strategies, while further validation and improvement are recommended to increase recall and capture more true positive cases.

INTERNSHIP Internship Experience

Furaha Centre Meru — Children's health and community livelihood project data

Dataset Used: Furaha Centre Meru; datasets included assessment data, health data, and Goat Project data. Data on children with developmental conditions and community goats, used to evaluate early interventions, therapeutic outcomes, and the impact of livelihood support initiatives.

METHODOLOGY

Collected, cleaned, and analyzed children's health records to assess developmental needs and intervention outcomes. Tracked the status of goats distributed to needy families, documenting survival, returns, and losses. Created visualizations and summaries to identify trends and gaps. Prepared inventory lists and automated payslip systems. Python and Excel were used for data processing, analysis, and reporting, supporting both health and livelihood projects.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Provided insights into developmental challenges and the effectiveness of early interventions for children with autism, developmental delays, cerebral palsy, and Down syndrome. Goat Project data revealed how families managed livestock, including survival rates and returns, informing sustainability and future community support. Developed technical skills in data analysis, visualization, and reporting, alongside soft skills in teamwork, communication, and problem-solving. Public health insights highlighted the importance of integrated child health services and community-based livelihood programs.

CURRENT STATUS

Not currently working

CAPSTONE Capstone Project*Predictive Modeling of Heart Disease in Kenya Using Machine Learning***BACKGROUND & METHODOLOGY**

Heart disease is a major public health concern in Kenya, accounting for approximately 10–15% of all deaths and ranking among the leading causes of mortality after infectious diseases. Cardiovascular disease (CVD) encompasses a range of conditions affecting the heart and blood vessels, often linked to risk factors such as high blood pressure, high cholesterol, diabetes, smoking, and physical inactivity. This study aims to identify the key risk factors of heart disease in Kenya and develop a predictive machine learning model to assist in early detection and prevention. Using secondary data obtained from Kaggle, logistic regression was employed for classification due to its effectiveness in analyzing relationships between medical factors and disease presence. Model evaluation was performed using confusion matrix, precision, recall, F1-score, and accuracy metrics. The model achieved an accuracy of 83%, demonstrating strong reliability in predicting heart disease outcomes. This research underscores the potential of machine learning in supporting healthcare professionals through early risk identification and improved allocation of limited health resources in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the presence of heart disease using medical and lifestyle-related risk factors. The study used secondary data from Kaggle and focused on cardiovascular disease indicators such as age, chest pain, blood pressure, maximum heart rate, abnormal heartbeat, exercise-induced angina, cholesterol-related factors, diabetes, smoking, physical inactivity, and stress. The model was evaluated using a confusion matrix, classification report, F1-score, and accuracy score, achieving an overall accuracy of 83%, with precision of 0.90, recall of 0.76, and F1-score of 0.82 for class 0. The outcome shows that machine learning can support early identification of individuals at risk of heart disease, improve clinical decision-making, and guide prevention and management strategies, while further validation and improvement are recommended to increase recall and capture more true positive cases.

INTERNSHIP Internship Experience*Furaha Center, Meru — Early childhood development and livelihood support for families of children with developmental conditions.*

Dataset Used: Children assessment records and Goat Project data from Furaha Center clinics in Meru, Mikinduri, and Laisamis. Data on children's developmental and health conditions, plus tracking of goats distributed to support needy families, was collected, cleaned, analyzed, and visualized to guide interventions and monitor project outcomes.

METHODOLOGY

Data from children's assessments was entered into Excel, cleaned using Python, and explored to identify patterns in developmental delays, behavioral issues, and health conditions across locations. Charts and graphs were created to visualize trends over time and by region. For the Goat Project, data on goat allocation, gender, and status (alive, dead, or returned) was collected from households, cleaned, and analyzed to monitor survival rates and track economic impact.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Assessment analysis revealed the most common developmental and health conditions among children, enabling Furaha management to plan targeted interventions and training programs. The Goat Project showed most goats survived and contributed to household income through milk or breeding, while data highlighted challenges such as goat health maintenance, informing strategies to improve sustainability and project support for families.

CURRENT STATUS

Administrator · Working / employed



CAPSTONE Capstone Project

Predicting the prevalence of polycystic ovary syndrome using machine learning

BACKGROUND & METHODOLOGY

This project developed a machine-learning pipeline to predict the prevalence of polycystic ovary syndrome (PCOS) among women of reproductive age. Using the PCOS_extended_dataset.csv from Google Drive, the team performed data cleaning (handling missing values, removing duplicates, and correcting types), engineered features (standardizing columns and creating dummy encodings for ?-hCG and AMH), and applied correlation-guided feature selection to remove weak predictors. A logistic regression classifier was then trained with a 70/30 train—test split (random_state = 42). Exploratory analyses and plots examined relationships between PCOS status and BMI, weight gain, AMH, hair growth, and skin darkening.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict the prevalence/status of Polycystic Ovary Syndrome (PCOS) among women using a PCOS dataset with 2,000 records and 44 variables. The study used health and clinical indicators such as age, weight, height, BMI, cycle regularity, hormonal measures, weight gain, hair growth, skin darkening, hair loss, pimples, fast food intake, regular exercise, blood pressure, follicle count, follicle size, and endometrium measurements. The data was cleaned, categorical variables were converted using dummy encoding, relevant features were selected, and the dataset was split into 70% training and 30% testing. The model was evaluated using a confusion matrix, classification report, and accuracy score, with a reported accuracy of 100%, showing strong classification performance in identifying PCOS status. The outcome demonstrates that machine learning can support early prediction of PCOS and help guide awareness, screening, and timely health interventions, although the perfect accuracy should be interpreted cautiously and validated further.

INTERNSHIP Internship Experience

Meru Hospice — Palliative care and cancer patient outcomes.

Dataset Used: Patient records from Meru Hospice covering demographics, diagnosis, treatment, and mortality data. Data from 175 cancer patients was collected, cleaned, analyzed, and modeled to explore factors influencing mortality and improve palliative care strategies.

METHODOLOGY

Entering patient records into Excel and cleaning data to ensure completeness and accuracy. Conducting exploratory analysis using Python in Google Colab to identify trends in demographics, cancer types, treatments, and mortality outcomes. Developing predictive models to assess factors affecting patient survival and creating visualizations to communicate insights to the hospice team.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis revealed key predictors of mortality, guiding improvements in palliative care and resource allocation. The work provided actionable insights into patient risk factors and informed evidence-based interventions. The process strengthened technical skills in data cleaning, analysis, and predictive modeling, while enhancing teamwork, time management, and professional etiquette.

CURRENT STATUS

Site Agent · Working / employed

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Predicting the prevalence of polycystic ovary syndrome using machine learning

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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INTERNSHIP Internship Experience

CIFORD Kenya — Water, sanitation, and financial health.

Dataset Used: Meru Baseline Survey, Tharaka Administrator Survey, and Water Tanks Project Data. Analysis of community water access, sanitation conditions, and financial practices to inform public health interventions and resource allocation.

METHODOLOGY

Cleaning and organizing survey and project data to correct inconsistencies and missing values, ensuring reliability for analysis. Conducting descriptive analysis in Google Colab, extracting trends in water access, sanitation practices, and financial management. Creating visualizations using Python libraries (Pandas, Matplotlib, Seaborn) to illustrate the impact of sanitation and water availability on health outcomes. Collaborating with team members during proposal meetings and community activities to validate findings and contextualize results.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed that inadequate water access significantly contributed to sanitation challenges, which in turn affected community health outcomes. Visualizations and reports provided actionable insights for CIFORD Kenya, highlighting priority areas for intervention and resource allocation. The internship enhanced technical skills in data cleaning, analysis, and visualization, strengthened teamwork and communication abilities, and deepened understanding of public health issues related to water and sanitation in Meru and Tharaka regions.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predicting the prevalence of polycystic ovary syndrome using machine learning

BACKGROUND & METHODOLOGY

This project developed a machine-learning pipeline to predict the prevalence of polycystic ovary syndrome (PCOS) among women of reproductive age. Using the PCOS_extended_dataset.csv from Google Drive, the team performed data cleaning (handling missing values, removing duplicates, and correcting types), engineered features (standardizing columns and creating dummy encodings for ?-hCG and AMH), and applied correlation-guided feature selection to remove weak predictors. A logistic regression classifier was then trained with a 70/30 train—test split (random_state = 42). Exploratory analyses and plots examined relationships between PCOS status and BMI, weight gain, AMH, hair growth, and skin darkening.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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INTERNSHIP Internship Experience

Furaha Centre, Meru — Child health and developmental assessment.

Dataset Used: Furaha Centre Meru; datasets included assessment data, health data, and Goat Project data. Data on children with developmental conditions and community goats, used to evaluate early interventions, therapeutic outcomes, and the impact of livelihood support initiatives.

METHODOLOGY

Entering and organizing children's assessment and goat project data from various sources to ensure it was structured and accessible. Cleaning and preparing the data by correcting inconsistencies, imputing missing values, and removing duplicates to maintain accuracy. Conducting exploratory data analysis using statistical techniques such as descriptive statistics, correlation analysis, and regression analysis to identify trends and patterns. Creating visualizations including bar charts, histograms, and scatter plots to clearly communicate findings. Drafting comprehensive reports summarizing insights, implications, and recommendations for management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed that cerebral palsy and Duchenne Muscular Dystrophy were most commonly diagnosed between ages 2–3, autism prevalence decreased with age, and Down syndrome was consistently distributed across all age groups. These findings highlighted critical areas needing targeted interventions to improve child health outcomes in Meru and Marsabit. The reports and visualizations provided actionable evidence for policy development and strategic planning at the Furaha Centre. Additionally, the internship strengthened technical skills in data analysis, enhanced communication and reporting abilities, and deepened understanding of public health challenges in child development.

CURRENT STATUS

Data Analyst · Working / employed

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predictive Modeling for Gestational Diabetes Using Machine Learning

BACKGROUND & METHODOLOGY

Gestational diabetes is a public health concern that occurs during pregnancy when the body cannot produce sufficient insulin to meet increased demands, leading to high blood sugar levels. Although it typically resolves after childbirth, it significantly increases the risk of developing Type 2 diabetes later in life. The study aims to identify risk factors for gestational diabetes among pregnant women and develop a predictive machine learning model for early detection. Using secondary data from Kaggle, logistic regression was employed due to its effectiveness in handling binary outcomes. Key variables included pregnancies, glucose levels, blood pressure, insulin, BMI, and age. The developed model achieved 77.9% accuracy, with precision of 0.81, recall of 0.86, and an F1-score of 0.83. This predictive model facilitates early identification of at-risk women, enabling timely monitoring and preventive interventions to improve maternal and fetal health outcomes and reduce healthcare costs associated with gestational diabetes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a logistic regression machine learning model to predict gestational diabetes among pregnant women using variables such as pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and outcome. The study used secondary data from Kaggle, applied model training and evaluation, and assessed performance using precision, recall, F1-score, and confusion matrix. The model achieved an accuracy of 77.9%, with precision of 0.81, recall of 0.86, and F1-score of 0.83 for identifying non-diabetic cases, showing good predictive performance. The outcome demonstrates that machine learning can support early detection of gestational diabetes, helping healthcare providers identify at-risk women, improve monitoring, guide preventive interventions, and promote better maternal and fetal health outcomes.

INTERNSHIP Internship Experience

CIFORD Kenya — Water access, sanitation, and community development.

Dataset Used: Meru Baseline Survey, Water Tanks Project, and Tharaka Administrators Survey. Data on water availability, sanitation practices, community financial management, and local infrastructure to assess public health and resource management challenges.

METHODOLOGY

Cleaning and organizing datasets from the Meru Baseline Survey, Water Tanks Project, and Tharaka Administrators Survey. Performing descriptive analysis using Python in Google Colab to identify trends in water access, sanitation challenges, and community practices. Developing visualizations with Matplotlib and Seaborn to communicate findings clearly. Collaborating with supervisors and peers to ensure analysis aligned with organizational goals, and supporting capacity-building insights for community training in record-keeping and financial management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Revealed that water scarcity and inadequate sanitation significantly impact community health. Highlighted the need for improved water management and maintenance of water tanks. Identified gaps in financial management and record-keeping skills among community members. The findings provided actionable insights for CIFORD to plan targeted interventions, improve public health outcomes, and prioritize resources for water and sanitation projects in Meru and Tharaka communities.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

A Machine Learning Approach to Predicting Fetal Health Outcomes

BACKGROUND & METHODOLOGY

Model performance was evaluated with a confusion matrix, classification report, and accuracy, and achieved 100% accuracy with perfect precision, recall, and F1-scores on the test set. These results suggested excellent separability in the dataset, while also indicating possible overfitting or data leakage. The study demonstrated that machine learning could support PCOS risk stratification and screening, and it recommended future work to validate generalizability with broader, balanced cohorts, richer clinical and behavioral features, and production-grade model packaging.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a supervised machine learning model using logistic regression to classify fetal health and support early detection of fetal well-being concerns. The study focused on fetal health problems such as preterm birth, fetal distress, fetal anemia, birth defects, macrosomia, and umbilical cord complications, with emphasis on improving traditional cardiotocography-based assessment, which can be time-consuming and unreliable. The team loaded and explored the dataset, visualized patterns, selected relevant features using correlation analysis and a heatmap, split the data for training and testing, fitted the logistic regression model, and evaluated its performance. The model reportedly achieved 100% performance, showing that machine learning can support faster and more reliable classification of fetal health status and assist healthcare providers in identifying at-risk pregnancies for timely intervention.

INTERNSHIP Internship Experience

LifeCare Hospital Meru — Hospital inpatient outcomes and insurance claim optimization.

Dataset Used: Inpatient, outpatient, and deceased patient files at LifeCare Hospitals. Patient records capturing demographics, diagnoses, duration of stay, treatment history, and insurance claim details, used to improve hospital operations and reduce claim rejections.

METHODOLOGY

Collecting and organizing inpatient and outpatient patient files while ensuring data accuracy. Cleaning and preparing data for analysis using Excel and Python, including identifying patterns in repeated treatments and prolonged hospital stays. Developing predictive models to estimate patient length of stay and automating identification of multiple treatments within a month. Creating visualizations with Streamlit to highlight trends and support decision-making in hospital resource allocation and claims management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified common diagnoses, such as hypertension and diabetes, that contribute to longer hospital stays, allowing for improved resource planning. Developed methods to flag repeated treatments, helping the hospital understand and reduce chronic disease claim rejections. Enhanced data quality and visualization techniques supported actionable insights, contributing to more efficient patient care and better-informed public health strategies.

CURRENT STATUS

Data Analyst · Working / employed



CAPSTONE Capstone Project

Predicting Treatment Outcomes for Tuberculosis (TB) Patients Using Machine Learning

BACKGROUND & METHODOLOGY

This project focused on predicting treatment outcomes for tuberculosis (TB) patients, categorized as successful or unsuccessful. Using machine learning techniques, the dataset was preprocessed and analyzed to identify key features influencing treatment response. A supervised learning model was developed and evaluated, with results demonstrating the potential of predictive analytics to enhance patient management. The findings provide insights for healthcare providers to optimize interventions and improve TB treatment success rates.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

This project explores the application of predictive analytics in healthcare using a structured dataset containing patient demographics, admission details, and medical records. The primary objective was to perform exploratory data analysis and evaluate predictive models, with a focus on identifying patterns associated with patient medical conditions, including cancer. Data preprocessing revealed no missing or duplicate values, and conditions were categorized into cancerous and non-cancerous outcomes. Statistical tests such as chi-square, odds ratios, and risk ratios were conducted to assess associations between gender, blood type, test results, and medical conditions, all of which showed no statistically significant relationships. Logistic regression was applied as the supervised machine learning framework, supported by tools such as Python, Pandas, Scikit-learn, Matplotlib, and Seaborn. Although model deployment was not completed due to dataset complexity, the findings highlight the potential of predictive analytics for risk assessment and targeted healthcare interventions. This work demonstrates how data-driven insights can guide more effective patient care strategies while underscoring the need for further refinement of predictive modeling approaches.

INTERNSHIP Internship Experience

Mbungoni Health Center — Mother and Child Health (MCH) factors and their impact on HIV treatment and viral load suppression.

Dataset Used: Data were accessed from K-EMR, KHIS, DWAPI, DATIM, 3PM entries, NishaUri portals, and manual collection during medical camps. Analysis of patient demographics, HIV care enrollment, treatment start dates, and viral load trends to identify key predictors of successful viral suppression in breastfeeding and pregnant mothers

METHODOLOGY

Generating and compiling reports from multiple reporting tools, data entry and validation, troubleshooting servers, recreating and refreshing data tables in K-EMR, enrolling clients in various programs (Ushauri, Nishauri, Nimeconfirm), data collection during medical camps, cross-checking facility reporting tools, and weekly reporting on patient appointments and viral load uptake. The intern collaborated with staff, ensured data accuracy, and analyzed trends to support healthcare decision-making.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The internship specifically identified key predictors of successful viral load suppression in breastfeeding and pregnant mothers, demonstrating that earlier treatment start dates and timely enrollment in HIV care significantly improve patient outcomes. Weekly viral load reports and validated datasets enabled healthcare providers to track progress, identify gaps in care, and make data-driven decisions to optimize HIV treatment strategies. Additionally, the work improved the accuracy and reliability of the facility's data, facilitated timely reporting to national health systems, and strengthened community health interventions through insights gained from medical camp data collection. These outcomes directly contributed to better patient care, informed policy decisions, and enhanced the hospital's overall public health monitoring capabilities.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Predicting Patient Re-admission using Machine Learning.

BACKGROUND & METHODOLOGY

This project explores the application of predictive analytics in healthcare using a structured dataset containing patient demographics, admission details, and medical records. The primary objective was to perform exploratory data analysis and evaluate predictive models, with a focus on identifying patterns associated with patient medical conditions, including cancer. Data preprocessing revealed no missing or duplicate values, and conditions were categorized into cancerous and non-cancerous outcomes. Statistical tests such as chi-square, odds ratios, and risk ratios were conducted to assess associations between gender, blood type, test results, and medical conditions, all of which showed no statistically significant relationships. Logistic regression was applied as the supervised machine learning framework, supported by tools such as Python, Pandas, Scikit-learn, Matplotlib, and Seaborn. Although model deployment was not completed due to dataset complexity, the findings highlight the potential of predictive analytics for risk assessment and targeted healthcare interventions. This work demonstrates how data-driven insights can guide more effective patient care strategies while underscoring the need for further refinement of predictive modeling approaches.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project analyzed a healthcare dataset to explore whether patient characteristics such as gender, blood type, test results, and medical condition could support prediction of health outcomes, with the medical condition grouped into cancer and not cancer. The dataset contained 9,100 records and 15 variables, including patient ID, age, gender, blood type, medical condition, admission type, medication, and test results. The team performed data cleaning, checked for missing values and duplicates, generated descriptive statistics, visualized distributions, and conducted chi-square tests, odds ratios, and risk ratios to examine relationships between cancer status and variables such as gender and test results. The findings showed no statistically significant relationship between gender and cancer status, and the odds and risk ratios were close to 1, indicating that males and females had almost the same likelihood of developing cancer in the dataset. The outcome demonstrates the value of exploratory data analysis in healthcare prediction projects, although model deployment was not completed due to dataset complexity.

INTERNSHIP Internship Experience

Moving the Goal Post — Maternal and reproductive health among young mothers and teen moms.

Dataset Used: Health datasets and annual registration records for girls (2022–2024) from clinics and dispensaries. Demographic information, program enrollment, contraceptive use, and health service utilization, providing insights into the reproductive health, education, and community engagement of young mothers and teen moms.

METHODOLOGY

Data collection was conducted manually and through health facility records, focusing on program enrollment, contraceptive use, and service utilization. The datasets were cleaned, modified, and organized for analysis using Excel and Google Colab. Exploratory data analysis and visualization were performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, creating clear insights into trends and patterns. Beyond data work, the internship included community engagement activities, such as group discussions with teen mothers and boys, health education sessions, first aid training, and reporting on football matches, all aimed at promoting reproductive health, gender equality, and general wellbeing.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis identified key trends in enrollment and contraceptive use, highlighting gaps in service delivery and areas for targeted intervention. Visualizations, such as pie charts, enabled clear communication of findings to stakeholders. Community engagement initiatives enhanced awareness of reproductive health, immunization practices, and first aid, empowering young mothers to make informed health decisions. These efforts contributed to improved health literacy, promoted gender-sensitive practices, and provided actionable insights that informed program planning and resource allocation within the community.

CURRENT STATUS

Not currently working



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INTERNSHIP Internship Experience

MEPHI Health Services (Kilifi) — Patient diagnosis patterns and predictive modeling to improve healthcare decision-making.

Dataset Used: Patient diagnosis data collected from multiple hospital departments including Health Records Office, Claims Office, Minor Theatre, Reception, Triage, and Secretary Office. Analysis of patient demographics, diagnoses, and correlations with age and gender to inform predictive models and support data-driven healthcare strategies.

METHODOLOGY

Collecting and organizing patient records across hospital departments, cleaning and preparing data to ensure accuracy and consistency, performing statistical analyses to uncover trends and correlations, visualizing insights through charts and graphs, and developing predictive models using Python, Pandas, Scikit-learn, Seaborn, and Matplotlib. These activities focused on understanding how age and gender influence susceptibility to specific health conditions and guiding preventive measures.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The success of the project led to identification and prediction of key patterns within the set of data, which could help taking an action to reduce the rate of infection of various diseases to the individuals. These valuable insights help in making effective decisions and policies to saving individuals life from attack by diseases. The data visualization offered a quick learn at a glance which offered effective communication between the health practitioners and health decision makers within the facility.

CURRENT STATUS

Claims Officer · Working / employed

Research & Analysis

Projects centred on statistical analysis, modelling, and reporting.

48

beneficiaries

DeKUT (10) · SEKU (8) · JOOUST (16) · Kabarak (4) · Pwani (10)



CAPSTONE **Capstone Project**
Unsupervised Analysis of Cancer Diagnosis Data

BACKGROUND & METHODOLOGY

This project explores the use of statistical methods and unsupervised machine learning, specifically K-means clustering, to analyze cancer diagnosis data derived from medical imaging measurements. By pre-processing the data, performing exploratory analysis, and applying clustering, the study seeks to identify inherent patterns in cancer diagnostic features such as radius, texture, and area, to enhance diagnostic accuracy and support targeted interventions. Results indicate distinct groupings correlating with malignant and benign diagnoses, validated by cluster quality metrics and visualizations.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully applied K-Means clustering to cancer imaging data with optimal k=3 clusters, achieving a Silhouette Score of 0.62 indicating moderate to strong cluster separation. Key imaging features including radius, area, and texture were identified as primary differentiators between benign and malignant cases. Cluster analysis revealed three distinct patient profiles: Cluster 1 with low radius and area (predominantly benign), Cluster 2 with medium values (mixed diagnosis), and Cluster 3 with larger measurements (mostly malignant). PCA visualization confirmed well-separated clusters in reduced-dimensional space. The model correctly classified 515 out of 569 cases with only 54 misclassifications, demonstrating strong potential for supporting clinical diagnostic workflows and targeted medical interventions.

INTERNSHIP **Internship Experience**

Mt. Kenya Sub County Hospital — redicting Ambulance Service Demand through Geospatial Hotspot Analysis | AI for Emergency Response Optimization | Machine Learning for EMS Resource Allocation

Dataset Used: Historical ambulance incident records from Nyeri County EMS (1,073 validated records containing geospatial coordinates, emergency type classifications, dispatch timestamps, resolution times, sub-county locations, and priority levels). roject applying geospatial clustering techniques (K-Means and DBSCAN) to historical ambulance dispatch data to identify high-demand emergency hotspots and optimize ambulance deployment in Nyeri County. Dataset includes variables such as latitude/longitude coordinates, emergency type (Obstetrics/Childbirth, Trauma, Medical, etc.), call timestamps, dispatch times, resolution durations, sub-county location, and priority levels, used to uncover spatial-temporal demand patterns and support evidence-based resource allocation for emergency medical services.

METHODOLOGY

Managed and validated the integrity of the EMS electronic register at the internship centre; conducted data cleaning and preprocessing using Python (pandas) including handling missing values and parsing timestamps; performed descriptive analysis of emergency types and frequency distributions; engineered time-based features (Duration_min, Response_time_sec) for temporal analysis; applied K-Means and DBSCAN geospatial clustering algorithms to identify high-demand hotspots; analyzed temporal demand patterns by hour, day, and week using heatmaps and trend plots; visualized clusters and trends using matplotlib and seaborn; developed predictive forecasting models for future demand in low-density areas; generated actionable recommendations for strategic ambulance placement and resource specialization

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified Nyeri Town as the primary emergency hotspot containing over 725 incidents, with Obstetrics/Childbirth emerging as the most frequent emergency type at 39.7% of all cases; revealed that while dispatch times were highly efficient (median 0.0 minutes), total resolution time averaged 60.2 minutes, pinpointing travel and transport as the primary operational bottleneck; identified secondary demand clusters in Othaya and Mathira East with consistent moderate activity; forecasted stable baseline demand in Kieni sub-county (~44 calls/month) justifying permanent service presence while predicting near-zero demand for Laikipia East; produced data-driven recommendations for strategic ambulance station placement within identified clusters, obstetric resource specialization for core ambulances, dynamic shift scheduling based on temporal demand patterns, and improved timestamp recording for accurate response analysis; enhanced practical skills in geospatial analytics, clustering algorithms, time series forecasting, and public health operations research

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Unsupervised Analysis of Cancer Diagnosis Data

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CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Interpretable Pattern Discovery in Mental Health Symptoms Using Statistical Tests and Association Rule Mining

BACKGROUND & METHODOLOGY

The capstone project explored symptom co-occurrence and diagnosis-related patterns in the Dataset_Mental_Disorders using descriptive statistics, chi-square tests of independence, and unsupervised Association Rule Mining (Apriori/FP-Growth). After cleaning and encoding ordinal/categorical responses (e.g., Seldom ? Most-Often scales), distributions and symptom co-occurrence were visualized via bar charts, correlation/association matrices, and heatmaps. The data were transformed into a transactional format and frequent itemsets and rules were mined, evaluated with support, confidence, and lift. The analysis surfaced interpretable clusters of symptoms (such as sadness, exhaustion, mood swings) and behaviors (such as respect for authority, aggression response) that may inform screening and triage. Key findings included frequent itemsets above min_support and high-lift rules, some of which aligned with clinical expectations. Implications, limitations (dataset scope, self-report bias, ordinal encoding choices), and recommendations for integrating rule insights into early screening workflows were discussed. The work was purely analytical and did not provide medical advice.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified significant symptom co-occurrence patterns in mental health data using chi-square tests and Apriori association rule mining. Key findings revealed mood swings as a central linking factor, showing significant associations with anorexia ($p=0.0336$), aggressive response ($p=0.0110$), and nervous breakdown ($p=0.0269$). Association rules with confidence >0.60 and lift >1.20 uncovered meaningful clusters, including rules linking nervous breakdown, mood swing, and aggressive response to anorexia and concentration problems (confidence 0.67, lift 1.74). Core symptoms including sadness, euphoric mood, exhaustion, and sleep disorder showed universal presence across all records. High-lift rules above 7.0 demonstrated strong interdependencies between behavioral and cognitive symptoms, offering valuable insights for early detection strategies, targeted interventions, and integration into diagnostic screening tools.

INTERNSHIP Internship Experience

Othaya Sub County Hospital — Statistical Analysis and Unsupervised Association Rule Mining for Morbidity Pattern Detection | AI for Comorbidity Identification | Data-Driven Health Planning

Dataset Used: Othaya Sub County Hospital morbidity dataset (January 2022–December 2024, 2,817 validated records containing age, sex, ICD codes/diagnosis text, sub-county, date of death, and mode of discharge). Project applying statistical analysis and Association Rule Mining (Apriori algorithm) to facility morbidity data to identify frequent diagnoses, comorbidity bundles, and demographic-outcome patterns. Dataset includes variables such as patient age, sex, diagnostic categories (Hypertension, Diabetes, HIV, TB, Malaria, Pneumonia, Asthma, Pregnancy), and discharge outcomes, used to uncover hidden relationships between demographic factors and clinical conditions for targeted public health interventions in a Level 4 rural-periurban facility.

METHODOLOGY

Data Cleaning and Preparation: Standardized age to years (converted days/months), imputed missing values where possible, dropped records with $>50\%$ missing essential fields, mapped ICD codes and free-text diagnoses into standardized diagnostic groups, created binary flags for diagnoses, one-hot encoded demographics for ARM | Statistical Analysis: Computed descriptive statistics (mean/median/range for age, frequencies for sex/diagnoses/outcomes), performed correlation analysis (age category — primary diagnosis), conducted Chi-square tests for categorical associations (sex — diagnosis, diagnosis — mode of discharge) at $p<0.05$ significance | Association Rule Mining: Applied Apriori algorithm via mlxtend with thresholds (support ≥ 0.02 , confidence ≥ 0.6 , lift >1), pruned rules for clinical relevance, generated visualizations (scatter plots, heatmaps, network graphs) | Clustering: Performed K-means clustering to segment patients into distinct clinical profiles based on demographic and diagnostic characteristics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified significant association between patient sex and top 10 diagnoses ($\chi^2=184.87$, $p<0.0001$), confirming gendered patterns in disease presentation (e.g., pregnancy-related conditions concentrated in females, hypertension more frequent in males) | Established significant association between primary diagnosis and mode of discharge ($\chi^2=35.44$, $p<0.0001$), indicating outcome patterns aligned with disease severity | Discovered strong age-diagnosis links: hypertension overwhelmingly concentrated in older patients (lift nearly 3–), two-thirds of diabetes cases in patients aged 51+ confirming age-related NCD risk | Generated clinically meaningful association rules: Age 19-35 strongly associated with female sex (maternal health burden), pediatric pneumonia frequent but rarely fatal (positive outcome insight), elderly males with diabetes showing favorable survival | K-means clustering identified four distinct patient profiles: Cluster 0 (young females with acute infections, high survival), Cluster 1 (reproductive-age females with pregnancy complications), Cluster 2 (older patients with chronic conditions, slightly higher mortality), Cluster 3 (young males with acute infections) | Provided actionable recommendations: establish integrated NCD clinics for Hypertension+Diabetes, strengthen under-five illness triage for malaria/pneumonia during seasonal peaks, improve maternal health pathways for ages 15-49, integrate dashboards into hospital Quality Improvement meetings | Demonstrated practical application of unsupervised learning and statistical testing in routine health data for evidence-based planning in resource-limited settings

CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Interpretable Pattern Discovery in Mental Health Symptoms Using Statistical Tests and Association Rule Mining***BACKGROUND & METHODOLOGY**

The capstone project explored symptom co-occurrence and diagnosis-related patterns in the Dataset_Mental_Disorders using descriptive statistics, chi-square tests of independence, and unsupervised Association Rule Mining (Apriori/FP-Growth). After cleaning and encoding ordinal/categorical responses (e.g., Seldom ? Most-Often scales), distributions and symptom co-occurrence were visualized via bar charts, correlation/association matrices, and heatmaps. The data were transformed into a transactional format and frequent itemsets and rules were mined, evaluated with support, confidence, and lift. The analysis surfaced interpretable clusters of symptoms (such as sadness, exhaustion, mood swings) and behaviors (such as respect for authority, aggression response) that may inform screening and triage. Key findings included frequent itemsets above min_support and high-lift rules, some of which aligned with clinical expectations. Implications, limitations (dataset scope, self-report bias, ordinal encoding choices), and recommendations for integrating rule insights into early screening workflows were discussed. The work was purely analytical and did not provide medical advice.

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Successfully identified significant symptom co-occurrence patterns in mental health data using chi-square tests and Apriori association rule mining. Key findings revealed mood swings as a central linking factor, showing significant associations with anorexia ($p=0.0336$), aggressive response ($p=0.0110$), and nervous breakdown ($p=0.0269$). Association rules with confidence >0.60 and lift >1.20 uncovered meaningful clusters, including rules linking nervous breakdown, mood swing, and aggressive response to anorexia and concentration problems (confidence 0.67, lift 1.74). Core symptoms including sadness, euphoric mood, exhaustion, and sleep disorder showed universal presence across all records. High-lift rules above 7.0 demonstrated strong interdependencies between behavioral and cognitive symptoms, offering valuable insights for early detection strategies, targeted interventions, and integration into diagnostic screening tools.

INTERNSHIP **Internship Experience***Karatina Sub-County Hospital — Analysis of Demographics, Comorbidities, and HIV Test Outcomes for Patient Segmentation and Association Rule Mining | AI for HIV Risk Stratification | Data-Driven Patient Profiling*

Dataset Used: KSCH ENGAGE DATASET cleaned.xls containing HIV testing information, demographics, and comorbidities from Karatina Sub-County Hospital. Dataset containing HIV testing records with demographic variables (age, sex), marital status, population type, testing history, comorbidity information (TB screening), testing strategy, entry point, and HIV test results used to apply association rule mining and clustering techniques for identifying patient segments and risk factors associated with HIV positivity

METHODOLOGY

Data entry and validation of HIV testing records in KenyaEMR system; cleaning and preprocessing of dataset by handling missing values through column dropping and mode imputation; feature engineering to create Reproductive Age, Elderly, Chronic Condition, and Age Group variables; descriptive statistics and correlation analysis; Chi-square tests for association between categorical variables; K-Means clustering for patient segmentation using elbow method; Association rule mining using Apriori/FP-Growth algorithm with support, confidence, and lift metrics; visualization using histograms, bar plots, heatmaps, and network graphs; Python tools including pandas, scikit-learn, mlxtend, matplotlib, seaborn in Google Colab environment

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Achieved 1.10% overall HIV positivity rate with statistically significant association between Sex and Final Result ($p < 0.05$); generated 102 association rules revealing strong links between age groups 50-59, female gender, insurance status and hypertension; K-Means clustering ($k=3$) identified distinct patient segments with Cluster 1 showing highest hypertension prevalence (~90%); demonstrated feasibility of applying association rule mining and unsupervised learning to HIV testing data for risk stratification; provided actionable insights for targeted interventions and resource allocation; enhanced practical skills in health data management, feature engineering, clustering algorithms, and association rule mining for public health applications

CURRENT STATUS**Recent Graduate · Unemployed — Actively searching**

Sector relevance: Partially related

SKILLS IN USE

Data collection · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predictive Analysis of Diabetes Risk Factors Using a Multivariate Logistic Regression Model

BACKGROUND & METHODOLOGY

This project addresses the critical need for a proactive diagnostic tool for diabetes, a chronic disease with a rising global prevalence. Utilizing the Diabetes Data dataset, we developed a multivariate logistic regression model to identify key risk factors and predict diabetes with high accuracy. The methodology involved comprehensive data preprocessing, including one-Hot Encoding of categorical variables, followed by a detailed statistical analysis. Our model, trained on key predictors like Polyuria and Polydipsia, demonstrating its effectiveness. The results suggest that these symptoms are the most significant indicators of diabetes in the dataset. This predictive model offers a valuable, data-driven tool that can assist healthcare professionals' in early, targeted screening, thereby improving patient outcomes and alleviating the public health burden of diabetes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and evaluated a multivariate logistic regression model for diabetes prediction achieving 94.23% accuracy, 98.33% precision, 92.19% recall, and 95.16% F1-score. Key risk factors identified were Polyuria and Polydipsia as the most significant predictors of diabetes. The model demonstrates strong reliability for early screening with minimal false positives, providing a valuable data-driven tool to assist healthcare professionals in targeted proactive diagnosis and improved patient outcomes.

INTERNSHIP Internship Experience

Karatina Sub County Hospital Health Information Department — Unsupervised Clustering and Risk Profiling of Inpatient Morbidity and Mortality Data | Machine Learning for Patient Stratification | AI for Healthcare Quality Improvement

Dataset Used: Inpatient morbidity and mortality dataset ("clean morbidity raw data(1).xls") from Karatina Hospital containing 15,707 unique inpatient episodes with variables including age, sex, diagnosis, ward, length of stay, and patient outcome. Capstone project applying data science techniques to routinely collected inpatient records to systematically identify distinct clinical profiles and high-risk patient groups. The analysis used descriptive statistics and Hierarchical Agglomerative Clustering (HAC) to segment patients based on age, length of stay, ward, and outcome, with the goal of providing data-driven evidence to prioritize resource allocation and develop targeted care pathways for vulnerable populations, particularly elderly patients.

METHODOLOGY

Data Preprocessing: Removed 223 duplicate inpatient records, conducted missing data analysis identifying 97-99% missingness in death details and surgery fields, handled missing values through imputation or exclusion | Feature Engineering: Calculated Length of Stay (LOS) from admission/discharge dates, created age group categories (Infants, Children, Youth, Adults, Elderly), aggregated granular Primary_Diagnosis into grouped categories to reduce dimensionality | Descriptive Analysis: Performed frequency analysis and cross-tabulations to quantify relationships between categorical variables (age vs. mortality, diagnosis frequency) | Clustering Analysis: Applied Agglomerative Hierarchical Clustering (HAC) with Z-score normalization of numerical features, Euclidean distance metric, and Ward's linkage method; determined optimal cluster count using dendrogram visualization and clinical interpretability | Visualization: Generated cluster profiles, dendrograms, and summary tables to interpret and present findings using Python

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified five distinct clinical profiles (clusters) based on age, LOS, ward, and outcome | Cluster 5 (Elderly, avg. age 68.21 years) exhibited highest inpatient mortality rate at 29.25%, identifying this as the critical priority group requiring focused geriatric care pathways | Clusters 2, 3, and 4 predominantly contained morbidity cases with 0% mortality, representing patients requiring efficient discharge planning | Cluster 1 (Infants/Children, avg. age 1.81 years) showed mixed outcomes highlighting pediatric vulnerability | Cluster 3 (Adults with exceptionally long LOS, avg. 31.1 days) identified patients requiring complex subacute care management | Descriptive analysis revealed elderly patients (60+ years) had mortality rate of 24.01%, approximately four times higher than youth/adult groups | Identified data quality issues including high-frequency use of "Cardiopulmonary arrest" as potential default diagnosis code requiring internal validation | Demonstrated practical application of unsupervised machine learning in public health surveillance for evidence-based resource allocation and targeted intervention planning

CURRENT STATUS

I don't have a job at the moment · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Using Statistical Analysis and K-means Clustering for Early Diagnosis of Breast Cancer from Medical Imaging Data

BACKGROUND & METHODOLOGY

This project investigates the use of unsupervised learning and statistical techniques to support early breast cancer detection. A medical imaging dataset was analyzed through exploratory data analysis to examine the distribution of key features and their relationship to tumor diagnosis (benign or malignant). The findings showed that several morphological and size-related characteristics strongly correlate with diagnosis outcomes. K-Means clustering was then applied, effectively separating most benign and malignant cases into two clusters, reflecting the natural structure of the data in alignment with clinical results. However, some overlap between groups was noted, indicating that unsupervised methods alone may be limited. Overall, the study demonstrates that statistical analysis and K-Means clustering can provide valuable insights for data exploration and preliminary grouping, laying a foundation for integrating supervised learning approaches to achieve more accurate and reliable early diagnosis of breast cancer.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully applied K-Means clustering to breast cancer imaging data, effectively separating benign and malignant cases into two distinct clusters with high alignment to clinical diagnoses. Key morphological features including radius_mean, perimeter_mean, area_mean, and concave points_mean were identified as strongly correlated with malignancy. The clustering achieved a Silhouette Score of 0.343, demonstrating meaningful separation despite some overlap between clusters. PCA visualization confirmed the two-cluster structure in reduced-dimensional space. Results provide a foundation for integrating supervised learning approaches to achieve more accurate and reliable early breast cancer diagnosis.

INTERNSHIP Internship Experience

Mukurweini Sub County Hospital — lustering Morbidity Patterns to Identify High-Risk Patient Segments and Operational Gaps | Machine Learning for Patient Stratification in Hospital Settings | AI for Healthcare Quality Improvement

Dataset Used: Retrospective inpatient morbidity dataset from Mukurweini District Hospital, Nyeri County, Kenya (2019–2022, 13,145 validated admission records containing demographic, diagnostic, ward, referral, length of stay, and outcome variables). Retrospective study applying hierarchical clustering analysis to inpatient morbidity data to identify high-risk patient segments and operational gaps. Dataset includes variables such as age, sex, primary diagnosis (ICD-coded), mode of discharge, admission/discharge dates, length of stay, ward name, and patient residence, used to uncover hidden morbidity patterns, assess mortality risks, evaluate seasonal trends, and inform targeted interventions for improving patient outcomes and resource allocation in a Level 4 facility.

METHODOLOGY

Internship Centre Work: Prepared summaries for cancer case reporting and outcomes tracking; Conducted data quality checks to ensure completeness and accuracy of registry entries | Capstone Project: Loaded and cleaned inpatient dataset from hospital records; Handled missing values via median/mode imputation and Winsorizing for outliers; Engineered features (Length of Stay, Age groups, Diagnosis categories, Seasonality); Performed descriptive statistics and frequency analysis; Applied Chi-square tests and Kruskal-Wallis hypothesis testing; Conducted Hierarchical Agglomerative Clustering (Ward linkage) on standardized variables; Generated dendrograms, cluster profiles, heatmaps, and PCA visualizations using Python (Pandas, Scikit-learn, Matplotlib, Seaborn); Performed operational segmentation and risk profiling by ward, diagnosis category, and season

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established that 75% of patients were diagnosed at late stages (III–IV), confirming late presentation as a major barrier to survival | Chi-square testing revealed stage at diagnosis as the only significant predictor of survival outcomes ($\chi^2 = 30.56$, $p < 0.001$), while age and marital status showed no significant associations | Hierarchical clustering identified three distinct patient subgroups: Cluster 1 (maternal/perinatal admissions with low mortality but longer stays), Cluster 2 (high-risk group dominated by non-communicable diseases with 12.3% mortality and mean LOS 11.5 days), Cluster 3 (small seasonally concentrated group with minimal operational impact) | Identified significant variation in Length of Stay across hospital wards and a weak negative correlation between LOS and mortality | Revealed under-five vulnerability to infectious diseases and sex-based differences in injury incidence | Highlighted data quality gaps including implausible age/LOS values and missing diagnoses requiring improved recording protocols | Provided actionable recommendations for targeted interventions including KMC strengthening for neonatal cases, 48-72h post-op checklists for surgical patients, trauma triage improvements, and comprehensive discharge planning for elderly patients | Demonstrated feasibility of applying unsupervised machine learning for patient stratification in resource-limited hospital settings | Enhanced practical skills in oncology data management, statistical testing, clustering algorithms, and result interpretation for public health applications

CURRENT STATUS

Academic writing · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Generative AI / LLMs

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Using Statistical Analysis and K-means Clustering for Early Diagnosis of Breast Cancer from Medical Imaging Data

BACKGROUND & METHODOLOGY

This project investigates the use of unsupervised learning and statistical techniques to support early breast cancer detection. A medical imaging dataset was analyzed through exploratory data analysis to examine the distribution of key features and their relationship to tumor diagnosis (benign or malignant). The findings showed that several morphological and size-related characteristics strongly correlate with diagnosis outcomes. K-Means clustering was then applied, effectively separating most benign and malignant cases into two clusters, reflecting the natural structure of the data in alignment with clinical results. However, some overlap between groups was noted, indicating that unsupervised methods alone may be limited. Overall, the study demonstrates that statistical analysis and K-Means clustering can provide valuable insights for data exploration and preliminary grouping, laying a foundation for integrating supervised learning approaches to achieve more accurate and reliable early diagnosis of breast cancer.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully applied K-Means clustering to breast cancer imaging data, effectively separating benign and malignant cases into two distinct clusters with high alignment to clinical diagnoses. Key morphological features including radius_mean, perimeter_mean, area_mean, and concave points_mean were identified as strongly correlated with malignancy. The clustering achieved a Silhouette Score of 0.343, demonstrating meaningful separation despite some overlap between clusters. PCA visualization confirmed the two-cluster structure in reduced-dimensional space. Results provide a foundation for integrating supervised learning approaches to achieve more accurate and reliable early breast cancer diagnosis.

INTERNSHIP Internship Experience

Mukurweini Sub County Hospital — lustering Morbidity Patterns to Identify High-Risk Patient Segments and Operational Gaps | Machine Learning for Patient Stratification in Hospital Settings | AI for Healthcare Quality Improvement

Dataset Used: Retrospective inpatient morbidity dataset from Mukurweini District Hospital, Nyeri County, Kenya (2019–2022, 13,145 validated admission records containing demographic, diagnostic, ward, referral, length of stay, and outcome variables). Retrospective study applying hierarchical clustering analysis to inpatient morbidity data to identify high-risk patient segments and operational gaps. Dataset includes variables such as age, sex, primary diagnosis (ICD-coded), mode of discharge, admission/discharge dates, length of stay, ward name, and patient residence, used to uncover hidden morbidity patterns, assess mortality risks, evaluate seasonal trends, and inform targeted interventions for improving patient outcomes and resource allocation in a Level 4 facility.

METHODOLOGY

Internship Centre Work: Prepared summaries for cancer case reporting and outcomes tracking; Conducted data quality checks to ensure completeness and accuracy of registry entries | Capstone Project: Loaded and cleaned inpatient dataset from hospital records; Handled missing values via median/mode imputation and Winsorizing for outliers; Engineered features (Length of Stay, Age groups, Diagnosis categories, Seasonality); Performed descriptive statistics and frequency analysis; Applied Chi-square tests and Kruskal-Wallis hypothesis testing; Conducted Hierarchical Agglomerative Clustering (Ward linkage) on standardized variables; Generated dendrograms, cluster profiles, heatmaps, and PCA visualizations using Python (Pandas, Scikit-learn, Matplotlib, Seaborn); Performed operational segmentation and risk profiling by ward, diagnosis category, and season

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established that 75% of patients were diagnosed at late stages (III–IV), confirming late presentation as a major barrier to survival | Chi-square testing revealed stage at diagnosis as the only significant predictor of survival outcomes ($\chi^2 = 30.56$, $p < 0.001$), while age and marital status showed no significant associations | Hierarchical clustering identified three distinct patient subgroups: Cluster 1 (maternal/perinatal admissions with low mortality but longer stays), Cluster 2 (high-risk group dominated by non-communicable diseases with 12.3% mortality and mean LOS 11.5 days), Cluster 3 (small seasonally concentrated group with minimal operational impact) | Identified significant variation in Length of Stay across hospital wards and a weak negative correlation between LOS and mortality | Revealed under-five vulnerability to infectious diseases and sex-based differences in injury incidence | Highlighted data quality gaps including implausible age/LOS values and missing diagnoses requiring improved recording protocols | Provided actionable recommendations for targeted interventions including KMC strengthening for neonatal cases, 48-72h post-op checklists for surgical patients, trauma triage improvements, and comprehensive discharge planning for elderly patients | Demonstrated feasibility of applying unsupervised machine learning for patient stratification in resource-limited hospital settings | Enhanced practical skills in oncology data management, statistical testing, clustering algorithms, and result interpretation for public health applications

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Analysis of Diabetes Risk Factors Using a Multivariate Logistic Regression Model

BACKGROUND & METHODOLOGY

This project addresses the critical need for a proactive diagnostic tool for diabetes, a chronic disease with a rising global prevalence. Utilizing the Diabetes Data dataset, we developed a multivariate logistic regression model to identify key risk factors and predict diabetes with high accuracy. The methodology involved comprehensive data preprocessing, including one-Hot Encoding of categorical variables, followed by a detailed statistical analysis. Our model, trained on key predictors like Polyuria and Polydipsia, demonstrating its effectiveness. The results suggest that these symptoms are the most significant indicators of diabetes in the dataset. This predictive model offers a valuable, data-driven tool that can assist healthcare professionals' in early, targeted screening, thereby improving patient outcomes and alleviating the public health burden of diabetes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and evaluated a multivariate logistic regression model for diabetes prediction achieving 94.23% accuracy, 98.33% precision, 92.19% recall, and 95.16% F1-score. Key risk factors identified were Polyuria and Polydipsia as the most significant predictors of diabetes. The model demonstrates strong reliability for early screening with minimal false positives, providing a valuable data-driven tool to assist healthcare professionals in targeted proactive diagnosis and improved patient outcomes.

INTERNSHIP Internship Experience

Karatina Subcounty Hospital — Non-Communicable Disease (NCD) Patient Profiling and Segmentation | Machine Learning for Chronic Disease Management | AI for Healthcare Quality Improvement

Dataset Used: Facility permanent NCD register from Karatina Subcounty Hospital (976 validated patient records containing demographic, clinical, insurance coverage, and follow-up variables for diabetes, hypertension, and related comorbidities). Project applying data science techniques to analyze NCD patient data at Karatina Subcounty Hospital to identify enrollment, follow-up, insurance coverage and utilization patterns among patients with diabetes, hypertension and related comorbidities. Dataset includes variables such as age, diagnosis, insurance status (SHIF/NHIF), duration since diagnosis, and visit counts, used to apply descriptive statistics, correlation analysis, and hierarchical clustering for patient segmentation and targeted intervention planning in chronic disease management.

METHODOLOGY

Internship Centre Work: Data entry and updating NCD registers; Validating and cleaning health records by checking for missing values, duplicates, and inconsistencies; Preparing monthly reports for submission to county health department; Supporting NCD data review meetings to discuss trends and discrepancies | Capstone Project Work: Extracted 976 patient records from facility NCD register; Cleaned and standardized column names, parsed dates, derived age and duration since diagnosis variables, recoded insurance and diagnosis flags; Performed descriptive statistics and correlation analysis using Python; Applied hierarchical clustering (Agglomerative Clustering with k=5) to segment patients based on demographic and clinical characteristics; Generated visualizations (gender distribution, age histograms, scatter plots, boxplots, cluster profiles) using matplotlib; Analyzed associations between SHIF coverage, age, duration since diagnosis, and visit patterns

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated and cleaned approximately 976 patient records, improving data completeness and standardization for NCD reporting | Identified positive correlation between age and duration since diagnosis, suggesting older patients have longer disease management histories | Established that SHIF coverage was recorded at 0.0% in available dataset, highlighting critical gap in insurance documentation | Recognized that visit counts were largely recorded as zero, limiting utilization analysis and indicating need for improved data capture protocols | Successfully segmented patients into five distinct clusters using hierarchical clustering: Cluster 0 (older patients with longer diagnosis duration, moderate visits - established patients needing regular follow-up), Cluster 2 (younger, newly diagnosed, low visits - requiring early engagement), Cluster 4 (older with short diagnosis duration, very low visits - possible low engagement needing outreach) | Generated actionable visualizations and cluster profiles to support facility decision-making and targeted outreach strategies | Provided evidence-based recommendations for standardizing diagnosis entries, ensuring SHIF/NHIF field completion, implementing SMS reminders and CHV follow-ups for patient retention, and conducting targeted SHIF enrollment drives | Demonstrated practical application of Python, pandas, sklearn, scipy, and matplotlib for health data analysis and chronic disease management in resource-limited settings

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · Ethics & bias in AI · Public health data analysis

Confidence applying ENGAGE skills: **3 / 5**



CAPSTONE Capstone Project

Statistical Insights and Association Rule Mining for Discovering Relationships in Pharmaceutical Product Transactions

BACKGROUND & METHODOLOGY

This project applies unsupervised machine learning techniques to analyze a transactional dataset of pharmaceutical products. The primary goal is to discover hidden patterns and associations between different drugs purchased together. The methodology involves a two-pronged approach: initial statistical analysis and data visualization to understand the distribution and co-occurrence of items, followed by the implementation of the Apriori algorithm for Association Rule Mining (ARM). The analysis successfully identifies strong association rules, quantified by metrics like support, confidence, and lift. These rules reveal significant product affinities, such as the strong association between Metronidazole and Ibuprofen. The insights derived are crucial for informing inventory management, designing targeted marketing strategies like bundle promotions, and enhancing customer service by understanding common treatment patterns.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified strong association rules between pharmaceutical products, with the strongest being between Cotrimoxazole, Metronidazole, and Ibuprofen (confidence 86.4%, lift 3.24). Key findings enable optimized inventory management, strategic product placement, and targeted bundle promotions to increase revenue and improve customer service.

INTERNSHIP Internship Experience

Othaya Sub County Hospital — Statistical Analysis and Unsupervised Association Rule Mining for Morbidity Pattern Detection | AI for Comorbidity Identification | Data-Driven Health Planning

Dataset Used: Othaya Sub County Hospital morbidity dataset (January 2022–December 2024, 2,817 validated records containing age, sex, ICD codes/diagnosis text, sub-county, date of death, and mode of discharge). Project applying statistical analysis and Association Rule Mining (Apriori algorithm) to facility morbidity data to identify frequent diagnoses, comorbidity bundles, and demographic-outcome patterns. Dataset includes variables such as patient age, sex, diagnostic categories (Hypertension, Diabetes, HIV, TB, Malaria, Pneumonia, Asthma, Pregnancy), and discharge outcomes, used to uncover hidden relationships between demographic factors and clinical conditions for targeted public health interventions in a Level 4 rural-periurban facility.

METHODOLOGY

Data Cleaning and Preparation: Standardized age to years (converted days/months), imputed missing values where possible, dropped records with >50% missing essential fields, mapped ICD codes and free-text diagnoses into standardized diagnostic groups, created binary flags for diagnoses, one-hot encoded demographics for ARM | Statistical Analysis: Computed descriptive statistics (mean/median/range for age, frequencies for sex/diagnoses/outcomes), performed correlation analysis (age category — primary diagnosis), conducted Chi-square tests for categorical associations (sex — diagnosis, diagnosis — mode of discharge) at $p < 0.05$ significance | Association Rule Mining: Applied Apriori algorithm via mlxtend with thresholds (support ≥ 0.02 , confidence ≥ 0.6 , lift > 1), pruned rules for clinical relevance, generated visualizations (scatter plots, heatmaps, network graphs) | Clustering: Performed K-means clustering to segment patients into distinct clinical profiles based on demographic and diagnostic characteristics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified significant association between patient sex and top 10 diagnoses ($\chi^2 = 184.87$, $p < 0.0001$), confirming gendered patterns in disease presentation (e.g., pregnancy-related conditions concentrated in females, hypertension more frequent in males) | Established significant association between primary diagnosis and mode of discharge ($\chi^2 = 35.44$, $p < 0.0001$), indicating outcome patterns aligned with disease severity | Discovered strong age-diagnosis links: hypertension overwhelmingly concentrated in older patients (lift nearly 3—), two-thirds of diabetes cases in patients aged 51+ confirming age-related NCD risk | Generated clinically meaningful association rules: Age 19-35 strongly associated with female sex (maternal health burden), pediatric pneumonia frequent but rarely fatal (positive outcome insight), elderly males with diabetes showing favorable survival | K-means clustering identified four distinct patient profiles: Cluster 0 (young females with acute infections, high survival), Cluster 1 (reproductive-age females with pregnancy complications), Cluster 2 (older patients with chronic conditions, slightly higher mortality), Cluster 3 (young males with acute infections) | Provided actionable recommendations: establish integrated NCD clinics for Hypertension+Diabetes, strengthen under-five illness triage for malaria/pneumonia during seasonal peaks, improve maternal health pathways for ages 15-49, integrate dashboards into hospital Quality Improvement meetings | Demonstrated practical application of unsupervised learning and statistical testing in routine health data for evidence-based planning in resource-limited settings

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Statistical Insights and Association Rule Mining for Discovering Relationships in Pharmaceutical Product Transactions

BACKGROUND & METHODOLOGY

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INTERNSHIP Internship Experience

Othaya Sub County Hospital — Statistical Analysis and Unsupervised Association Rule Mining for Morbidity Pattern Detection | AI for Comorbidity Identification | Data-Driven Health Planning

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METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Breast Cancer Mortality Rate in Kenya Using Naïve Bayes

BACKGROUND & METHODOLOGY

This capstone project focuses on predicting breast cancer mortality using a Naive Bayes classification model, addressing a critical issue in global health. Breast cancer remains one of the leading causes of cancer-related deaths among women, and the ability to accurately predict mortality risk can significantly improve patient outcomes by enabling timely and personalized interventions. The model analyzes key medical and lifestyle factors such as tumor size and stage, lymph node involvement, patient age, hormone receptor status, genetic markers, previous treatments, and lifestyle habits. Data was collected from clinical databases and public health sources, followed by data preprocessing, exploratory data analysis, model training, evaluation, and deployment. Ethical considerations, including data privacy and bias mitigation, were carefully addressed to ensure responsible use of health data. This project advances the application of machine learning in healthcare and provides a meaningful tool for clinicians and healthcare systems to reduce breast cancer mortality, enhance patient care, and alleviate the emotional and social burdens faced by patients and their families.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Naive Bayes classification model to predict breast cancer mortality among women in Kenya, using clinical and demographic variables including age, tumor characteristics, hormone receptor status, BMI, and lifestyle factors. The model achieved an accuracy of approximately 81%, with precision of 81%, recall of 78.6%, an F1-score of 79.9%, and a ROC-AUC score of 0.87, demonstrating strong discriminatory ability between mortality outcomes. The model was trained, validated, and deployed for real-world predictions, providing a fast, lightweight, and clinically applicable tool to support early risk detection, treatment prioritization, and resource allocation in breast cancer care

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Breast Cancer Mortality Rate in Kenya Using Naïve Bayes

BACKGROUND & METHODOLOGY

This capstone project focuses on predicting breast cancer mortality using a Naive Bayes classification model, addressing a critical issue in global health. Breast cancer remains one of the leading causes of cancer-related deaths among women, and the ability to accurately predict mortality risk can significantly improve patient outcomes by enabling timely and personalized interventions. The model analyzes key medical and lifestyle factors such as tumor size and stage, lymph node involvement, patient age, hormone receptor status, genetic markers, previous treatments, and lifestyle habits. Data was collected from clinical databases and public health sources, followed by data preprocessing, exploratory data analysis, model training, evaluation, and deployment. Ethical considerations, including data privacy and bias mitigation, were carefully addressed to ensure responsible use of health data. This project advances the application of machine learning in healthcare and provides a meaningful tool for clinicians and healthcare systems to reduce breast cancer mortality, enhance patient care, and alleviate the emotional and social burdens faced by patients and their families.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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INTERNSHIP Internship Experience

Jordan Hospital — Monitoring Child Growth Using Random Forest Classifier | Machine Learning for Malnutrition Risk Prediction | AI for Pediatric Health Analytics

Dataset Used: Child health records from Jordan Hospital, Kitui County (2,616 records collected between 2018–2025 from routine growth monitoring registers). Dataset containing child growth monitoring records with variables including age, gender, height, weight, height categories, and weight categories, used to classify children into nutritional status groups (underweight, normal, overweight, obese) using Random Forest Classifier to support early detection of malnutrition risks and improve clinical decision-making in maternal and child health services.

METHODOLOGY

Internship Centre Work: Entered and validated 2,616 child health records into the DHIS2 system; conducted data quality checks by comparing paper registers with electronic records | Capstone Project: Extracted child growth monitoring records from 2018–2025 registers; cleaned and recoded variables (Age, Gender, Height, Weight, Height categories, Weight categories); performed descriptive analysis using Python; trained Random Forest Classifier to predict nutritional status categories; generated visualizations (distribution charts, feature importance bar graphs); evaluated model performance using accuracy metrics.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest Classifier achieved 84% overall accuracy in predicting child weight categories; model performed well in distinguishing between normal weight and underweight children but showed moderate performance for overweight/obese categories due to class imbalance; identified height, weight, and their categorized versions as key predictors of nutritional status; revealed that nearly 80% of children in dataset were classified as normal weight; demonstrated feasibility of using routinely collected clinical measurements for predictive analytics in hospital settings; enhanced practical skills in data cleaning, machine learning modeling, and health data management; provided evidence-based recommendations for integrating predictive tools into routine child growth monitoring to support early intervention for at-risk children.

CURRENT STATUS

Learner and searching for internship or job. · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · Generative AI / LLMs · Freelancing in data science / building a portfolio · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Malaria Predictions Based on Climate Variables in Kenya with Machine Learning

BACKGROUND & METHODOLOGY

This study applied machine learning to predict malaria outcomes in Kenya using climate variables and mosquito density per trap. A Random Forest classifier was trained and evaluated on a dataset of 276,976 records. The model achieved an accuracy of 71% and a ROC-AUC of 0.732. Results indicate the model is more effective in predicting non-malaria cases than malaria-positive cases, with rainfall and humidity emerging as strong predictors. While the classifier demonstrated fair predictive ability, further model optimization is needed to reduce false negatives and improve detection of malaria-positive cases for effective early warning systems.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classifier to predict malaria outcomes in Kenya using climate variables including rainfall, humidity, and mosquito density per trap, trained and tested on a dataset of 276,976 records. The model achieved an overall accuracy of 71% and a ROC-AUC of 0.73, with strong performance in detecting non-malaria cases (precision: 0.75, recall: 0.84, F1-score: 0.79) but limited sensitivity for malaria-positive cases (recall: 0.48, F1-score: 0.54). Feature importance analysis identified rainfall and humidity as the most influential predictors, with recommendations to apply SMOTE for class balancing, incorporate additional climate variables such as altitude and temperature, and develop a user-friendly early warning dashboard for malaria surveillance and resource allocation by health authorities in Kenya.

INTERNSHIP Internship Experience

SEKU University Health Clinic — Predicting Patient Sex Using Age and Five Common Diagnoses | Gaussian Naïve Bayes for Clinical Classification | AI for Demographic Pattern Analysis

Dataset Used: Manual outpatient clinic records from South Eastern Kenya University Health Clinic (779 validated outpatient cases with age, sex, and five most common diagnosis variables). Study applying machine learning to predict patient sex using age and five common outpatient diagnoses at the SEKU University Clinic. The project aimed to assess whether simple demographic and diagnostic information could serve as effective predictors in clinical classification tasks, demonstrating the application of AI and data science in healthcare training and internship clinic settings.

METHODOLOGY

Converted manual clinic records from books into Excel format; cleaned data by handling missing values and removing unnecessary columns; performed exploratory data analysis with visualizations (pie charts, histograms) to understand data distribution; applied Gaussian Naïve Bayes classifier using Python and Jupyter Notebook; trained and tested model with standard train-test split; evaluated performance using accuracy, precision, recall, and F1-score metrics; generated classification reports and confusion matrix for model interpretation

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Gaussian Naïve Bayes model achieved 61.7% accuracy, 61.8% precision, 61.7% recall, and 61.2% F1-score for predicting patient sex; Class 0 (n=408) showed stronger performance with 72% recall and 0.66 F1-score versus Class 1 (n=371) with 50% recall and 0.56 F1-score, indicating class imbalance in predictive performance; demonstrated moderate but meaningful predictive power beyond random guessing; enhanced practical skills in health data management, AI model application, and result interpretation; identified key challenges including limited dataset size, few predictive features, and class imbalance; provided foundation for future studies to expand features and test alternative models for improved clinical classification

CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Machine Learning Model for Predicting Polio Vaccination Status in Kenya***BACKGROUND & METHODOLOGY**

Poliomyelitis remains a major public health concern in some parts of the world despite extensive eradication efforts. Vaccination is the only effective preventive measure, yet disparities in coverage persist due to socioeconomic, demographic, and health-system factors. This study explores the application of machine learning techniques in predicting polio vaccination status, with the goal of identifying individuals and communities at risk of incomplete or missed immunization. Using secondary health data including demographic and socioeconomic indicators, a Bernoulli Naive Bayes model was developed to classify vaccination outcomes. The results indicated high overall accuracy (~92%), but the model performed poorly in detecting vaccinated individuals due to class imbalance, highlighting limitations in the current dataset. Nonetheless, the findings demonstrate the potential of machine learning to support public health planning through targeted interventions, resource allocation, and early warning systems. By improving data balance, incorporating additional features, and exploring advanced algorithms, predictive models can significantly contribute to achieving global polio eradication goals.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Bernoulli Naive Bayes model to predict polio vaccination status in Kenya using secondary data from Kaggle, incorporating demographic and socioeconomic variables such as age, gender, maternal education, clinic visits, and exposure to health awareness campaigns. The model achieved an overall accuracy of approximately 92%, with strong performance in identifying unvaccinated individuals (92% precision) but limited recall for the vaccinated class due to significant class imbalance in the dataset. The findings demonstrate the potential of machine learning for public health surveillance and targeted immunization campaigns, with recommendations to address class imbalance via SMOTE, explore advanced algorithms such as Random Forest and Gradient Boosting, and integrate additional predictors to improve minority class detection.

INTERNSHIP **Internship Experience***SEKU University Health Clinic — Predicting Patient Sex Using Age and Five Common Diagnoses | Gaussian Naïve Bayes for Clinical Classification | AI for Demographic Pattern Analysis*

Dataset Used: Manual outpatient clinic records from South Eastern Kenya University Health Clinic (779 validated outpatient cases with age, sex, and five most common diagnosis variables). Study applying machine learning to predict patient sex using age and five common outpatient diagnoses at the SEKU University Clinic. The project aimed to assess whether simple demographic and diagnostic information could serve as effective predictors in clinical classification tasks, demonstrating the application of AI and data science in healthcare training and internship clinic settings.

METHODOLOGY

Converted manual clinic records from books into Excel format; cleaned data by handling missing values and removing unnecessary columns; performed exploratory data analysis with visualizations (pie charts, histograms) to understand data distribution; applied Gaussian Naïve Bayes classifier using Python and Jupyter Notebook; trained and tested model with standard train-test split; evaluated performance using accuracy, precision, recall, and F1-score metrics; generated classification reports and confusion matrix for model interpretation

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Gaussian Naïve Bayes model achieved 61.7% accuracy, 61.8% precision, 61.7% recall, and 61.2% F1-score for predicting patient sex; Class 0 (n=408) showed stronger performance with 72% recall and 0.66 F1-score versus Class 1 (n=371) with 50% recall and 0.56 F1-score, indicating class imbalance in predictive performance; demonstrated moderate but meaningful predictive power beyond random guessing; enhanced practical skills in health data management, AI model application, and result interpretation; identified key challenges including limited dataset size, few predictive features, and class imbalance; provided foundation for future studies to expand features and test alternative models for improved clinical classification

CURRENT STATUS**I don't have any · Unemployed — Actively searching**

Sector relevance: Partially related

SKILLS IN USE

Python · Data collection · Data visualization · ML modeling · Model evaluation · Ethics & bias in AI · Public health data analysis

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Machine Learning Model for Predicting Polio Vaccination Status in Kenya

BACKGROUND & METHODOLOGY

Poliomyelitis remains a major public health concern in some parts of the world despite extensive eradication efforts. Vaccination is the only effective preventive measure, yet disparities in coverage persist due to socioeconomic, demographic, and health-system factors. This study explores the application of machine learning techniques in predicting polio vaccination status, with the goal of identifying individuals and communities at risk of incomplete or missed immunization. Using secondary health data including demographic and socioeconomic indicators, a Bernoulli Naive Bayes model was developed to classify vaccination outcomes. The results indicated high overall accuracy (~92%), but the model performed poorly in detecting vaccinated individuals due to class imbalance, highlighting limitations in the current dataset. Nonetheless, the findings demonstrate the potential of machine learning to support public health planning through targeted interventions, resource allocation, and early warning systems. By improving data balance, incorporating additional features, and exploring advanced algorithms, predictive models can significantly contribute to achieving global polio eradication goals.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Bernoulli Naive Bayes model to predict polio vaccination status in Kenya using secondary data from Kaggle, incorporating demographic and socioeconomic variables such as age, gender, maternal education, clinic visits, and exposure to health awareness campaigns. The model achieved an overall accuracy of approximately 92%, with strong performance in identifying unvaccinated individuals (92% precision) but limited recall for the vaccinated class due to significant class imbalance in the dataset. The findings demonstrate the potential of machine learning for public health surveillance and targeted immunization campaigns, with recommendations to address class imbalance via SMOTE, explore advanced algorithms such as Random Forest and Gradient Boosting, and integrate additional predictors to improve minority class detection.

INTERNSHIP Internship Experience

Equity Afya Medical Centre — Kitui Branch — Predicting T2D Complication Risk Using Machine Learning | AI for Clinical Decision Support in Diabetes Care | Multi-Layer Perceptron for Patient Risk Stratification

Dataset Used: 7,000 anonymized patient records from Equity Afya Medical Centre (Kitui) containing demographic, clinical, and treatment variables including age, gender, diet quality, BMI, LDL, HDL, blood glucose, and complication status. Capstone project applying machine learning to predict likelihood of complication occurrence among Type 2 Diabetes patients. Dataset includes clinical measurements, lifestyle factors, and treatment history used to train an MLP classifier for early risk detection and targeted intervention planning in resource-limited healthcare settings.

METHODOLOGY

Internship Centre Work: Data entry and updating digital patient registers; Cleaning and validating patient records; Preparing visual summaries of complication trends; Supporting reporting and data review sessions with health staff | Capstone Project: Data preprocessing (handling missing values, encoding categorical variables, scaling numerical features); Descriptive analysis and visualization of glucose, BMI, and age distributions using Python (Pandas, Matplotlib); Training and testing Multi-Layer Perceptron (MLP) classifier using Scikit-learn; Model evaluation using accuracy, precision, recall, F1-score, and ROC-AUC metrics; Developing Streamlit web application for real-time clinical risk prediction and decision support

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully entered and validated 7,000 patient records, improving data quality for facility reporting | Generated interactive dashboards and charts on patient demographics and complication occurrence for health worker review | MLP classifier achieved exceptional performance: Accuracy = 96.7%, Precision = 96.0%, Recall = 96.9%, F1-score = 96.5%, ROC-AUC = 0.996 | Identified key predictors of complication risk: elevated blood glucose, high BMI, older age, and longer treatment duration | Model demonstrated strong balance between detecting true complication cases and minimizing false negatives, critical for clinical utility | Successfully deployed Streamlit web app providing clinicians with real-time risk stratification for proactive patient management | Enhanced practical skills in end-to-end data science pipeline from data cleaning to model deployment in public health settings | Demonstrated feasibility of using routine clinical data for predictive analytics in diabetes care | Provided actionable recommendations for integrating AI-driven tools into clinical workflows to support early intervention and personalized care strategies

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Malaria Predictions Based on Climate Variables in Kenya with Machine Learning

BACKGROUND & METHODOLOGY

This study applied machine learning to predict malaria outcomes in Kenya using climate variables and mosquito density per trap. A Random Forest classifier was trained and evaluated on a dataset of 276,976 records. The model achieved an accuracy of 71% and a ROC-AUC of 0.732. Results indicate the model is more effective in predicting non-malaria cases than malaria-positive cases, with rainfall and humidity emerging as strong predictors. While the classifier demonstrated fair predictive ability, further model optimization is needed to reduce false negatives and improve detection of malaria-positive cases for effective early warning systems.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classifier to predict malaria outcomes in Kenya using climate variables including rainfall, humidity, and mosquito density per trap, trained and tested on a dataset of 276,976 records. The model achieved an overall accuracy of 71% and a ROC-AUC of 0.73, with strong performance in detecting non-malaria cases (precision: 0.75, recall: 0.84, F1-score: 0.79) but limited sensitivity for malaria-positive cases (recall: 0.48, F1-score: 0.54). Feature importance analysis identified rainfall and humidity as the most influential predictors, with recommendations to apply SMOTE for class balancing, incorporate additional climate variables such as altitude and temperature, and develop a user-friendly early warning dashboard for malaria surveillance and resource allocation by health authorities in Kenya.

INTERNSHIP Internship Experience

Equity Afya, Kitui — Identifying Antibiotic Resistance Using Random Forest Classifier | Machine Learning for Antimicrobial Stewardship | AI for Clinical Decision Support in Infectious Disease Management

Dataset Used: Urinary tract infection patient dataset containing demographic and clinical information along with antibiotic susceptibility test results. Study applying machine learning to identify resistance patterns in urinary tract infection cases. Data preprocessing and feature selection were performed before applying a Random Forest classifier model to predict antibiotic resistance, supporting evidence-based prescription practices and antimicrobial stewardship programs in resource-limited healthcare settings.

METHODOLOGY

Data preprocessing involving cleaning, encoding categorical variables, and scaling numerical features | Applied Random Forest classifier model using Python and scikit-learn | Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix | Conducted feature selection to identify key predictors of antibiotic resistance | Generated classification reports and visualizations for result interpretation

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest classifier model achieved accuracy of 53-54%, demonstrating moderate predictive capability for antibiotic resistance patterns | Model performed better in identifying sensitive cases (recall ~47%) compared to resistant cases (recall ~53%), highlighting class imbalance challenges | Findings confirm potential of machine learning to support clinical decision-making and guide targeted treatment strategies | Identified key predictors associated with resistant bacterial strains | Demonstrated feasibility of applying AI-driven approaches to antimicrobial stewardship in Kenyan healthcare settings | Highlighted need for larger, balanced datasets and inclusion of behavioral/clinical predictors to improve model performance | Provided actionable recommendations for integrating predictive analytics into hospital diagnostic workflows and strengthening antimicrobial resistance surveillance

CURRENT STATUS

Not yet reported



CAPSTONE **Capstone Project**

Predicting Teenage Pregnancy Using Random Forest Classifier

BACKGROUND & METHODOLOGY

Teenage pregnancy remains a pressing public health issue in Kenya, affecting education, health outcomes, and socioeconomic development. This study aimed to understand the existing problems related to teenage pregnancy and applied a Random Forest classifier to predict factors associated with it. Secondary data from the Kenya Demographic and Health Survey (KDHS) 2022 was used, focusing on girls aged 14–18 years. Key predictors identified include poverty quantile, age, education level, residence, and access to contraceptives. The model achieved an overall accuracy of 93%, with precision of 1.00 and recall of 0.91 for non-pregnant cases, and precision of 0.75 and recall of 0.99 for pregnant cases, yielding a weighted average F1-score of 0.93. The study recommends policy action, community sensitization, and improved access to youth-friendly reproductive health services to address the complex determinants of teenage pregnancy in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classifier to predict teenage pregnancy risk among girls aged 14–18 in Kenya using secondary data from the KDHS 2022, incorporating variables such as age, education level, poverty quantile, residence, region, and access to contraceptives. The model achieved an overall accuracy of 93%, a recall of 0.99 for identifying pregnant teens, and a weighted average F1-score of 0.93, demonstrating high sensitivity in detecting at-risk individuals with minimal missed cases. The model was deployed for practical use in predicting teenage pregnancy risk, with recommendations to improve precision for the positive class, incorporate longitudinal and qualitative data, and integrate the tool into existing health information systems to enable real-time risk monitoring and support evidence-based adolescent reproductive health interventions.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Teenage Pregnancy Using Random Forest Classifier

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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INTERNSHIP Internship Experience

Equity Afya, Kitui — Identifying Antibiotic Resistance Using Random Forest Classifier | Machine Learning for Antimicrobial Stewardship | AI for Clinical Decision Support in Infectious Disease Management

★ LED TO A JOB OFFER

Dataset Used: Urinary tract infection patient dataset containing demographic and clinical information along with antibiotic susceptibility test results. Study applying machine learning to identify resistance patterns in urinary tract infection cases. Data preprocessing and feature selection were performed before applying a Random Forest classifier model to predict antibiotic resistance, supporting evidence-based prescription practices and antimicrobial stewardship programs in resource-limited healthcare settings.

METHODOLOGY

Data preprocessing involving cleaning, encoding categorical variables, and scaling numerical features | Applied Random Forest classifier model using Python and scikit-learn | Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix | Conducted feature selection to identify key predictors of antibiotic resistance | Generated classification reports and visualizations for result interpretation

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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CURRENT STATUS

SALONIST · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · Ethics & bias in AI · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Impact of Socioeconomic Status on Child Growth

BACKGROUND & METHODOLOGY

This study investigates the critical influence of socioeconomic factors—specifically maternal education and household wealth—on child physical development as measured by body weight. Utilizing a dataset of 799 records from child health surveys, the research employs Multiple Linear Regression to quantify the relationship between these socioeconomic determinants and nutritional outcomes. Exploratory data analysis revealed significant growth disparities: children from the wealthiest households averaged 15.8 kg, while those from the poorest averaged 12.5 kg; similarly, children of mothers with secondary education or higher showed improved weight outcomes compared to those of mothers with only primary education. The resulting predictive model achieved an R^2 of 0.75 on the test set, demonstrating that socioeconomic status is a potent predictor of child growth. These findings provide a data-driven foundation for public health policies, suggesting that targeted economic support and maternal education initiatives are essential to mitigating undernutrition and fostering equitable health outcomes for children.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The study successfully established a robust predictive framework for child physical development, demonstrating that socioeconomic determinants serve as significant predictors of growth. By applying a Multiple Linear Regression model to the dataset, the research achieved an R^2 value of approximately 0.75 on the test set, indicating that maternal education and household wealth account for a substantial portion of the variance in child weight. Specifically, the findings revealed that children of mothers with secondary education or higher, as well as those from the wealthiest quintile, exhibited significantly higher average weights compared to those from lower socioeconomic brackets.

INTERNSHIP Internship Experience

Kisumu County Referral Hospital — Emergency Operations Centre — Malaria Case Forecasting and Disease Surveillance Using Machine Learning

Dataset Used: Historical monthly malaria incidence data extracted from Kisumu County Referral Hospital DHIS2 digital records (January—September 2024). Monthly malaria case count dataset extracted from DHIS2 covering suspected malaria cases, mRDT positive cases, and microscopy positive cases across multiple months in Kisumu County, used to identify seasonal trends and forecast future malaria incidences.

METHODOLOGY

Performed data cleaning and consistency checks on health records at the Emergency Operations Centre; assisted in creating visual dashboards summarizing malaria cases by month; cleaned and validated malaria records to reduce inconsistencies; preprocessed the DHIS2 malaria dataset including missing value handling via imputation and removal of incomplete entries; conducted exploratory data analysis to detect seasonal trends using Matplotlib and Seaborn; trained and evaluated three machine learning models (Linear Regression, Random Forest, and Support Vector Regression) on an 80/20 train-test split; and developed and deployed an interactive Streamlit web application for malaria trend visualization and forecasting.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a malaria forecasting web application using historical hospital data from Kisumu County, with linear regression achieving the best predictive performance ($R = 0.953$, $MSE = 870$, $RMSE = 1,024$), significantly outperforming Random Forest ($R = -0.106$) and Support Vector Regression ($R = -0.942$) on the limited dataset; exploratory analysis identified March as the peak malaria month, aligning with the onset of long rains; the application was deployed publicly at <https://knaxjtkgbsoudrjla3vt7a.streamlit.app/> providing interactive visualization of historical trends and future projections for suspected cases, mRDT positives, and microscopy positives, with recommendations to incorporate climate and environmental variables such as rainfall and temperature for improved accuracy, consistently update data sources, and integrate AI-based forecasting into county disease surveillance frameworks.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Gender Based Analysis of BMI Distribution

BACKGROUND & METHODOLOGY

This research assesses the association between participation in polio campaigns and diarrheal status in children, addressing the premise that vaccination alone is insufficient for holistic child health. Using a 'child health data' dataset, the study employed a logistic regression model to predict diarrhea status based on campaign participation. The model achieved an accuracy of 89.47%, and hypothesis testing via a Chi-squared test yielded a p-value of 0.576. Because the p-value exceeded the 0.05 significance level, the study failed to reject the null hypothesis, concluding that no statistically significant association exists between polio campaign participation and diarrheal status within this specific dataset.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully completed a gender-stratified statistical analysis of BMI distribution using 400 records from the bmi_train.csv dataset. Key findings revealed that a significant proportion of the population falls into severe and morbid obesity categories (BMI Index 4 and 5), indicating critical public health concerns. Males demonstrated slightly higher mean BMI, weight, and height compared to females, highlighting gender-based differences in anthropometric measurements. The analysis employed descriptive statistics, count plots for BMI category distribution, and scatter plots for height-weight relationships colored by gender. The study confirms the value of stratified analysis over aggregate approaches in public health research, providing evidence-based insights for gender-specific health interventions, targeted messaging, and personalized health promotion strategies. Recommendations include expanding future analysis to incorporate age, socioeconomic status, diet, and physical activity variables for deeper causal understanding.

INTERNSHIP Internship Experience

New Busia Maternity & Nursing Home — Analysis of outpatient diagnosis by age, gender, and sub-county demographics

Dataset Used: Outpatient department records at New Busia Maternity & Nursing Home. Dataset containing patient demographic details (age, gender, sub-county) and outpatient diagnostic information used to analyze disease trends and healthcare patterns

METHODOLOGY

Data entry and validation, cleaning outpatient records, handling missing values and duplicates, standardizing variables, descriptive analysis using Excel and Python, creating visualizations (bar charts, violin plots, heat maps), preparing monthly reports, participating in data review meetings

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Improved data accuracy and reporting reliability, identified disease distribution trends across demographics, generated visual insights for healthcare decision-making, strengthened public health surveillance, enhanced technical skills in data analysis and visualization, supported evidence-based healthcare planning

CURRENT STATUS

Data Analyst · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP) · Prompt engineering · Deep Learning · Ethics & bias in AI · Freelancing in data science / building a portfolio · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Drug Prescription Prediction Using Machine Learning

BACKGROUND & METHODOLOGY

The use of artificial intelligence (AI)-based tools to guide prescribing decisions offers a promising pathway to enhance patient outcomes, yet current clinical decision support systems (CDSS) often suffer from "alert fatigue" due to overwhelming and irrelevant notifications. This project addresses this clinical gap by proposing an AI-assisted system designed to provide physicians with data-driven recommendations for drug prescriptions. By training a machine learning model on a structured dataset (drug200.csv) containing patient attributes such as age, sex, blood pressure, and sodium-to-potassium ratios, the study validates the premise that AI can accurately learn the complex relationships required for optimal drug selection. The results demonstrate a significant potential to improve patient safety by reducing medication errors and adverse drug events (ADEs), which are leading causes of patient harm.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a supervised machine learning framework for predicting optimal drug prescriptions, achieving a high accuracy rate of approximately 98.3% using Random Forest and Decision Tree models. The project served as a critical proof-of-concept, identifying the sodium-to-potassium (Na-to-K) ratio and blood pressure (BP) as the most influential factors in determining appropriate medication. Beyond technical validation, the study established a foundation for "augmented intelligence" in healthcare, proving that data-driven tools can effectively streamline clinical workflows and enhance patient safety. The findings recommend the future integration of Explainable AI (XAI) to provide transparent justifications for predictions, thereby bridging the "trust deficit" and facilitating the adoption of AI tools by healthcare professionals in high-stakes environments.

INTERNSHIP Internship Experience

Centre for Health Solutions-Kenya (CHS Kenya) — Predicting high viral load for enhanced patient care

Dataset Used: Clinical records and viral load dataset (–VL DATASET.csv–) from CHS Kenya health information system. Dataset containing patient demographics, treatment history, appointment dates, viral load results, regimen information, and clinical indicators used to predict high viral load risk among patients

METHODOLOGY

Data loading and exploration, cleaning and preprocessing data, handling missing values, feature engineering (appointment adherence), exploratory data analysis, visualization, training machine learning models (Logistic Regression, Decision Tree, Random Forest), evaluating models using precision/recall/F1-score, building Streamlit prototype for deployment

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed predictive models for identifying high viral load risk, identified key risk factors and demographic patterns, achieved high overall model accuracy though limited minority-class recall, demonstrated feasibility of AI-driven patient risk prediction, improved skills in machine learning, preprocessing, model evaluation, and public health analytics

CURRENT STATUS

Data entry assistant at watu credit limited · Unemployed — Actively searching

Sector relevance: Partially related

SKILLS IN USE

Data collection · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE **Capstone Project** *Gender Based Analysis of BMI Distribution*

BACKGROUND & METHODOLOGY

This report presents a comprehensive analysis of a Body Mass Index (BMI) dataset, stratified by gender. The primary objectives were to perform a statistical analysis, create insightful visualizations, and generate a report that documents the key findings. The dataset, `bmi_train.csv`, contains 400 records with features for gender, height, weight, and BMI index. The analysis revealed that a significant portion of the dataset falls into the severe and morbid obesity categories (BMI Index 4 and 5). A key finding from the gender-stratified approach was that males in the dataset have a slightly higher mean BMI, weight, and height compared to females. The project's methodology focused on data preprocessing, exploratory data analysis, and the creation of visualizations, without employing complex machine learning models. The findings demonstrate the value of a nuanced approach to public health data and provide recommendations for more targeted health interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully completed a gender-stratified statistical analysis of BMI distribution using 400 records from the `bmi_train.csv` dataset. Key findings revealed that a significant proportion of the population falls into severe and morbid obesity categories (BMI Index 4 and 5), indicating critical public health concerns. Males demonstrated slightly higher mean BMI, weight, and height compared to females, highlighting gender-based differences in anthropometric measurements. The analysis employed descriptive statistics, count plots for BMI category distribution, and scatter plots for height-weight relationships colored by gender. The study confirms the value of stratified analysis over aggregate approaches in public health research, providing evidence-based insights for gender-specific health interventions, targeted messaging, and personalized health promotion strategies. Recommendations include expanding future analysis to incorporate age, socioeconomic status, diet, and physical activity variables for deeper causal understanding.

INTERNSHIP **Internship Experience**

New Busia Maternity and Nursing Home — Analysis of patient demographics, diagnoses, medicines, and payment methods using healthcare data visualization

Dataset Used: Patient outpatient records from New Busia Maternity and Nursing Home. Dataset containing patient age, gender, village, payment method, diagnoses, and prescribed medicines used to analyze healthcare utilization patterns and disease trends

METHODOLOGY

Data entry and updating patient registers, cleaning and structuring outpatient data, handling missing values, converting data types, exploratory data analysis, descriptive statistics, creating visualizations (bar plots, pie charts, heatmaps, violin plots), analyzing insurance usage and disease patterns, compiling reports

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Improved patient data accuracy and organization, identified common diagnoses and medicines, revealed trends in insurance/payment methods and village-level healthcare utilization, strengthened skills in Python-based data analysis and visualization, enhanced understanding of AI/ML applications in healthcare systems, generated insights for targeted public health interventions and healthcare planning

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Effectiveness of Polio Campaigns on Child Health Outcomes

BACKGROUND & METHODOLOGY

This research assesses the association between participation in polio campaigns and diarrheal status in children, addressing the premise that vaccination alone is insufficient for holistic child health. Using a 'child health data' dataset, the study employed a logistic regression model to predict diarrhea status based on campaign participation. The model achieved an accuracy of 89.47%, and hypothesis testing via a Chi-squared test yielded a p-value of 0.576. Because the p-value exceeded the 0.05 significance level, the study failed to reject the null hypothesis, concluding that no statistically significant association exists between polio campaign participation and diarrheal status within this specific dataset.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully completed a statistical and machine learning analysis investigating the link between the August 2013 Polio campaign and child diarrhea status. Key findings indicated that while the logistic regression model showed high overall accuracy, it was ineffective at identifying positive cases of diarrhea due to class bias. Statistical testing confirmed no direct association between the campaign and diarrhea prevalence, though visualizations suggested that participating children had higher average weight and height. The study highlights the complexity of assessing indirect vaccine benefits and underscores the necessity of a multi-faceted public health approach. Recommendations include integrating nutrition, sanitation, and clean water access with vaccination efforts and expanding future research to include more comprehensive socioeconomic and environmental variables.

INTERNSHIP Internship Experience

Jaramogi Oginga Odinga University of Science and Technology (JOOUST) Medical Clinic — Individual Health Status Prediction Based on Vital Signs Using Machine Learning

Dataset Used: JOOUST Bondo Medical Clinic EMR system and manual patient registration documents (JOOUST Clinical Data 2025, 923 records). Clinical dataset of 923 patient records containing age, sex, weight, temperature, blood pressure (split into systolic and diastolic), pulse rate, and diagnosis, with health status engineered as a three-class target variable (healthy, at risk, ill) based on WHO vital sign thresholds, used to train and evaluate machine learning models for automated health status prediction.

METHODOLOGY

Registered 923+ patients in the facility's EMR and clinical registry; performed triage including measurement of weight, temperature, pulse rate, and blood pressure; checked patient eligibility and made financial claims on the SHA portal; prepared monthly TB reports; cleaned and preprocessed the clinical dataset by imputing missing values, splitting blood pressure into systolic and diastolic columns, and engineering a three-class health status target variable based on WHO criteria; conducted exploratory data analysis with histograms, scatter plots, box plots, and count plots; trained and evaluated logistic regression, random forest, and decision tree models; and deployed the best-performing model via Streamlit and Flask-ngrok.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Decision Tree Classifier to predict individual health status (healthy, at risk, ill) based on vital signs, achieving the highest performance alongside Random Forest with accuracy of 99.28%, precision of 99.29%, recall of 99.28%, and F1-score of 99.27%, significantly outperforming logistic regression (84.84% accuracy); analysis of 923 patient records classified 488 as at risk, 241 as healthy, and 194 as ill, confirming vital signs as highly reliable predictors of health status; the deployed Streamlit web application enables real-time health status classification by inputting temperature, pulse rate, systolic BP, and diastolic BP, with recommendations for the facility to install wireless internet, incorporate machine learning into routine diagnosis, and ensure reliable power supply to support digital health operations.

CURRENT STATUS

Not yet reported

**CAPSTONE** Capstone Project*Impact of Socioeconomic Status on Child Growth***BACKGROUND & METHODOLOGY**

This study investigates the critical influence of socioeconomic factors—specifically maternal education and household wealth—on child physical development as measured by body weight. Utilizing a dataset of 799 records from child health surveys, the research employs Multiple Linear Regression to quantify the relationship between these socioeconomic determinants and nutritional outcomes. Exploratory data analysis revealed significant growth disparities: children from the wealthiest households averaged 15.8 kg, while those from the poorest averaged 12.5 kg; similarly, children of mothers with secondary education or higher showed improved weight outcomes compared to those of mothers with only primary education. The resulting predictive model achieved an R^2 of 0.75 on the test set, demonstrating that socioeconomic status is a potent predictor of child growth. These findings provide a data-driven foundation for public health policies, suggesting that targeted economic support and maternal education initiatives are essential to mitigating undernutrition and fostering equitable health outcomes for children.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The study successfully established a robust predictive framework for child physical development, demonstrating that socioeconomic determinants serve as significant predictors of growth. By applying a Multiple Linear Regression model to the dataset, the research achieved an R^2 value of approximately 0.75 on the test set, indicating that maternal education and household wealth account for a substantial portion of the variance in child weight. Specifically, the findings revealed that children of mothers with secondary education or higher, as well as those from the wealthiest quintile, exhibited significantly higher average weights compared to those from lower socioeconomic brackets.

INTERNSHIP Internship Experience*St. Paul's Methodist Health Centre — Predicting the Effectiveness of Antiretroviral Therapy Regimens Among HIV Patients Using Machine Learning | AI for HIV Treatment Monitoring | Data-Driven ART Program Management*

Dataset Used: Active on ART Patients Linelist from St. Paul's Methodist Health Centre (August 2025, 161 validated test records after preprocessing containing demographic, clinical, regimen type, adherence, and viral load variables). Capstone project applying machine learning to predict viral load suppression status among patients on Antiretroviral Therapy (ART). Dataset included variables such as current regimen type, age, sex, weight, time on ART, WHO stage, TB status, AHD client status, and viral load results. Viral suppression was defined programmatically as <1000 copies/ml or "undetected/suppressed" (coded as 1) versus ≥1000 copies/ml or "detected" (coded as 0). The project aimed to support healthcare providers in identifying patients at risk of treatment failure through predictive analytics, enabling timely adherence support and regimen optimization in resource-limited HIV care settings.

METHODOLOGY

Internship Centre Work: Reviewed and updated ART and TB patient registers; conducted data entry and verification for newly enrolled patients in the Taifacare portal; supported preparation of monthly reports for submission to Sub-County Health Records Office; assisted in data cleaning and cross-checking between manual registers and electronic health systems; participated in routine data validation meetings and field data audits; engaged in discussions on treatment adherence tracking and viral load monitoring indicators | Capstone Project: Removed personally identifiable information and columns with >60% missing values; defined viral suppression target variable programmatically; performed exploratory data analysis and feature engineering (Time on ART calculation, WHO stage, TB status encoding); applied OneHotEncoder and StandardScaler for preprocessing; trained Random Forest Classifier (200 estimators) with SMOTE for class imbalance handling using 80/20 stratified train-test split; evaluated model using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix; extracted feature importance values; serialized model using joblib and developed Streamlit web application for real-time clinical risk prediction with user-friendly interface for healthcare workers

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest model achieved 98.1% overall accuracy but demonstrated severe limitation in identifying unsuppressed patients due to extreme class imbalance (158 suppressed vs. 3 unsuppressed in test set); suppressed class performance: precision=0.98, recall=1.00, F1=0.99; unsuppressed class: precision=0.00, recall=0.00; ROC-AUC=0.50 indicating limited discriminative ability; key predictors identified: Current Regimen (DTG-based regimens strongly associated with suppression), Time on ART (longer duration correlated with higher suppression), Age (younger patients slightly less likely to achieve suppression), Sex (males marginally lower suppression rates), Clinical Indicators (WHO stage, TB status, AHD status); successfully deployed Streamlit web app demonstrating feasibility of integrating predictive analytics into routine HIV program surveillance; highlighted critical challenge of class imbalance in predictive modeling for rare adverse outcomes; provided actionable recommendations for data quality improvement, enhanced monitoring of unsuppressed patients, continued DTG scale-up, and advanced modeling techniques (SMOTE variants, cost-sensitive learning) for future research

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Modelling of Cholera Test Result from Risk Factors Using Supervised Machine Learning Mode

BACKGROUND & METHODOLOGY

Cholera remains a significant global public health challenge, disproportionately affecting communities with limited access to clean water and adequate sanitation. This study addresses this challenge by analyzing a dataset to predict cholera positivity based on key risk factors like water source, sanitation type, income, and informal settlement status. Using Logistic Regression and Random Forest models, the research aims to move beyond reactive outbreak management toward a proactive, evidence-based approach to disease control. The findings revealed that the Random Forest model was the most effective predictive tool due to its superior recall, which is critical for identifying actual positive cases in a public health context. The analysis confirmed strong correlations between cholera positivity and factors such as living in informal settlements, relying on public wells, and having low income levels.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a Supervised Machine Learning framework specifically designed to predict cholera risk in vulnerable communities. By leveraging a Random Forest model, the study achieved a recall rate of 45.45%, prioritizing the identification of actual positive cases to prevent overlooked outbreaks in public health settings. The project successfully shifted the narrative from reactive emergency management to proactive, evidence-based disease control by quantifying the impact of critical socio-economic drivers. The findings established that residents of informal settlements, individuals with low income levels, and those relying on public wells face the highest risk of infection. Ultimately, this framework provides health officials with a precise, data-driven tool to optimize the distribution of life-saving resources, such as vaccines and water treatment kits, directly to the populations that need them most.

CURRENT STATUS

Not yet reported



CAPSTONE **Capstone Project**

Effectiveness of Polio Campaigns on Child Health Outcomes

BACKGROUND & METHODOLOGY

This research assesses the association between participation in polio campaigns and diarrheal status in children, addressing the premise that vaccination alone is insufficient for holistic child health. Using a 'child health data' dataset, the study employed a logistic regression model to predict diarrhea status based on campaign participation. The model achieved an accuracy of 89.47%, and hypothesis testing via a Chi-squared test yielded a p-value of 0.576. Because the p-value exceeded the 0.05 significance level, the study failed to reject the null hypothesis, concluding that no statistically significant association exists between polio campaign participation and diarrheal status within this specific dataset.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully completed a statistical and machine learning analysis investigating the link between the August 2013 Polio campaign and child diarrhea status. Key findings indicated that while the logistic regression model showed high overall accuracy, it was ineffective at identifying positive cases of diarrhea due to class bias. Statistical testing confirmed no direct association between the campaign and diarrhea prevalence, though visualizations suggested that participating children had higher average weight and height. The study highlights the complexity of assessing indirect vaccine benefits and underscores the necessity of a multi-faceted public health approach. Recommendations include integrating nutrition, sanitation, and clean water access with vaccination efforts and expanding future research to include more comprehensive socioeconomic and environmental variables.

CURRENT STATUS

assistant manager · Internship

Sector relevance: Partially related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Socioeconomic Determinants of Diarrhea Prevalence

BACKGROUND & METHODOLOGY

This project investigates the relationship between key socioeconomic factors—specifically household wealth and maternal education level—and the prevalence of diarrhea among children. Utilizing the `childhealthdata.csv` dataset, the study employs a quantitative approach and Logistic Regression to model and predict the likelihood of illness, emphasizing the critical role social conditions play in child health outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified a statistically significant correlation between lower socioeconomic status and a higher incidence of diarrhea, with the model achieving an accuracy of 89.2%. However, the researchers noted a "failure" in predictive utility due to significant class imbalance, as the model predicted "No diarrhea" for every test instance. The project concludes with recommendations to address these imbalances using techniques like SMOTE or non-linear models like Random Forest to better identify at-risk populations.

INTERNSHIP Internship Experience

New Busia Maternity and Nursing Home — Patient Readmission Risk Prediction Within 30 Days Using Inpatient Clinical Data

Dataset Used: Inpatient health records from New Busia Maternity and Nursing Home, recorded in an Excel worksheet (502 patient records). Inpatient dataset of 502 records containing patient demographics (age, gender), admission source (self, referred, ER), insurance type (SHA, cash, other), vital signs (blood pressure, temperature, pulse, respiration rate), healthcare interventions (surgical, medical, non-medical, diagnostic), number of admissions in the past year, length of stay, ICD-10 primary and secondary diagnosis codes, and a binary target variable for readmission within 30 days, with derived features including diagnosis chapters and abnormal vital sign flags.

METHODOLOGY

Completed facility orientation and reviewed admission and discharge records; collected and consolidated inpatient variables including demographics, diagnoses, vitals, and interventions; performed thorough data cleaning including handling missing values, removing duplicates, normalizing continuous variables, and outlier treatment; created derived variables such as diagnosis chapters and abnormal vital flag features; balanced the dataset using SMOTE; conducted descriptive and exploratory data analysis using correlation heatmaps and distribution charts; trained and evaluated Logistic Regression, Random Forest, Decision Tree, and Support Vector Machine models; developed a real-time patient readmission risk prediction app; and presented findings and the predictive app to hospital management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a patient readmission risk prediction model within 30 days using inpatient data from New Busia Maternity and Nursing Home, with Logistic Regression achieving the best overall performance at 92% accuracy, precision of 0.97, recall of 0.86, and F1-score of 0.91, closely followed by Random Forest (91% accuracy, recall 0.92, F1-score 0.91); the confusion matrix confirmed 88 true negatives, 72 true positives, 12 false negatives, and 2 false positives; age, blood pressure, insurance type, and diagnosis chapters were identified as the strongest predictors of readmission risk; the model was integrated into a real-time predictive app presented to hospital management, with recommendations to implement the tool in hospital workflows, strengthen post-discharge follow-up programs, improve EHR data completeness, and expand the dataset with social and behavioral health factors for improved model generalizability.

CURRENT STATUS

I'm currently not working · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project Drug Prescription Prediction Using Machine Learning

BACKGROUND & METHODOLOGY

The use of artificial intelligence (AI)-based tools to guide prescribing decisions offers a promising pathway to enhance patient outcomes, yet current clinical decision support systems (CDSS) often suffer from "alert fatigue" due to overwhelming and irrelevant notifications. This project addresses this clinical gap by proposing an AI-assisted system designed to provide physicians with data-driven recommendations for drug prescriptions. By training a machine learning model on a structured dataset (drug200.csv) containing patient attributes such as age, sex, blood pressure, and sodium-to-potassium ratios, the study validates the premise that AI can accurately learn the complex relationships required for optimal drug selection. The results demonstrate a significant potential to improve patient safety by reducing medication errors and adverse drug events (ADEs), which are leading causes of patient harm.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a supervised machine learning framework for predicting optimal drug prescriptions, achieving a high accuracy rate of approximately 98.3% using Random Forest and Decision Tree models. The project served as a critical proof-of-concept, identifying the sodium-to-potassium (Na-to-K) ratio and blood pressure (BP) as the most influential factors in determining appropriate medication. Beyond technical validation, the study established a foundation for "augmented intelligence" in healthcare, proving that data-driven tools can effectively streamline clinical workflows and enhance patient safety. The findings recommend the future integration of Explainable AI (XAI) to provide transparent justifications for predictions, thereby bridging the "trust deficit" and facilitating the adoption of AI tools by healthcare professionals in high-stakes environments.

INTERNSHIP Internship Experience

Kisumu County Referral Hospital — Development of Integrated Digital Health Solutions for Operational Efficiency | AI and Data Analytics for Healthcare Management | Digital Health Innovation

Dataset Used: Hospital operational data from Kisumu County Referral Hospital including patient records, appointment schedules, inventory data, dispatch logs, and queue management information stored in SQLite databases, CSV files, and Excel spreadsheets. Project developing five Streamlit-based digital health applications to improve operational efficiency at Kisumu County Referral Hospital. Applications include an Intelligent Dispatch Dashboard for ambulance logistics, Inventory and Pharmacy Tracker for drug stock monitoring, Hospital Appointment Booking System for patient scheduling, Patient Flow Analytics Dashboard for tracking admissions and discharge trends, and Smart Queue System Management platform for optimizing patient flow. Data sources contained patient demographics, clinical diagnoses, service utilization metrics, vehicle tracking information, drug inventory records, and operational indicators used to automate core hospital operations and support evidence-based decision-making in resource-limited healthcare settings.

METHODOLOGY

Adopted project-based learning and agile development approach with iterative testing and continuous improvement | Developed five Streamlit-based applications using Python for backend development, SQLite for database management, and Pandas/Plotly for analytics and visualization | Built Intelligent Dispatch Dashboard with centralized dispatch log, status-based vehicle tracking, response-time analytics, utilization metrics, and exportable reports | Created Patient Flow Analytics Dashboard with disease trend visualization, age/gender analysis, interactive filters, and data summaries | Developed Smart Queue System with digital ticketing, queue tracking dashboard, department-based queuing, and automatic updates | Built Appointment Booking App with registration forms, department/doctor selection, appointment summaries, and search/filter options | Created Inventory and Pharmacy Tracker with drug registration, real-time stock tracking, expiry management, inventory dashboard, and export functions | Deployed all applications on Streamlit Cloud for web accessibility | Conducted requirement gathering, system design, coding, debugging, testing, and user feedback collection throughout development cycle

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed five functional web-based applications addressing key hospital workflow challenges including ambulance dispatch, patient booking, queue management, and inventory tracking | Intelligent Dispatch Dashboard digitized dispatch records and provided response-time metrics enabling faster evidence-based decisions for ambulance allocation and emergency response planning | Patient Flow Analytics Dashboard enabled hospital management to quickly identify disease trends across age groups and time periods enhancing data-driven decision-making for resource planning | Smart Queue System streamlined patient flow and reduced congestion through organized digital queue processes improving service coordination and transparency | Appointment Booking App automated patient scheduling reducing waiting times and improving clinic schedule organization | Inventory and Pharmacy Tracker improved stock management efficiency and drug accountability with real-time visibility preventing shortages and supporting informed procurement planning | All applications demonstrated feasibility of implementing low-cost data-driven digital tools in local healthcare environments using open-source technologies | Enhanced technical proficiency in Python programming, database management, dashboard design, and machine learning integration | Provided practical insights into AI applications for improving healthcare outcomes in low-resource settings | Applications simplified clinical workflows, enhanced data quality, and improved disease monitoring capabilities supporting the hospital's goal of achieving efficient digital-first healthcare experience

CURRENT STATUS

Unemployed · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · Public health data analysis

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

Relationship Between Lifestyle Factor and Drug Type

BACKGROUND & METHODOLOGY

The connection between lifestyle behaviors and drug prescriptions is poorly understood, particularly how the sodium-to-potassium (Na to K) ratio affects clinical choices. This research investigates how dietary electrolyte balance, patient demographics like age, and lifestyle habits interplay to inform medication strategies. By analyzing prescription patterns and patient data, the study aims to elucidate these multifactorial influences. The goal is to improve therapeutic outcomes and foster precision medicine by providing a framework for personalized drug selection based on individual physiological markers.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a high-performing predictive model using a Decision Tree Classifier to determine drug prescriptions based on patient characteristics. The model achieved an overall accuracy of 97.4%, with a precision of 97.7% and an F1-score of 97.3%, demonstrating its reliability in handling both numerical and categorical data. Analysis identified the Na_to_K ratio as the most significant predictor, particularly for patients prescribed "Drug Y," followed by blood pressure, age, and cholesterol levels. While the model outperformed simpler algorithms like Logistic Regression and offered better interpretability than Support Vector Machines, the study notes limitations regarding the small dataset size (200 entries). Recommendations include expanding the dataset for better generalizability and integrating these predictive markers into clinical decision-support tools to enhance personalized patient care.

INTERNSHIP Internship Experience

New Busia Maternity and Nursing Home — Neonatal Clinical Concern Prediction Using Maternal and Delivery Variables

Dataset Used: Maternity case records from New Busia Maternity and Nursing Home, loaded from a CSV file. : Maternity dataset containing maternal demographics (age, HIV status), antenatal care visits, duration of labor, gestational age at birth, mode of delivery, clinical diagnoses, neonatal birth weight (grams), Apgar score at one minute, and a derived binary target variable 'clinical concern' (minimal concern vs. observation) engineered from child birth weight health and Apgar health categories, with class imbalance addressed using RandomOverSampler.

METHODOLOGY

Loaded and explored maternity data using pandas including structure checks, missing value assessment, and descriptive statistics; cleaned data by dropping rows with missing values and coercing non-numeric entries in the maternal age column; engineered target variable 'clinical concern' from derived 'child health' and 'apgar health' features; conducted exploratory data analysis using box plots, count plots, and correlation heatmaps across maternal age, ANC visits, labor hours, mode of delivery, diagnosis, and gestational age; applied RandomOverSampler to address class imbalance; encoded categorical variables using one-hot encoding; trained and evaluated a Random Forest Classifier; saved the model using joblib; developed a Streamlit prediction application; and deployed it externally via ngrok.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier to predict neonatal clinical concern (minimal concern vs. observation) from maternity case data at New Busia Maternity and Nursing Home, achieving an overall accuracy of 85%, precision of 84%, recall of 84%, and F1-score of 83%, with strong performance on the majority minimal concern class (precision 0.86, recall 0.96) but limited recall of 0.46 for the minority observation class; key predictors identified included gestational age, maternal age, labor hours, mode of delivery, clinical diagnosis, HIV status, number of ANC visits, and parity; the model outperformed Logistic Regression (75% accuracy) and SVM alternatives; it was deployed as a Streamlit web application for real-time neonatal risk prediction, with recommendations to improve data quality through robust imputation, incorporate additional clinical variables, apply advanced class imbalance techniques, and conduct external clinical validation to strengthen generalizability.

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP) · Prompt engineering · Deep Learning · Generative AI / LLMs · Ethics & bias in AI · Freelancing in data science / building a portfolio · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Predictive Model for Child's Weight in Early Childhood

BACKGROUND & METHODOLOGY

Monitoring child growth is essential for detecting health deviations, yet traditional methods in resource-constrained settings are often reactive and hindered by data gaps. This study develops a predictive machine learning model to estimate child weight using demographic, socio-economic, and anthropometric features such as age, sex, mother's education, and household wealth. By analyzing these complex, multi-factorial determinants, the research aims to provide a proactive tool for early health assessment and risk stratification. The study evaluates several algorithms—including Random Forest, Support Vector Machines (SVM), and Linear Regression—to identify the most accurate approach for enhancing evidence-based decision-making in child health management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a machine learning framework for child weight prediction, with the Support Vector Machine (SVM) model emerging as the top performer, achieving the highest R^2 of 0.72 and the lowest error metrics (Test MSE: 3.57). Linear Regression also demonstrated strong utility with an R^2 of 0.71, offering a more interpretable baseline for clinical use. The analysis identified age, maternal education, and wealth index as significant predictors of growth outcomes. The project recommends deploying the SVM model into user-friendly applications for healthcare workers in low-resource settings to facilitate real-time identification of children at risk for malnutrition. Future work will focus on integrating Explainable AI (XAI) to improve model transparency and incorporating longitudinal data to track individual growth trajectories over time.

INTERNSHIP Internship Experience

Westlands Health Care — Development of Predictive and Analytical Dashboards for Healthcare Data Management | AI for Non-Communicable Disease Prediction | Digital Health Innovation

Dataset Used: linical patient health records from Westlands Health Care containing demographic data, vital parameters (blood sugar levels, blood pressure, BMI), and diagnostic information for diabetes and hypertension screening. Project developing four Streamlit-based digital health applications for clinical data management and disease prediction at Westlands Health Care. The applications include a Clinical Data Entry Dashboard for patient record management, a Diabetes Prediction App using machine learning models (Random Forest/Logistic Regression), a Hypertension Prediction App for risk assessment, and an integrated Analytics Dashboard for analyzing co-occurrence patterns between diabetes and hypertension. The system aims to streamline clinical workflows, enable early disease screening, and support evidence-based decision-making in resource-limited healthcare settings.

METHODOLOGY

Adopted project-based and agile methodology with iterative development and user-centered design | Developed Clinical Data Entry Form Dashboard with form-based data entry, real-time record display, and export functionality | Built Diabetes Prediction App using machine learning models trained on health data with interactive Streamlit interface | Created Hypertension Prediction App for vital health indicator-based risk assessment | Designed integrated Diabetes + Hypertension Analysis Dashboard with combined visualization and interactive filters | Utilized Python for backend development, Streamlit for web dashboards, Pandas & Plotly for data analysis and visualization, Scikit-learn for model training, and Excel/CSV for data storage | Conducted requirement gathering, system design, model training, dashboard integration, and deployment with user testing

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed four Streamlit-based applications using Streamlit Cloud | Improved healthcare data management through automated patient screening and enhanced accuracy in data entry | Introduced data-driven approach to preventive healthcare with early identification of high-risk patients for diabetes and hypertension | Demonstrated feasibility of integrating machine learning and digital dashboards in clinical settings | Enhanced technical proficiency in Python, data visualization, and machine learning integration | Provided practical insights into AI applications for improving healthcare outcomes in low-resource settings | Applications simplified clinical workflows, enhanced data quality, and improved disease monitoring capabilities for non-communicable disease management

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Relationship Between Lifestyle Factor and Drug Type

BACKGROUND & METHODOLOGY

The connection between lifestyle behaviors and drug prescriptions is poorly understood, particularly how the sodium-to-potassium (Na to K) ratio affects clinical choices. This research investigates how dietary electrolyte balance, patient demographics like age, and lifestyle habits interplay to inform medication strategies. By analyzing prescription patterns and patient data, the study aims to elucidate these multifactorial influences. The goal is to improve therapeutic outcomes and foster precision medicine by providing a framework for personalized drug selection based on individual physiological markers.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a high-performing predictive model using a Decision Tree Classifier to determine drug prescriptions based on patient characteristics. The model achieved an overall accuracy of 97.4%, with a precision of 97.7% and an F1-score of 97.3%, demonstrating its reliability in handling both numerical and categorical data. Analysis identified the Na_to_K ratio as the most significant predictor, particularly for patients prescribed "Drug Y," followed by blood pressure, age, and cholesterol levels. While the model outperformed simpler algorithms like Logistic Regression and offered better interpretability than Support Vector Machines, the study notes limitations regarding the small dataset size (200 entries). Recommendations include expanding the dataset for better generalizability and integrating these predictive markers into clinical decision-support tools to enhance personalized patient care.

INTERNSHIP Internship Experience

Centre for Health Solutions (CHS), Siaya County — Socioeconomic and Demographic Predictors of ART Adherence Among HIV Patients

Dataset Used: Secondary patient end-line dataset (patientdata_endline1.csv) from CHS Siaya County HIV clients, covering facilities in Gem and Rarieda sub-counties (102,182 records pre-cleaning, 52,342 records post-cleaning). Patient-level ART dataset containing demographic variables (sex, age at ART start, age at last visit, village), socioeconomic variables (education level, marital status, occupation, migratory status, migratory category, population type), clinical variables (viral load result, WHO stage, last regimen line, patient type), and behavioral variables (all appointments, missed appointments), with a three-class target variable of ART adherence categorized as good, poor, or inadequate.

METHODOLOGY

Participated in monthly data collection from HIV and TB facilities across Gem and Rarieda sub-counties; performed data entry into Kenya Electronic Medical Records (EMR) and CRIS system; validated and cleaned monthly reports including ANC HIV testing statistics and ART initiation records; cross-validated data between Kenya EMR and CRIS; cleaned and preprocessed the 52,342-record dataset including missing value handling, categorical encoding, and class balancing; conducted exploratory data analysis using bar charts and heatmaps; developed and evaluated multinomial logistic regression, CatBoost Classifier, and Decision Tree models; generated feature importance plots and decision tree visualizations; and deployed the CatBoost model as an ART Adherence Prediction Streamlit web application with CSV upload and batch prediction functionality.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a CatBoost Classifier to predict ART adherence categories (good, poor, inadequate) among HIV patients in Siaya County, with the multinomial logistic regression model achieving an overall accuracy of 98% on the test dataset; feature importance analysis identified education level, marital status, and occupation as the strongest predictors of adherence, while migratory status and sex contributed significantly to patterns of inadequate adherence; findings revealed that viral load suppression strongly mirrored good adherence behavior, and that patients in formal employment and with higher education levels consistently demonstrated better adherence; the model was deployed as a Streamlit web application with real-time prediction, confidence scoring, and CSV batch processing capabilities, with recommendations to address class imbalance for minority adherence classes through SMOTE, explore alternative missing data imputation strategies, and integrate socioeconomic data into routine HIV patient monitoring systems for targeted counselling and intervention.

CURRENT STATUS

research and data nurse · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**

**CAPSTONE** Capstone Project*Predictive Modelling of Cholera Test Result from Risk Factors Using Supervised Machine Learning Mode***BACKGROUND & METHODOLOGY**

Cholera remains a significant global public health challenge, disproportionately affecting communities with limited access to clean water and adequate sanitation. This study addresses this challenge by analyzing a dataset to predict cholera positivity based on key risk factors like water source, sanitation type, income, and informal settlement status. Using Logistic Regression and Random Forest models, the research aims to move beyond reactive outbreak management toward a proactive, evidence-based approach to disease control. The findings revealed that the Random Forest model was the most effective predictive tool due to its superior recall, which is critical for identifying actual positive cases in a public health context. The analysis confirmed strong correlations between cholera positivity and factors such as living in informal settlements, relying on public wells, and having low income levels.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a Supervised Machine Learning framework specifically designed to predict cholera risk in vulnerable communities. By leveraging a Random Forest model, the study achieved a recall rate of 45.45%, prioritizing the identification of actual positive cases to prevent overlooked outbreaks in public health settings. The project successfully shifted the narrative from reactive emergency management to proactive, evidence-based disease control by quantifying the impact of critical socio-economic drivers. The findings established that residents of informal settlements, individuals with low income levels, and those relying on public wells face the highest risk of infection. Ultimately, this framework provides health officials with a precise, data-driven tool to optimize the distribution of life-saving resources, such as vaccines and water treatment kits, directly to the populations that need them most.

INTERNSHIP Internship Experience*St. Paul's Methodist Health Centre — Advanced HIV Disease (AHD) Risk Prediction Using Calibrated Random Forest Machine Learning*

Dataset Used: Retrospective de-identified patient records from St. Paul's Methodist Health Centre facility database (816 HIV-positive patient records). Dataset of 816 HIV-positive patient records containing demographic variables (age, gender, weight), clinical variables (CD4 count, viral load, WHO clinical stage), and ART-related variables (ART regimen, months on ART), with an engineered binary target variable (AHD vs. non-AHD) defined by CD4 count below 200 cells/?L or WHO clinical stage 3 or 4, exhibiting significant class imbalance (2.1% AHD vs. 97.9% non-AHD) and approximately 30% missingness in key variables.

METHODOLOGY

Performed data entry and validation of patient records at the outpatient department and EMR system across OPD, MCH, ANC, and CCC departments; participated in weekly data review and reporting meetings; compiled monthly performance reports on HIV testing, ART uptake, and maternal health indicators; assisted in cleaning and verifying ART cohort data for quarterly reporting; generated visual summaries for the facility dashboard using DHIS2 and Excel; cleaned and preprocessed the 816-patient dataset including statistical imputation, categorical encoding, outlier detection via IQR, data normalization, and SMOTE application; trained and calibrated a Random Forest classifier using Platt Scaling on a three-way train/calibration/test split; applied SHAP for global and local model interpretability; and deployed the model as a three-tab Streamlit web application with a prediction tab, analytics dashboard, and HIV expert chatbot.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a calibrated Random Forest classifier to predict Advanced HIV Disease (AHD) risk among 816 HIV-positive patients, achieving outstanding performance with 99.4% accuracy, 100% recall (zero missed AHD cases), 99.3% specificity, and a ROC-AUC of 98.88%; Platt Scaling calibration improved probability alignment with observed outcomes, enabling clinically trustworthy risk scores; SHAP analysis identified CD4 count (40–50% contribution), viral load (20–25%), WHO clinical stage, and months on ART as the top predictors of AHD risk, fully aligning with WHO 2017 AHD management guidelines; the model was deployed as a comprehensive Streamlit web application featuring a real-time AHD risk prediction tab with SHAP explanation plots, an analytics dashboard for facility-level HIV service trend analysis, and a rule-based HIV expert chatbot for healthcare worker guidance, with recommendations to strengthen EMR data quality, extend internship durations for deeper practical engagement, and validate the model externally across multiple facilities.

CURRENT STATUS**Health systems Manager/ Data analyst · Unemployed — Actively searching**

Sector relevance: Partially related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · ML modeling · Model evaluation · Generative AI / LLMs · Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Predictive Model for Child's Weight in Early Childhood

BACKGROUND & METHODOLOGY

Monitoring child growth is essential for detecting health deviations, yet traditional methods in resource-constrained settings are often reactive and hindered by data gaps. This study develops a predictive machine learning model to estimate child weight using demographic, socio-economic, and anthropometric features such as age, sex, mother's education, and household wealth. By analyzing these complex, multi-factorial determinants, the research aims to provide a proactive tool for early health assessment and risk stratification. The study evaluates several algorithms—including Random Forest, Support Vector Machines (SVM), and Linear Regression—to identify the most accurate approach for enhancing evidence-based decision-making in child health management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated a machine learning framework for child weight prediction, with the Support Vector Machine (SVM) model emerging as the top performer, achieving the highest R^2 of 0.72 and the lowest error metrics (Test MSE: 3.57). Linear Regression also demonstrated strong utility with an R^2 of 0.71, offering a more interpretable baseline for clinical use. The analysis identified age, maternal education, and wealth index as significant predictors of growth outcomes. The project recommends deploying the SVM model into user-friendly applications for healthcare workers in low-resource settings to facilitate real-time identification of children at risk for malnutrition. Future work will focus on integrating Explainable AI (XAI) to improve model transparency and incorporating longitudinal data to track individual growth trajectories over time.

INTERNSHIP Internship Experience

St. Paul Methodist Health Centre, Ugunja Town, Siaya County — HIV Likelihood Predictive Model Using Logistic Regression | Machine Learning for HIV Risk Stratification | AI for Targeted Testing Interventions

Dataset Used: HIV testing dataset "HIV likelihood.xlsx - Sheet5.csv" containing 172 patient records with demographic, behavioral, and geographic variables (age, marital status, sexual partners, condom use, village, ward, period since last test). Project applying logistic regression to predict likelihood of HIV positive test results using patient demographic, behavioral, and location-specific factors. Dataset included variables such as current age, marital status, number of sexual partners in last 12 months, condom use with last partner, sub-county name, village, and period since last HIV test. The highly imbalanced dataset (92% Negative, 8% Positive) was analyzed using class weighting to optimize performance on the minority class, with the goal of identifying high-risk individuals and geographic hotspots for targeted HIV testing and prevention interventions in resource-limited settings.

METHODOLOGY

Internship Centre Work: Supported data management in DHIS2 system including data entry, validation, and cleaning of patient records; corrected inconsistencies, missing values, and entry errors | Capstone Project: Data preprocessing including target encoding (Negative?0, Positive?1), dropping unique identifiers, handling severe class imbalance using `class_weight='balanced'`, one-hot encoding categorical predictors resulting in 202 features from 16 original variables | Trained Logistic Regression model with stratified 80/20 train-test split using `solver='liblinear'` and `max_iter=1000` | Evaluated model using ROC AUC score and classification metrics (precision, recall, F1-score, confusion matrix) | Conducted feature importance analysis by interpreting model coefficients to identify key demographic, behavioral, and geographic risk factors | Generated visualizations and interpreted results for public health insights and targeted intervention planning

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic Regression model achieved ROC AUC score of 0.7083, demonstrating moderate discriminative ability significantly better than random guessing | Model performed well on majority class (Negative: Precision=0.91, Recall=0.91, F1=0.91) but failed to identify positive cases in small test set (Recall=0.00) due to severe class imbalance and limited positive samples (n=14 total) | Key risk factors identified: testing more than one year ago significantly increased risk ($\beta=+1.46$) while testing within 3 months was strongly protective ($\beta=-1.84$) | Geographic hotspots identified: villages Masanga ($\beta=+1.51$), SIREMBE ($\beta=+1.48$), and Siholo showed highest positive correlation with HIV positivity, suggesting localized transmission networks | Marital status analysis showed Single status associated with lower risk ($\beta=-1.13$) compared to Married/Cohabiting baseline | Findings provide actionable evidence for prioritizing testing for individuals not tested recently and directing resources to geographically defined high-risk areas | Highlighted critical need for larger, more balanced datasets and advanced techniques (SMOTE, regularization) to improve model reliability for operational case identification in HIV prevention programs

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Socioeconomic Determinants of Diarrhea Prevalence

BACKGROUND & METHODOLOGY

This project investigates the relationship between key socioeconomic factors—specifically household wealth and maternal education level—and the prevalence of diarrhea among children. Utilizing the `childhealthdata.csv` dataset, the study employs a quantitative approach and Logistic Regression to model and predict the likelihood of illness, emphasizing the critical role social conditions play in child health outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified a statistically significant correlation between lower socioeconomic status and a higher incidence of diarrhea, with the model achieving an accuracy of 89.2%. However, the researchers noted a "failure" in predictive utility due to significant class imbalance, as the model predicted "No diarrhea" for every test instance. The project concludes with recommendations to address these imbalances using techniques like SMOTE or non-linear models like Random Forest to better identify at-risk populations.

INTERNSHIP Internship Experience

Kisumu County Referral Hospital — Digital Health Systems Development for Emergency Operations and Community Health Management

Dataset Used: Simulated and operational health facility data from Kisumu County Referral Hospital, stored locally using SQLite databases within HOSCON and TIBASASA applications. Hospital emergency operations data (emergency case logs, departmental roles, staff assignments, incident status updates, response times) and community health data (patient demographics, visit history, health outcomes, follow-up tracking), both stored in offline-capable SQLite databases designed to function without continuous internet connectivity.

METHODOLOGY

Identified operational challenges in manual data systems at Kisumu County Referral Hospital including communication delays and data inefficiencies; conceptualized and designed two digital health applications; developed HOSCON (Hospital Operations and Coordination) using Python, Streamlit, and SQLite in Google Colab as an offline emergency operations dashboard with modules for case logging, staff assignment, and response time tracking; developed TIBASASA (Treatment and Information-Based Application for Sustainable Access to Services and Awareness) as a companion community health and patient management system with patient registration, follow-up tracking, and basic analytics; designed both applications to operate offline with data syncing capability upon connectivity restoration; and evaluated usability and functionality of both systems within a simulated hospital setting.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and demonstrated two functional digital health applications — HOSCON and TIBASASA — deployed as Streamlit web applications with offline SQLite data storage, addressing critical gaps in hospital emergency coordination and community health data management at Kisumu County Referral Hospital; HOSCON enabled real-time recording, updating, and monitoring of emergency operations through a structured dashboard interface, while TIBASASA provided structured community health data collection and patient tracking for use by community health extension workers and mobile health units; both applications demonstrated feasibility in low-resource, offline-capable settings, with recommendations to integrate HOSCON into hospital emergency preparedness frameworks, adopt TIBASASA in community health programs for standardized data collection, develop synchronization capabilities with national health databases, and continue ENGAGE mentorship to advance scalable digital health innovations in Kenya.

CURRENT STATUS

Data scientist · Unemployed — Actively searching

Sector relevance: Partially related

SKILLS IN USE

Ethics & bias in AI

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE **Capstone Project**
Predicting Acute Kidney Disease Using Machine Learning Model

BACKGROUND & METHODOLOGY

Acute Kidney Injury (AKI) is a sudden decline in renal function that frequently occurs in hospitalized patients and, if untreated, can progress to Chronic Kidney Disease (CKD). CKD may ultimately require dialysis or transplantation, both of which are costly and significantly reduce quality of life. Early detection of AKI is therefore critical for improving patient outcomes and reducing healthcare burdens. This project aimed to develop and evaluate machine learning models capable of predicting AKI using patient clinical and laboratory data. Two models---Logistic Regression and Random Forest---were implemented and compared. The dataset consisted of 43 demographic and clinical features, including vital signs, biochemical parameters, and comorbidities. Preprocessing involved handling missing values, encoding categorical variables, and addressing class imbalance using the Synthetic Minority Oversampling Technique (SMOTE). Model performance was assessed using accuracy, precision, recall, and F1-score. Logistic Regression achieved an accuracy of 70%, with limited predictive ability. In contrast, the Random Forest model demonstrated significantly stronger performance, achieving 84% accuracy, with a recall of 78% and precision of 89% for AKI cases. These findings indicate that Random Forest provides a robust predictive framework for early AKI detection, outperforming traditional baseline models. The study highlights the potential of machine learning approaches in clinical decision support, with the promise of enabling earlier interventions to prevent CKD progression and reduce reliance on costly dialysis treatments.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and evaluated machine learning models for early prediction of Acute Kidney Injury using 32,864 patient records with 43 clinical and laboratory features. Random Forest significantly outperformed Logistic Regression, achieving 84% accuracy versus 54%, with 89% precision and 78% recall for AKI cases. SMOTE was applied to balance the dataset, resulting in equal representation of AKI and non-AKI cases. The Random Forest model effectively captured complex nonlinear relationships between risk factors, making it suitable for clinical decision support. Key clinical implications include enabling timely interventions before irreversible kidney damage, reducing progression to Chronic Kidney Disease, and lowering the financial burden of dialysis costs (KES 60,000-180,000 monthly) on Kenyan households. The study recommends future validation on multi-center datasets across Kenya, integration of time-series laboratory trends, and development of user-friendly EHR-linked decision support tools to facilitate clinical adoption and strengthen renal care in low-resource settings.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Early Detection of Cervical Cancer Risk Using Clinical and Behavioral Predictors: Machine Learning Model Approach

BACKGROUND & METHODOLOGY

Cervical cancer remains a leading cause of preventable deaths among women in Kenya, due to late detection and limited screening resources. This study developed machine learning model using data from 300 women to predict cervical cancer risk. The model grouped women into high and low risks categories. Its implementation could save lives while reducing screening costs. An app was designed to enable community health promoters (CHPs) to perform risk assessments, providing scalable prevention for resource limited settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a machine learning model for early cervical cancer risk prediction using synthetic data from 300 women with clinical and behavioral features. Logistic Regression was selected as the final model achieving 93.6% accuracy, 91.7% precision, 95.7% recall, and 0.936 F1-Score, outperforming Random Forest (91.5% accuracy) and XGBoost. The model demonstrated exceptional recall with only 1 false negative out of 23 actual positive cases, critical for medical screening where missing at-risk women has severe consequences. SMOTE was applied to address class imbalance. Key risk predictors included age, number of sexual partners, number of pregnancies, age at first sexual intercourse, smoking status, hormonal contraceptive use, family history, and STD history. An interactive app was designed for community health promoters to perform household risk assessments, enabling targeted screening programs, optimal resource allocation, and scalable prevention in resource-limited settings. The model addresses the challenge of late detection by identifying high-risk women before symptoms emerge, supporting Kenya's goal of reducing the 3,211 annual cervical cancer deaths.

INTERNSHIP Internship Experience

Bondeni Hospital / Bondeni Sub-County Hospital, Nakuru — Maternal health; anemia risk screening in pregnant women; antenatal care risk prediction

Dataset Used: Antenatal care records from Bondeni Hospital covering 2024–2025. ANC dataset containing health records for pregnant women, including hemoglobin levels and anemia-related predictors such as parity, gravidity, HIV status, blood pressure, malaria infection, BMI, previous history, and hemoglobin concentration. The model classified women into low-risk and high-risk anemia categories based mainly on hemoglobin thresholds

METHODOLOGY

Cleaned and preprocessed ANC data; explored anemia patterns using visualizations; trained and compared machine learning models including Logistic Regression, Random Forest, and XGBoost; classified anemia risk into low-risk and high-risk groups; deployed the trained model for clinical/community screening use

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a machine-learning screening model for early anemia risk detection among pregnant women; identified a high proportion of women at risk of anemia; Random Forest and XGBoost were reported as the best-performing models; created an AI-powered screening tool to support healthcare providers and CHPs in identifying high-risk mothers early and enabling timely intervention

CURRENT STATUS

Unemployed · Unemployed — Actively searching

Sector relevance: Partially related

Confidence applying ENGAGE skills: **3 / 5**



CAPSTONE Capstone Project

Early Detection of Cervical Cancer Risk Using Clinical and Behavioral Predictors: Machine Learning Model Approach

BACKGROUND & METHODOLOGY

Cervical cancer remains a leading cause of preventable deaths among women in Kenya, due to late detection and limited screening resources. This study developed machine learning model using data from 300 women to predict cervical cancer risk. The model grouped women into high and low risks categories. Its implementation could save lives while reducing screening costs. An app was designed to enable community health promoters (CHPs) to perform risk assessments, providing scalable prevention for resource limited settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a machine learning model for early cervical cancer risk prediction using synthetic data from 300 women with clinical and behavioral features. Logistic Regression was selected as the final model achieving 93.6% accuracy, 91.7% precision, 95.7% recall, and 0.936 F1-Score, outperforming Random Forest (91.5% accuracy) and XGBoost. The model demonstrated exceptional recall with only 1 false negative out of 23 actual positive cases, critical for medical screening where missing at-risk women has severe consequences. SMOTE was applied to address class imbalance. Key risk predictors included age, number of sexual partners, number of pregnancies, age at first sexual intercourse, smoking status, hormonal contraceptive use, family history, and STD history. An interactive app was designed for community health promoters to perform household risk assessments, enabling targeted screening programs, optimal resource allocation, and scalable prevention in resource-limited settings. The model addresses the challenge of late detection by identifying high-risk women before symptoms emerge, supporting Kenya's goal of reducing the 3,211 annual cervical cancer deaths.

INTERNSHIP Internship Experience

Bondeni Hospital / Bondeni Sub-County Hospital, Nakuru — Maternal health; anemia risk screening in pregnant women; antenatal care risk prediction

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METHODOLOGY

Cleaned and preprocessed ANC data; explored anemia patterns using visualizations; trained and compared machine learning models including Logistic Regression, Random Forest, and XGBoost; classified anemia risk into low-risk and high-risk groups; deployed the trained model for clinical/community screening use

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a machine-learning screening model for early anemia risk detection among pregnant women; identified a high proportion of women at risk of anemia; Random Forest and XGBoost were reported as the best-performing models; created an AI-powered screening tool to support healthcare providers and CHPs in identifying high-risk mothers early and enabling timely intervention

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Acute Kidney Disease Using Machine Learning Model

BACKGROUND & METHODOLOGY

Acute Kidney Injury (AKI) is a sudden decline in renal function that frequently occurs in hospitalized patients and, if untreated, can progress to Chronic Kidney Disease (CKD). CKD may ultimately require dialysis or transplantation, both of which are costly and significantly reduce quality of life. Early detection of AKI is therefore critical for improving patient outcomes and reducing healthcare burdens. This project aimed to develop and evaluate machine learning models capable of predicting AKI using patient clinical and laboratory data. Two models---Logistic Regression and Random Forest---were implemented and compared. The dataset consisted of 43 demographic and clinical features, including vital signs, biochemical parameters, and comorbidities. Preprocessing involved handling missing values, encoding categorical variables, and addressing class imbalance using the Synthetic Minority Oversampling Technique (SMOTE). Model performance was assessed using accuracy, precision, recall, and F1-score. Logistic Regression achieved an accuracy of 70%, with limited predictive ability. In contrast, the Random Forest model demonstrated significantly stronger performance, achieving 84% accuracy, with a recall of 78% and precision of 89% for AKI cases. These findings indicate that Random Forest provides a robust predictive framework for early AKI detection, outperforming traditional baseline models. The study highlights the potential of machine learning approaches in clinical decision support, with the promise of enabling earlier interventions to prevent CKD progression and reduce reliance on costly dialysis treatments.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and evaluated machine learning models for early prediction of Acute Kidney Injury using 32,864 patient records with 43 clinical and laboratory features. Random Forest significantly outperformed Logistic Regression, achieving 84% accuracy versus 54%, with 89% precision and 78% recall for AKI cases. SMOTE was applied to balance the dataset, resulting in equal representation of AKI and non-AKI cases. The Random Forest model effectively captured complex nonlinear relationships between risk factors, making it suitable for clinical decision support. Key clinical implications include enabling timely interventions before irreversible kidney damage, reducing progression to Chronic Kidney Disease, and lowering the financial burden of dialysis costs (KES 60,000-180,000 monthly) on Kenyan households. The study recommends future validation on multi-center datasets across Kenya, integration of time-series laboratory trends, and development of user-friendly EHR-linked decision support tools to facilitate clinical adoption and strengthen renal care in low-resource settings.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Medical Condition Capstone Project: Predicting Asthma, Pneumonia, and Hypertension Using Naive Bayes

BACKGROUND & METHODOLOGY

This project focuses on the prediction of common medical conditions — asthma, pneumonia, and hypertension — based on age, occupation, and gender using a Naive Bayes classification model. Early detection and proactive management of these conditions are vital in reducing healthcare burdens and improving patient outcomes. The Naive Bayes algorithm was chosen for its simplicity, efficiency, and strong performance on classification tasks with categorical input features. The model was trained on a labeled dataset containing anonymized patient records with age, occupation, gender, and diagnosis information. After preprocessing and exploratory data analysis, the dataset was split into training and testing sets to evaluate model performance. The model achieved an overall accuracy of 74%, precision of 64%, recall of 74%, and an F1-score of 68%, demonstrating solid predictive performance using minimal demographic inputs. The study underscores the potential of integrating lightweight machine learning models into early disease detection frameworks, particularly in resource-constrained settings, to assist clinicians and enable timely interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Naive Bayes classification model to predict three common medical conditions — asthma, pneumonia, and hypertension — using age, occupation, and gender as input features, trained on anonymized hospital patient records. The model achieved an overall accuracy of 74%, precision of 64%, recall of 74%, and an F1-score of 68%, with strongest classification performance observed for pneumonia cases. A user-friendly deployment interface was designed to enable real-time predictions using minimal demographic input, supporting early medical screening in resource-limited environments, with recommendations to expand the feature set with clinical symptoms and medical history, and to explore hybrid modeling approaches for improved discrimination between overlapping conditions.

INTERNSHIP Internship Experience

Mikindani Catholic CBHC — Child Welfare Clinic (CWC) Service Utilization and Preventive Care Association Analysis for Under-5 Children

Dataset Used: Secondary data from the Maternal and Child Health Care department CWC register at Mikindani Catholic CBHC (manually collected and digitized). CWC register dataset for children under 5 years containing variables including date, OPD number, revisit status, age, sex, village/estate, weight, exclusive breastfeeding status, deworming, any disability, any danger signs, immunization status, LLIN given (under one year), and follow-up type (nutrition services or rehabilitation services), used to analyze associations between preventive and follow-up care services.

METHODOLOGY

Entered and validated CWC register records into Excel; tracked patients who missed appointments; fed patient details into the Kenya EMR system for refill and visit updates; generated departmental reports in the system; assisted with photocopying, printing, and scanning of documents; collected, cleaned, and encoded binary categorical variables; and applied Association Rule Learning using the Apriori algorithm in Python, training and evaluating rules using support, confidence, and lift metrics.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully applied the Apriori Association Rule Learning model to identify meaningful patterns in child health service delivery, achieving an average confidence of 78% across all strong rules with lift values greater than 1.0 confirming non-random associations; the strongest rule identified was between disability and rehabilitation follow-up (confidence 88%, lift 2.10), indicating excellent targeting of vulnerable children; additional strong rules revealed that 82% of exclusively breastfed children were also immunized, 79% of LLIN recipients were fully immunized, and 74% of children with danger signs were not exclusively breastfed; findings support recommendations to systematically track danger sign and disability cases for follow-up services and to leverage immunization campaigns to reinforce nutrition screening enrollment.

CURRENT STATUS

unemployed · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Data collection · Data cleaning & wrangling · Data visualization

Confidence applying ENGAGE skills: **4 / 5**

**CAPSTONE** Capstone Project*Medical Condition Capstone Project: Predicting Asthma, Pneumonia, and Hypertension Using Naive Bayes***BACKGROUND & METHODOLOGY**

This project focuses on the prediction of common medical conditions — asthma, pneumonia, and hypertension — based on age, occupation, and gender using a Naive Bayes classification model. Early detection and proactive management of these conditions are vital in reducing healthcare burdens and improving patient outcomes. The Naive Bayes algorithm was chosen for its simplicity, efficiency, and strong performance on classification tasks with categorical input features. The model was trained on a labeled dataset containing anonymized patient records with age, occupation, gender, and diagnosis information. After preprocessing and exploratory data analysis, the dataset was split into training and testing sets to evaluate model performance. The model achieved an overall accuracy of 74%, precision of 64%, recall of 74%, and an F1-score of 68%, demonstrating solid predictive performance using minimal demographic inputs. The study underscores the potential of integrating lightweight machine learning models into early disease detection frameworks, particularly in resource-constrained settings, to assist clinicians and enable timely interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Naive Bayes classification model to predict three common medical conditions — asthma, pneumonia, and hypertension — using age, occupation, and gender as input features, trained on anonymized hospital patient records. The model achieved an overall accuracy of 74%, precision of 64%, recall of 74%, and an F1-score of 68%, with strongest classification performance observed for pneumonia cases. A user-friendly deployment interface was designed to enable real-time predictions using minimal demographic input, supporting early medical screening in resource-limited environments, with recommendations to expand the feature set with clinical symptoms and medical history, and to explore hybrid modeling approaches for improved discrimination between overlapping conditions.

INTERNSHIP Internship Experience*Viwa Level 4 Healthcare Services — Predicting Diagnosis of Under-Five Outpatients based on Age, Weight, Temperature, and Gender / Machine Learning for Pediatric Disease Classification*

Dataset Used: Under-five outpatient registers from Viwa Level 4 Healthcare Services (435 initial records, filtered to 124-126 records focusing on pneumonia and upper respiratory tract infection cases). Dataset containing outpatient records for children under five years with variables including age, weight, temperature, gender, and diagnosis outcomes (pneumonia and upper respiratory tract infection - URTI), used to develop a predictive model for classifying common childhood illnesses and supporting clinical decision-making in primary healthcare settings.

METHODOLOGY

Entered outpatient under-five data into Microsoft Excel from manually collected registers; recorded patient vital signs; prepared monthly outpatient summaries for facility reporting; preprocessed and cleaned data in Excel and Python; handled missing values and removed irrelevant entries; computed descriptive statistics for patient characteristics; applied logistic regression model to predict diagnosis outcomes; generated visualizations (pie charts, bar charts, scatter plots, stacked bar charts) using Python libraries; evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 95% accuracy (21 out of 22 test samples correctly predicted), 96% precision, 95% recall, and 95% F1-score; identified temperature, age, weight, and gender as significant predictors of diagnosis; revealed gender distribution of 54.8% female and 45.2% male patients; established URTI as significantly more common (~60 cases) than pneumonia (~10 cases); model achieved exceptional URTI performance (95% precision, 100% recall) but moderate pneumonia detection (100% precision, 50% recall); demonstrated positive correlation between age and weight; identified Mnarani as leading in URTI cases and Ganze in pneumonia cases; revealed 60.3% new patients versus 39.7% revisits; highlighted potential of AI and data science in supporting clinical decision-making and improving healthcare efficiency for under-five children.

CURRENT STATUS**Juice vendor · Unemployed — Actively searching**

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Analyzing and Predicting Outcomes in Tuberculosis Contact Tracing in Makueni, Kenya

BACKGROUND & METHODOLOGY

This project analyzes a tuberculosis contact-tracing dataset from Makueni, Kenya, to evaluate the effectiveness of Tuberculosis Preventive Therapy (TPT) and identify associated risk factors. The analysis focuses on three key areas: demographics, TPT outcomes, and the relationship between BMI, HIV status, and TB outcomes. Descriptive statistics reveal a diverse cohort with a mean age of 29.69 years and a near-balanced gender distribution of 52.2% female and 47.8% male. A significant portion of contacts (32.6%) initiated TPT, but 66.8% had incomplete initiation data. Of those who started therapy, the completion rate was 48.75%, with the remainder having unrecorded outcomes. Multiple linear regression modeling revealed that weight is a strong and statistically significant predictor of BMI, while contact age showed no significant linear relationship. The study concludes that while TPT completion presents a significant challenge, robust data collection is crucial for more conclusive findings, and the models developed serve as a foundational framework for future research and public health interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a multiple linear regression model to predict BMI as the dependent variable using weight and contact age as independent variables, applied to a TB contact-tracing dataset of 722 individuals from Makueni, Kenya. The model demonstrated a strong fit with weight identified as a statistically significant predictor of BMI ($p < 0.05$), while contact age showed no significant linear relationship, and VIF analysis confirmed no multicollinearity between predictors. Key public health findings included a TPT completion rate of only 48.75% among initiators and significant data gaps in 66.8% of initiation records, highlighting critical deficiencies in program monitoring. A reproducible Python script was provided as a deployable analytical artifact, with recommendations to strengthen data collection systems, investigate barriers to TPT adherence, and develop predictive models for TB risk with more complete datasets.

INTERNSHIP Internship Experience

Mombasa Catholic Community Based Health Centre — Comprehensive Analysis of Tuberculosis Cases Using Facility HTS/TB Register Data | Predictive Modeling for TB Treatment Outcomes

Dataset Used: Facility HTS/TB Register from Mombasa Catholic Community Based Health Centre (July 2025, 360 registered client records). Dataset containing TB patient records with variables including age, sex, HIV status, TB test results, diagnostic modality, and treatment outcomes, used to analyze epidemiological patterns, identify predictors of unfavourable treatment outcomes, and support evidence-based TB program management through descriptive statistics and machine learning classification models.

METHODOLOGY

Data Extraction and Cleaning: Parsed date fields to standard YYYY-MM format, coerced non-numeric ages, normalized categorical free-text entries (sex, HIV, outcome) | Descriptive Analysis: Computed counts, proportions, median and IQR for demographic and clinical variables | Predictive Modeling: Trained Random Forest classifier to identify patients at risk of unfavourable outcomes using age, sex, and HIV status as predictors (75/25 train-test split) | Model Evaluation: Assessed performance using accuracy, precision, recall, F1-score, and AUC | Visualization: Generated figures and tables to visualize trends in age distribution, sex ratios, HIV status, and treatment outcomes

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified median age of 31 years among 360 registered clients, with females comprising 58.6% and males 41.4% | Established TB disproportionately affects adults aged 25-44 years (63.6% of cases) | Revealed strong association between HIV co-infection and adverse treatment outcomes, highlighting need for integrated TB/HIV management | Demonstrated feasibility of using basic demographic and clinical data to flag high-risk patients for proactive follow-up | Identified critical gaps in data recording, particularly for HIV status and treatment outcome variables | Provided actionable recommendations for strengthening TB surveillance, improving adherence counselling, and integrating predictive tools into facility dashboards

CURRENT STATUS

no job · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Health Care Data Analysis Report

BACKGROUND & METHODOLOGY

This report presents an exploratory data analysis of a healthcare dataset containing 1,311 patient records with demographic, medical, administrative, and billing information. The primary aim was to identify patterns in medical conditions, patient demographics, and relationships between age and billing amounts. The analysis applied descriptive statistics including frequency distributions and mode analysis, alongside bar charts and scatter plots. A Gaussian Naïve Bayes classifier was built to predict medical conditions using age, gender, blood type, test results, admission type, medication, billing amount, and insurance provider as independent variables. The model achieved an accuracy of 85.75%, precision of 1.0, recall of 0.83, and an F1-score of 0.91, demonstrating reliable predictive performance. Key findings revealed that arthritis, cancer, and diabetes were the most prevalent conditions, with older age groups more commonly affected by arthritis and hypertension. The insights gained support improved resource allocation, early disease detection, and evidence-based clinical decision-making in healthcare settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Gaussian Naïve Bayes classifier to predict medical conditions from patient records in a dataset of 1,311 entries, using demographic and clinical variables including age, gender, blood type, test results, admission type, medication, and billing amount. The model achieved an accuracy of 85.75%, perfect precision of 1.0, recall of 0.83, and an F1-score of 0.91, demonstrating strong predictive reliability with minimal false positives. Statistical analysis identified arthritis, cancer, and diabetes as the most prevalent conditions, with age emerging as a key determinant for chronic disease occurrence. A deployment framework was designed to allow healthcare staff to input patient data and receive real-time medical condition predictions, with recommendations to integrate the model into hospital decision-support systems and explore ensemble methods to further improve precision.

INTERNSHIP Internship Experience

New Wanainchi Maternity and Nursing Home — Child Growth Monitoring and Weight Category Prediction Using CWC Data

Dataset Used: Child Welfare Clinic (CWC) hard copy records from New Wanainchi Maternity and Nursing Home, digitized into Excel by the intern. CWC dataset containing child growth indicators including age, sex, height, weight, village, vitamin A supplementation status, immunization status, and deworming records for children under five years, used to analyze nutritional growth patterns and predict weight categories.

METHODOLOGY

Entered and validated CWC hard copy records into Excel; cleaned data by handling missing values, removing unnecessary columns, and encoding categorical variables; analyzed diagnosis register data and computed summary statistics; created visualizations using Excel and Python including scatterplots and histograms; built, trained, tested, and deployed a machine learning-based child growth prediction app; and participated as part of a pair in the capstone project analyzing hospital admission records using a Gaussian Naïve Bayes and Random Forest Classifier.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier to predict child weight categories from CWC data, achieving an accuracy of 83% and F1-score of 0.91, demonstrating strong potential for AI-supported child health assessment at community level; capstone project analysis of hospital admission records identified arthritis (28%), diabetes (22%), and cancer (15%) as the most prevalent conditions, with female patients comprising 55% of admissions and older patients associated with higher billing amounts; the Gaussian Naïve Bayes model achieved 85.75% overall accuracy with perfect recall for minority class detection; outcomes included a deployed child growth prediction app and recommendations for regular CWC data updates, capacity building for health workers in digital data management, and community sensitization campaigns on child nutrition and growth monitoring.

CURRENT STATUS

Still searching · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Not yet working

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project *Health Care Data Analysis Report*

BACKGROUND & METHODOLOGY

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INTERNSHIP Internship Experience

Moving the Goal Posts (MTG), Kilifi County — Community Health and Empowerment Program Implementation Analysis

Dataset Used: MTG internal activity reports and attendance registers, Kilifi County. Program dataset containing variables including county, department, project name, activity area, topics covered, and data source, drawn from MTG's monitoring records across departments including C.E.F., Football, and Tafisa initiatives, covering thematic areas such as menstrual hygiene, mental health, life skills, teamwork, and leadership.

METHODOLOGY

Performed data entry and validation of program records; reviewed weekly activity and attendance reports; assisted in preparation of performance summaries and charts for monitoring meetings; cleaned and organized MTG program data including handling duplicates, missing values, and inconsistent naming; conducted descriptive analysis using frequency counts and distributions; applied a Gaussian Naïve Bayes (GaussianNB) classifier for predictive modeling of program outcomes; evaluated model performance using a confusion matrix; and produced visualizations including histograms and bar charts using Python, Pandas, Matplotlib, and Seaborn.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully analyzed and visualized MTG program implementation data, identifying Kilifi County and the C.E.F. department as the highest activity areas, with Football and Tafisa as the most active projects and menstrual hygiene, mental health, teamwork, and leadership as the most frequently covered topics; the Gaussian Naïve Bayes classifier achieved approximately 100% accuracy, correctly classifying 231 out of 232 instances, confirming dominant implementation patterns; findings highlighted underrepresentation of Tana River and Mombasa counties as priority gaps for equitable coverage expansion, with recommendations to standardize data collection tools, integrate data science capacity building into program M&E, and increase outreach to underserved counties.

CURRENT STATUS

Job seeker · Unemployed — Actively searching

Sector relevance: Not related

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Analyzing and Predicting Outcomes in Tuberculosis Contact Tracing in Makueni, Kenya

BACKGROUND & METHODOLOGY

This project analyzes a tuberculosis contact-tracing dataset from Makueni, Kenya, to evaluate the effectiveness of Tuberculosis Preventive Therapy (TPT) and identify associated risk factors. The analysis focuses on three key areas: demographics, TPT outcomes, and the relationship between BMI, HIV status, and TB outcomes. Descriptive statistics reveal a diverse cohort with a mean age of 29.69 years and a near-balanced gender distribution of 52.2% female and 47.8% male. A significant portion of contacts (32.6%) initiated TPT, but 66.8% had incomplete initiation data. Of those who started therapy, the completion rate was 48.75%, with the remainder having unrecorded outcomes. Multiple linear regression modeling revealed that weight is a strong and statistically significant predictor of BMI, while contact age showed no significant linear relationship. The study concludes that while TPT completion presents a significant challenge, robust data collection is crucial for more conclusive findings, and the models developed serve as a foundational framework for future research and public health interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a multiple linear regression model to predict BMI as the dependent variable using weight and contact age as independent variables, applied to a TB contact-tracing dataset of 722 individuals from Makueni, Kenya. The model demonstrated a strong fit with weight identified as a statistically significant predictor of BMI ($p < 0.05$), while contact age showed no significant linear relationship, and VIF analysis confirmed no multicollinearity between predictors. Key public health findings included a TPT completion rate of only 48.75% among initiators and significant data gaps in 66.8% of initiation records, highlighting critical deficiencies in program monitoring. A reproducible Python script was provided as a deployable analytical artifact, with recommendations to strengthen data collection systems, investigate barriers to TPT adherence, and develop predictive models for TB risk with more complete datasets.

INTERNSHIP Internship Experience

Viwa Healthcare Services — Analysis of Gynecologist Clinical Data and Diagnostic Trends | AI for Reproductive Health Insights

Dataset Used: VIWA Gynecologist Clinic patient records (112 patient encounters) from the 2024-2025 period, obtained from the digital patient register. Dataset containing clinical records with variables including age, gender, weight, blood pressure, diagnosis (DDX), and management plan (PLAN), used to assess prevalence of key reproductive health issues, identify data quality gaps, and generate actionable insights for resource allocation and clinical decision-making through descriptive statistics and Python-based visualization

METHODOLOGY

Internship Centre Work: Data entry and updating patient registers; Validating and cleaning health records in the physical system; Assisting in patient flow management and basic triage; Supporting preparation of monthly patient summary reports | Capstone Project: Data cleaning and standardization of diagnoses (DDX) and management plans (PLAN); Applied descriptive statistics to continuous variables (AGE, WEIGHT) and frequency analysis to categorical variables (DDX, GENDER, PLAN) using Python (Pandas) and Matplotlib for visualization; Generated horizontal bar charts for ranked diagnoses, pie charts for gender proportions, and histograms for age distribution; Identified and grouped low-frequency diagnoses into 'Other Rare Diagnoses' category

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated and cleaned approximately 112 patient records, ensuring data integrity for analysis; Identified patient demographic as overwhelmingly female (90.2%), with mean age of 33.2 years and primary demographic within 25-34 reproductive age bracket (33.9% of cases); Most prevalent specific diagnoses: Subfertility/Infertility (24.1%) and Vulvovaginal Candidiasis (16.1%); Drugs/Medication was leading management plan (49.1%); Identified data quality gaps: 6.3% missing WEIGHT entries and 8 records with missing diagnosis highlighting need for standardized data entry; Developed weekly summary reports on patient flow and clinical activity; Contributed to improvement of data quality by identifying non-standardized entries in DDX field leading to process review; Provided evidence-based recommendations for structured data entry forms, targeted subfertility screening, and consistent medication stock management

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Partially related

SKILLS IN USE

Data collection · Data cleaning & wrangling · Data visualization · Natural Language Processing (NLP) · Prompt engineering · Generative AI / LLMs · Ethics & bias in AI · Freelancing in data science / building a portfolio · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Maternal and Newborn Insights: Exploring the Relationship Between Maternal Health and Newborn Outcomes

BACKGROUND & METHODOLOGY

This study contains information about mothers and their newborn babies to help understand the factors that affect a baby's health at birth. It includes details about the mother such as age, weight, blood pressure, and mode of delivery, as well as information about the baby including weight, height, and gender. By applying data analysis and machine learning techniques, specifically the Naïve Bayes model, the study cleaned, analyzed, and modeled hospital and clinic records of 140 mother-newborn pairs to uncover patterns and relationships. The dataset was cleaned to retain four key variables: maternal age, maternal weight, baby gender, and newborn weight. A Gaussian Naïve Bayes classifier was trained to predict newborn weight category, achieving 100% accuracy on the test set, with perfect precision, recall, and F1-scores across all classes. Analysis revealed a moderate positive correlation between maternal weight and newborn weight, with most babies born to mothers aged 16–34 years falling within a healthy weight range of 3.0–3.5 kg. The study highlights the potential of machine learning to support prenatal care decision-making and improve maternal and child health outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Gaussian Naïve Bayes classifier to predict newborn birth weight categories (low, normal, and high) using maternal age and weight from hospital and clinic records of 140 mother-newborn pairs. The model achieved a perfect accuracy of 100%, with precision, recall, and F1-scores of 1.00 across all three classes, correctly classifying every baby in the test set without misclassification. Exploratory analysis identified a moderate positive correlation between maternal weight and newborn weight, with maternal age showing minimal independent effect. The findings provide a foundation for targeted prenatal interventions, with recommendations to expand the dataset, incorporate additional clinical variables such as nutrition and pregnancy complications, and explore advanced models to improve generalizability for broader populations.

INTERNSHIP Internship Experience

Pwani University Health Unit — Drug Dispensing Patterns and Co-Occurrence Analysis | Association Rule Learning for Pharmaceutical Inventory Management | AI for Drug Demand Forecasting

Dataset Used: Drug dispensing records from Pwani University Health Unit (2024, ~50,000 transaction records in Excel format containing Patient Number, Drug Name, Quantity Dispensed, Date Dispensed, Unit of Measure, Cost per Unit, and Total Cost). Study investigating drug dispensation patterns at the PU Health Unit to identify medicines that are often prescribed or dispensed together using Association Rule Learning (ARL). Dataset comprised transactional records with drug name, quantity, unit cost, and total cost, used to apply exploratory analysis and Apriori algorithm for finding frequent drug combinations and supporting inventory management, rational prescribing, and stock control decisions.

METHODOLOGY

Internship Centre Work: Patient registrations, arranging medical records and student reports, facility inventory management, pharmacy organization, monthly report analysis, and data entry | Capstone Project: Data cleaning and standardization of drug names, removal of duplicates, handling missing values | Defined "transactions" as all medicines a single patient received on a single day for basket analysis | Applied descriptive statistics to identify most frequently dispensed drugs | Applied Association Rule Learning (Apriori algorithm) using Python's mlxtend library to find drug co-occurrence rules | Evaluated rules using Support, Confidence, and Lift metrics | Generated visualizations using bar graphs, network diagrams, and co-occurrence plots | Developed Streamlit-based web interface for staff to access predictive rules | Used Python with Pandas, mlxtend, Matplotlib, and Seaborn for analysis and visualization

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified top dispensed drugs: Paracetamol, Amoxicillin, Metronidazole, Ibuprofen, and Diclofenac as most frequently issued medications indicating pain relief and infection management as dominant treatment patterns | Found 8 strong predictive rules with Support ≥ 0.01 , Lift ≥ 1.06 after applying strict filters | Key associations: Paracetamol $\rightarrow$ Amoxicillin (support=0.12, confidence=0.78, lift=1.9), Metronidazole $\rightarrow$ Amoxicillin (lift=1.6), Paracetamol $\rightarrow$ Ibuprofen (lift=1.8) | High Lift values (1.5–1.9 for common pairs; up to 21.02 for specialized combinations like Normax + Cital liquid) confirmed meaningful prescribing relationships beyond random chance | Identified fungal/bacterial infection treatment bundles: Griseofil 500mg tablet + CLOB CREAM (lift=14.90), Normax 400mg tablets + Cital liquid 100mls (lift=21.02) | Provided actionable insights for inventory management: ensuring frequently co-dispensed drugs are stocked together to prevent stockouts | Enabled rational drug use support by identifying recurring prescribing trends for clinical decision support | Successfully deployed Streamlit web app providing pharmacy staff with real-time access to co-dispensing rules | Enhanced practical skills in data cleaning, association rule mining, predictive analytics, and dashboard deployment for healthcare applications

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Maternal and Newborn Insights: Exploring the Relationship Between Maternal Health and Newborn Outcomes

BACKGROUND & METHODOLOGY

This study contains information about mothers and their newborn babies to help understand the factors that affect a baby's health at birth. It includes details about the mother such as age, weight, blood pressure, and mode of delivery, as well as information about the baby including weight, height, and gender. By applying data analysis and machine learning techniques, specifically the Naïve Bayes model, the study cleaned, analyzed, and modeled hospital and clinic records of 140 mother-newborn pairs to uncover patterns and relationships. The dataset was cleaned to retain four key variables: maternal age, maternal weight, baby gender, and newborn weight. A Gaussian Naïve Bayes classifier was trained to predict newborn weight category, achieving 100% accuracy on the test set, with perfect precision, recall, and F1-scores across all classes. Analysis revealed a moderate positive correlation between maternal weight and newborn weight, with most babies born to mothers aged 16–34 years falling within a healthy weight range of 3.0–3.5 kg. The study highlights the potential of machine learning to support prenatal care decision-making and improve maternal and child health outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Gaussian Naïve Bayes classifier to predict newborn birth weight categories (low, normal, and high) using maternal age and weight from hospital and clinic records of 140 mother-newborn pairs. The model achieved a perfect accuracy of 100%, with precision, recall, and F1-scores of 1.00 across all three classes, correctly classifying every baby in the test set without misclassification. Exploratory analysis identified a moderate positive correlation between maternal weight and newborn weight, with maternal age showing minimal independent effect. The findings provide a foundation for targeted prenatal interventions, with recommendations to expand the dataset, incorporate additional clinical variables such as nutrition and pregnancy complications, and explore advanced models to improve generalizability for broader populations.

INTERNSHIP Internship Experience

Pwani University Health Unit — Dispensation Trend Analysis and Predictive Modeling for Pharmaceutical Inventory Management | AI for Drug Demand Forecasting

Dataset Used: Drug dispensation records from Pwani University Health Unit (January–September 2025), originally in PDF format, converted to Excel for analysis. Study investigating drug dispensation patterns at the PU Health Unit to identify the most frequently dispensed pharmaceutical products as indicators of prevalent infections in the university community. Dataset comprised transactional records with drug name, quantity, unit cost, and total cost, used to apply exploratory analysis and ARIMA time series forecasting for predicting future drug demand and supporting inventory management decisions.

METHODOLOGY

Internship Centre Work: Patient registrations, arranging medical records and student reports, facility inventory management, pharmacy organization, monthly report analysis, and data entry | Capstone Project: Converted raw data from PDF to Excel format; inspected dataset for inconsistencies and missing values; dropped unnecessary columns; handled missing data using

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified top dispensed drugs: Griseoful 500mg tablet, Amoxicillin+Cloxacillin Capsules, and Amoxicillin 500mg Capsule as most frequently issued medications, indicating prevalent treatment of parasitic infections, bacterial infections, and related conditions | ARIMA model achieved MAE = 0.1332, MSE = 0.0178, RMSE = 0.1332, indicating high prediction accuracy and reliable forecasts for future drug demand | Detected seasonal patterns in dispensation (e.g., REMIFLU spikes indicating recurring flu seasons) enabling proactive public health response | Correlation analysis revealed weak negative correlation between Quantity and Unit Cost (-0.26), weak positive between Quantity and Total Cost (0.23), moderate positive between Unit Cost and Total Cost (0.56) | Provided actionable insights for inventory management, procurement planning, budgeting, and targeted public health interventions | Demonstrated practical application of time series forecasting in healthcare settings for demand prediction and resource allocation | Enhanced technical skills in data cleaning, exploratory analysis, visualization, and predictive modeling using Python tools

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Cancer Dataset Analysis Report: Predicting HIV Status Using Categorical Naïve Bayes

BACKGROUND & METHODOLOGY

The integration of machine learning in healthcare has become increasingly significant in supporting disease prediction, early diagnosis, and effective decision-making. This project explores the application of the Categorical Naïve Bayes (CNB) algorithm to predict HIV status outcomes using demographic and clinical features including sex, county of residence, and age. The dataset was divided into training and testing subsets, and categorical features were handled using CNB with a predefined `min_categories` parameter to accommodate potential unseen categories during prediction. The model was trained to learn conditional probabilities between features and diagnostic outcomes and evaluated using accuracy metrics. Results indicated that the model captured statistical relationships between the categorical variables and diagnosis labels, achieving an accuracy of 60%. Exploratory analysis revealed that age, sex, and county of residence — particularly Kakamega — were key influencing factors in HIV status prediction. This work demonstrates how probabilistic models can provide clinicians and policymakers with simple, robust tools for early screening and targeted interventions in low-resource environments.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Categorical Naïve Bayes classifier to predict HIV status outcomes using patient demographic features including sex, age, and county of residence, with the model trained and evaluated on split training and testing subsets. The model achieved an accuracy of 60%, effectively capturing statistical relationships between demographic variables and HIV diagnosis labels, with a confusion matrix showing a high proportion of correct predictions and minimal false negatives. A simple deployment interface was developed allowing healthcare practitioners to input patient details and receive predicted HIV status outcomes, with recommendations to expand the feature set with clinical and lifestyle variables, validate on larger datasets, and explore ensemble methods to reduce false negatives and improve overall predictive reliability.

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

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INTERNSHIP Internship Experience

World Friends Kenya — Optimizing Malaria Service Delivery: Retrospective Analysis of Patient Flow and Retention in Kilifi North (2022–2025) | Machine Learning for Malaria Trend Prediction

Dataset Used: Routinely collected malaria clinic records from Kilifi North health facilities (November 2022–May 2025, 2,200 entry records; 100+ malaria record files reviewed). Dataset containing malaria patient flow data tracking suspected cases, tested cases, and confirmed cases across the care pathway (consultation ? laboratory testing ? pharmacy). Variables include IPT dosage (1st, 2nd, 3rd dose), AL distribution by weight band (5–14 kg, 15–24 kg, 25–34 kg, 35+ kg), and laboratory test results (RDT and blood smear), used to analyze service delivery trends, retention patterns, and predictors of malaria confirmation using ridge regression modeling.

METHODOLOGY

Supported maternal and child health (MCH) data review and monthly report preparation; conducted data quality checks on clinic registers; extracted and cleaned 2,200 malaria records from Nov 2022–May 2025; standardized variables for IPT dosage, AL weight-band distribution, and lab test results (RDT/BS); performed descriptive analysis of malaria trends using Excel and Python; applied ridge regression to assess collinearity among RDT, IPT, and blood smear features; generated visualizations (pie charts, line graphs, bar charts) to illustrate service flow from consultation to pharmacy and case confirmation trends; evaluated model performance using accuracy metrics.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established IPT dose coverage: 37.4% received 1st dose, 33.2% 2nd dose, 29.4% 3rd dose, revealing dropout patterns across the prevention cascade; ridge regression model achieved 62% accuracy, demonstrating potential for predicting malaria confirmation trends; identified significant gaps between tested and confirmed cases, highlighting data quality and reporting challenges; visualized patient flow showing higher laboratory referrals from consultation and seasonal variation in confirmed cases; distinguished confirmation trends between under-5 and 5+ age groups; enhanced practical skills in data cleaning, statistical analysis, and predictive modeling for public health; provided actionable recommendations for strengthening malaria follow-up, improving data entry accuracy, and targeting interventions during peak transmission seasons.

CURRENT STATUS

Searching · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Data visualization · Public health data analysis

Confidence applying ENGAGE skills: **5 / 5**